
LF3: MACHINE-VERIFIED SINGLET KERNEL AND PROJECTIVE POINTER-SECTOR DECOMPOSITION WITH STRUCTURAL LEAKAGE BOUNDS

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ABSTRACT

LF3 formalises in Lean4 the finite-dimensional singlet-kernel and projective pointer-sector decomposition used in the Bell-singlet measurement layer of Constraint-Surface Dynamics. It is the third layer of the CSD formalisation programme. LF1 formalises repeated-trial frequency convergence, LF2 formalises the measure bridge and Born-weight wrapper, and LF3 applies the LF2 operator-form weight interface to a finite-dimensional projective readout model for the two-qubit singlet sector.

The purpose of LF3 is not to rederive typicality, the Fubini–Study measure bridge, or the Born-form probability assignment. Nor does LF3 derive the pointer-readout dynamics from a concrete microscopic finite-dimensional pointer generator. Instead, the Hamiltonian-control and branch-separation layers are encoded as explicit structure-field interfaces. Against those interfaces, LF3 verifies the projective pointer-sector decomposition, the branch-weight formula, the singlet probability kernel, the correlation identity, the marginal identities, and structural finite-leakage stability.

The formalisation is carried out for a two-qubit system coupled to two finite-dimensional pointer sectors. The joint system–apparatus Hilbert space is

$$H_{SA}^{(N)} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes K_A^{(d_A)} \otimes K_B^{(d_B)}, \quad N = 4d_A d_B.$$

For detector settings $a, b \in S^2$, the pointer-sector readout is represented by four projectors

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \widehat{Q}_s^A \otimes \widehat{Q}_t^B, \quad s, t \in \{\pm 1\}.$$

LF3 verifies that these projectors form a complete orthogonal projective readout and that, when applied to the branch-separated singlet-pointer state supplied by the structural interface, their operator-form weights reproduce the singlet kernel.

The main formalised calculation is

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

Consequently,

$$E(a, b) = \sum_{s,t=\pm 1} st P_{st}(a, b) = -a \cdot b,$$

and the local marginals are

$$P_A(s \mid a, b) = \frac{1}{2}, \quad P_B(t \mid a, b) = \frac{1}{2}.$$

Thus operational no-signalling follows by direct marginal summation.

LF3 also formalises structural finite-leakage control. The parameters ϵ_A and ϵ_B are treated as explicit pointer-readout compatibility bounds. They encode the compatibility between the selected pointer

generators, translated pointer states, and readout subspaces. They are not derived in LF3 from a concrete microscopic pointer construction. Under these structural bounds,

$$\left| P_{st}(a, b; T) - \frac{1 - st a \cdot b}{4} \right| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

It follows that

$$|E_T(a, b) + a \cdot b| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

and

$$\left| P_A(s \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

with the analogous bound for P_B .

LF3 therefore supplies the machine-checked singlet-kernel and projective pointer-sector layer of the CSD formalisation branch. Its scope is deliberately restricted. It does not formalise the Ashtekar–Schilling correspondence, the full ontic basin dynamics on Σ , general POVMs, mixed states, subsystem reduction, arbitrary multipartite Bell scenarios, or a derivation of small leakage from an explicit finite pointer model. Within its finite-dimensional singlet setting, LF3 establishes a complete theorem package for projective readout, singlet correlations, no-signalling marginals, structural leakage stability, and context-indexed outcome maps.

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1 Introduction

1.1 Purpose of LF3

The purpose of LF3 is to formalise, in Lean4, the finite-dimensional singlet-kernel and projective pointer-sector decomposition used in the Bell-singlet measurement layer of Constraint-Surface Dynamics. The target is deliberately narrow. LF3 does not attempt to derive the Born rule, reconstruct the Fubini–Study measure, prove repeated-trial typicality, or classify all physically admissible measurement partitions. Those results and assumptions belong to earlier or later layers of the programme.

The central object of LF3 is the singlet probability kernel

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}, \quad s, t \in \{\pm 1\}.$$

LF3 verifies this kernel as the projective pointer-sector weight obtained by applying the LF2 operator-form Born-weight interface to a finite-dimensional readout decomposition. The relevant projectors are

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \widehat{Q}_s^A \otimes \widehat{Q}_t^B.$$

These projectors act on the finite-dimensional system–apparatus Hilbert space

$$H_{SA}^{(N)} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes K_A^{(d_A)} \otimes K_B^{(d_B)}, \quad N = 4d_A d_B.$$

The Hamiltonian-control and branch-separation layers are included in LF3 as explicit structural interfaces. In particular, LF3 assumes the data required to connect a measurement context (a, b) to a branch-separated final state,

$$|\Psi_T\rangle = \sum_{s,t \in \{\pm 1\}} c_{st}(a, b) |s_a, t_b\rangle \otimes |\phi_s^A(T)\rangle \otimes |\phi_t^B(T)\rangle.$$

The formal contribution of LF3 is then to verify the projective readout structure, the branch-weight formula, the singlet kernel, the correlation, the marginals, and the finite-leakage bounds conditional on that explicit interface.

This distinction is important. LF3 does not derive the smallness of the leakage parameters from a concrete microscopic pointer generator. The parameters

$$\epsilon_A, \epsilon_B$$

are structural pointer-readout compatibility bounds. They encode how well the translated pointer states are resolved by the chosen readout subspaces. A later refinement may derive such bounds from a specific finite pointer construction, such as a cyclic shift on a discrete position lattice or a finite wavepacket model. That derivation is outside LF3 v1.00.

The ideal strong-readout result is

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

From this,

$$E(a, b) = \sum_{s,t \in \pm 1} st P_{st}(a, b) = -a \cdot b$$

follows directly. The local marginals satisfy

$$P_A(s \mid a, b) = \frac{1}{2}, \quad P_B(t \mid a, b) = \frac{1}{2}.$$

Thus the formalised pointer-sector construction reproduces the Bell-singlet correlation and the no-signalling marginals at the level of operator-form branch weights.

At finite separation, the branch weights satisfy the explicit bound

$$\left| P_{st}(a, b; T) - \frac{1 - st a \cdot b}{4} \right| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

The corresponding correlation and marginal formulae therefore remain stable under controlled structural leakage. This finite-leakage layer prevents the formalisation from depending only on an exact limiting readout while making clear which compatibility assumptions are imported into LF3.

1.2 Position within the Lean Formalisation Programme

The Lean formalisation branch of Constraint-Surface Dynamics is organised as a sequence of narrow theorem layers rather than as a single monolithic reconstruction. LF1 formalises the repeated-trial theorem underlying volume typicality. Its role is to show that, under a repeated-preparation model, empirical frequencies converge to normalised ontic weights. LF1 is therefore a frequency theorem, not a Born-rule theorem.

LF2 formalises the next layer. It isolates the finite-dimensional measure-and-weight interface between deterministic volume typicality and operational quantum weights. In particular, LF2 treats the pushforward from ontic Liouville measure to projective-state weights and packages the Born-weight layer using explicitly stated external theorem inputs. LF2 is therefore the appropriate upstream dependency for LF3: it supplies the formal probability interface that LF3 uses when assigning weights to projective pointer-sector outcomes.

LF3 occupies the next position in this sequence:

$$\text{LF1} \longrightarrow \text{LF2} \longrightarrow \text{LF3} \longrightarrow \text{LF4} \longrightarrow \text{LF5}.$$

Within this ordering, LF3 is the first formalisation layer concerned with an explicit system-apparatus measurement model. It does not generalise the operational dictionary to mixed states, POVMs, or subsystem reduction. Those topics belong to LF4. It also does not formalise outcome-conditioned update or sequential experiments. Those topics belong to LF5. LF3 is instead the projective-readout layer: it shows how the finite-dimensional Born-weight interface is instantiated in a concrete pointer-sector measurement for the Bell-singlet case.

This division of labour is essential. If LF3 attempted simultaneously to formalise typicality, the measure bridge, Born-weight uniqueness, pointer dynamics, mixed states, POVMs, and sequential update, the result would obscure the logical dependencies of the programme. The aim of LF3 is more disciplined: to make the TN3 constructive singlet layer machine-checkable while preserving the upstream and downstream theorem boundaries.

1.3 Relation to the TN3 Quantum-Pointer Construction

TN3 provides the mathematical and physical construction that LF3 formalises. It shows that the standard von Neumann measurement Hamiltonian embeds cleanly into the finite-dimensional geometric architecture used by Constraint-Surface Dynamics. The role of TN3 is constructive rather than foundational. It does not introduce a new measurement Hamiltonian, derive the Born rule, or solve the general measurement problem. Instead, it supplies the explicit system-apparatus realisation of the singlet probability kernel obtained upstream.

The central TN3 construction is the following. A two-qubit singlet state is coupled to two finite-dimensional pointer sectors. The detector settings $a, b \in S^2$ determine the Hamiltonian $\hat{H}_{a,b}$. The resulting unitary evolution separates the initial state into four pointer branches. The readout is then represented by pointer-sector projectors $\widehat{M}_{st}^{(a,b)}$, and the corresponding branch weights are operator-form quantities

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

Under the branch-separation assumptions, these weights approximate the squared singlet amplitudes:

$$P_{st}(a, b; T) = |c_{st}(a, b)|^2 + O(\epsilon_A + \epsilon_B).$$

In the strong-readout limit this becomes

$$P_{st}(a, b) = \frac{1 - st \cdot a \cdot b}{4}.$$

LF3 translates this structure into a Lean-compatible theorem map. The formalisation is not intended to reproduce every analytic detail of the continuum von Neumann pointer model. Instead, it works directly with the finite-dimensional truncation used by TN3. This is the correct target for Lean because all relevant spaces are finite-dimensional, the pointer projectors are algebraic objects, the singlet calculation is finite, and the leakage estimates can be stated as explicit inequalities.

The formalisation also records a key interpretive distinction. The construction gives definite outcomes per realised measurement context, but it does not define a Bell-local factorisation of outcomes into functions of the form

$$A(a, \lambda), \quad B(b, \lambda).$$

Nor does it define a single global assignment of outcomes across all incompatible CHSH contexts. The relevant outcome map is context-indexed:

$$F^{(a,b)} : \Sigma_{SA}^{(N)} \longrightarrow \{\pm 1\} \times \{\pm 1\}.$$

This is why the construction can reproduce Bell-singlet correlations without assuming the global joint assignment structure excluded by Fine-type arguments. In LF3, this Bell-consistency layer is treated as a context-map theorem boundary. The direct singlet correlation and marginal results are formalised explicitly, while the broader Fine-Bell obstruction may be imported or stated as an external structural theorem.

1.4 Main Results

The main formal results of LF3 are organised into six theorem groups.

First, LF3 formalises the singlet probability kernel:

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

This is the central algebraic result of the paper. It is obtained by applying the LF2 operator-form weight interface to the finite-dimensional projective pointer-sector readout and evaluating the two-qubit singlet amplitudes.

Second, LF3 formalises the Bell-singlet correlation:

$$E(a, b) = \sum_{s, t \in \{\pm 1\}} st P_{st}(a, b) = -a \cdot b.$$

This follows by finite summation over the two-element sign type.

Third, LF3 formalises the no-signalling marginal identities:

$$P_A(s \mid a, b) = \frac{1}{2}, \quad P_B(t \mid a, b) = \frac{1}{2}.$$

These identities follow by summing the singlet kernel over the remote outcome.

Fourth, LF3 formalises the projective pointer-sector decomposition. For each measurement context (a, b) , the four readout projectors

$$\widehat{M}_{st}^{(a, b)} = I_{AB} \otimes \widehat{Q}_s^A \otimes \widehat{Q}_t^B$$

are mutually orthogonal and complete:

$$\widehat{M}_{st}^{(a, b)} \widehat{M}_{s't'}^{(a, b)} = 0 \quad \text{when} \quad (s, t) \neq (s', t'),$$

and

$$\sum_{s, t \in \{\pm 1\}} \widehat{M}_{st}^{(a, b)} = I_{SA}.$$

Fifth, LF3 formalises structural finite-leakage stability. Given pointer-readout compatibility bounds ϵ_A, ϵ_B , the finite-time branch weights obey

$$\left| P_{st}(a, b; T) - \frac{1 - st a \cdot b}{4} \right| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

The corresponding finite-leakage correlation and marginal bounds are

$$|E_T(a, b) + a \cdot b| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

and

$$\left| P_A(s \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

with the analogous bound for P_B .

Sixth, LF3 formalises the context-indexed outcome-map structure. It distinguishes outcome maps for realised measurement contexts from a global CHSH assignment across incompatible contexts. This ensures that LF3 reproduces the singlet correlation and no-signalling marginals without assuming Bell-local factorisation or a global Fine-style joint assignment.

1.5 Scope of the Formalisation

The scope of LF3 is finite-dimensional, projective, and singlet-specific. The formalisation is intentionally restricted to the two-qubit Bell-singlet measurement with finite-dimensional pointer sectors. This restriction is not a weakness of the paper. It is the reason the formalisation is well-posed and achievable.

LF3 does not formalise the Ashtekar-Schilling correspondence between Schrödinger evolution and Hamiltonian flow on projective Hilbert space. That correspondence remains part of the external geometric quantum mechanics background. LF3 uses the finite-dimensional Hilbert-space measurement model directly.

LF3 does not formalise the full ontic basin dynamics on Σ . The pointer-sector construction can be interpreted within the CSD ontic picture, but the Lean target is the finite-dimensional system-apparatus readout structure, not a complete classification of ontic basins or physically realised measurement partitions.

LF3 does not formalise general POVMs, mixed states, or subsystem reduction. Those topics require the broader operational machinery developed in TN4 and are assigned to LF4.

LF3 does not formalise sequential measurement update. The outcome-conditioned update and sequential-experiment layer is developed in TN6 and is assigned to LF5.

LF3 also does not prove a general theorem about all Bell scenarios. It treats the two-qubit singlet as the canonical finite-dimensional case in which the pointer-sector construction can be made explicit. The broader no-go-theorem analysis is supplied by companion Fine-Bell work and is used here only to state the correct boundary between context-indexed deterministic outcomes and forbidden global joint assignments.

1.6 Structure of the Paper

Section 2 defines the finite-dimensional system-apparatus setup. It introduces the two-qubit system, finite-dimensional pointer sectors, detector settings, spin projectors, and the joint Hilbert space used throughout the formalisation.

Section 3 defines the impulsive measurement Hamiltonian and proves the commutation and factorisation results needed for the measurement unitary.

Section 4 treats branch separation. It defines the initial singlet-pointer state, expands it in the joint measurement basis, states the branch-separated final state, and introduces the leakage parameters ϵ_A and ϵ_B .

Section 5 defines the pointer-sector projectors and the corresponding branch weights. It proves orthogonality, completeness, and the finite-leakage branch-weight estimate.

Section 6 specialises the construction to the two-qubit singlet. It proves the squared-amplitude identity

$$|c_{st}(a, b)|^2 = \frac{1 - st \, a \cdot b}{4}.$$

Section 7 derives the correlation and marginal stability results:

$$E(a, b) = -a \cdot b, \quad P_A(s \mid a, b) = P_B(t \mid a, b) = \frac{1}{2},$$

together with their finite-leakage corrections.

Section 8 treats context-indexed outcome maps and Bell consistency. It explains why the formalised construction supplies deterministic outcomes per realised context without implying Bell-local factorisation or a global joint assignment across incompatible settings.

Section 9 gives the Lean4 formalisation architecture. It lists the modules, theorem groups, imported results, and intended top-level export structure.

Section 10 situates LF3 within the CSD dependency stack, especially its relation to Paper B, TN2, TN3, LF1, LF2, and the later LF4 and LF5 layers.

Section 11 records the assumptions and non-claims of the paper.

Section 12 concludes by summarising what LF3 establishes and how it prepares the next stages of the Lean formalisation programme.

2 Finite-Dimensional System-Apparatus Setup

2.1 Purpose of the Setup

This section defines the finite-dimensional objects used throughout LF3. The aim is to isolate the system, apparatus, detector settings, and readout subspaces before introducing the measurement Hamiltonian. The setup is algebraic and finite-dimensional, because LF3 is intended to be formalised in Lean4 without requiring infinite-dimensional spectral theory, unbounded operators, or the full analytic structure of continuous von Neumann pointers.

The measured system consists of two spin- $\frac{1}{2}$ degrees of freedom. Each measurement wing is coupled to a finite-dimensional pointer sector. The resulting system-apparatus Hilbert space is

$$H_{SA}^{(N)} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes K_A^{(d_A)} \otimes K_B^{(d_B)}, \quad N = 4d_A d_B.$$

All exact statements in LF3 are made inside this finite-dimensional space. The corresponding ray space is

$$\mathbb{CP}^{N-1}.$$

The Lean formalisation works primarily at the Hilbert-space and operator level, while the projective interpretation is retained at the paper level.

2.2 Two-Qubit System Space

Let

$$H_A \cong \mathbb{C}^2, \quad H_B \cong \mathbb{C}^2$$

denote the two spin- $\frac{1}{2}$ Hilbert spaces. The joint two-qubit system space is

$$H_{AB} = H_A \otimes H_B \cong \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4.$$

The standard computational basis for each qubit is written

$$\{|+\rangle_z, |-\rangle_z\}.$$

The singlet state is

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} (|+\rangle_z \otimes |-\rangle_z - |-\rangle_z \otimes |+\rangle_z).$$

This state is introduced here only as part of the system data. Its rotational invariance and probability consequences are used later in the singlet-specialisation section.

2.3 Finite-Dimensional Pointer Sectors

Each measurement wing is coupled to a pointer sector. In the ideal continuous model, a pointer is represented by a position degree of freedom. LF3 does not formalise that infinite-dimensional construction. Instead, each pointer is replaced by a finite-dimensional truncation:

$$K_A^{(d_A)}, \quad K_B^{(d_B)}, \quad d_A, d_B < \infty.$$

These finite spaces are interpreted as detector windows with sufficiently many resolved modes to distinguish the two readout outcomes during the measurement interval.

For the A -pointer, assume two orthogonal readout subspaces

$$K_A^{(+)}, \quad K_A^{(-)} \subseteq K_A^{(d_A)}.$$

For the B -pointer, assume two orthogonal readout subspaces

$$K_B^{(+)}, \quad K_B^{(-)} \subseteq K_B^{(d_B)}.$$

The corresponding orthogonal projectors are

$$\hat{Q}_+^A, \quad \hat{Q}_-^A, \quad \hat{Q}_+^B, \quad \hat{Q}_-^B.$$

They satisfy

$$\hat{Q}_+^A + \hat{Q}_-^A = I_{K_A}, \quad \hat{Q}_+^A \hat{Q}_-^A = 0,$$

and

$$\widehat{Q}_+^B + \widehat{Q}_-^B = I_{K_B}, \quad \widehat{Q}_+^B \widehat{Q}_-^B = 0.$$

For signs $s, t \in \{\pm 1\}$, define

$$\widehat{Q}_s^A = \begin{cases} \widehat{Q}_+^A, & s = +1, \\ \widehat{Q}_-^A, & s = -1, \end{cases} \quad \widehat{Q}_t^B = \begin{cases} \widehat{Q}_+^B, & t = +1, \\ \widehat{Q}_-^B, & t = -1. \end{cases}$$

These pointer readout projectors are the primitive outcome-sector objects used later in LF3. They replace any need for a Fubini-Study Voronoi-cell construction in this formalisation layer.

2.4 Joint System-Apparatus Hilbert Space

The full finite-dimensional system-apparatus Hilbert space is

$$H_{SA}^{(N)} = H_{AB} \otimes K_A^{(d_A)} \otimes K_B^{(d_B)}.$$

Since

$$\dim_{\mathbb{C}} H_{AB} = 4, \quad \dim_{\mathbb{C}} K_A^{(d_A)} = d_A, \quad \dim_{\mathbb{C}} K_B^{(d_B)} = d_B,$$

we have

$$\dim_{\mathbb{C}} H_{SA}^{(N)} = 4d_A d_B = N.$$

The pure-state space associated with this finite-dimensional Hilbert space is

$$\mathbb{P}(H_{SA}^{(N)}) \cong \mathbb{CP}^{N-1}.$$

In the geometric interpretation of CSD, this projective manifold is the operational state space for the finite system-apparatus model. In the Lean implementation, however, the primary objects are vectors, finite-dimensional inner-product spaces, linear maps, and projectors.

A generic system-apparatus state is written

$$|\Psi\rangle \in H_{SA}^{(N)}.$$

The initial state used later in the measurement calculation is

$$|\Psi_0\rangle = |\psi^-\rangle \otimes |\phi_0^A\rangle \otimes |\phi_0^B\rangle,$$

where

$$|\phi_0^A\rangle \in K_A^{(d_A)}, \quad |\phi_0^B\rangle \in K_B^{(d_B)}$$

are initially neutral pointer states.

2.5 Detector Settings on S^2

A detector setting is a unit vector in $\mathbb{R}^3$. Define

$$S^2 = \{v \in \mathbb{R}^3 : \|v\| = 1\}.$$

The two detector settings are written

$$a, b \in S^2.$$

The Pauli triple is

$$\sigma = (\sigma_x, \sigma_y, \sigma_z),$$

with

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For a setting $a = (a_x, a_y, a_z) \in S^2$, define

$$\sigma_a = \sigma \cdot a = a_x \sigma_x + a_y \sigma_y + a_z \sigma_z.$$

Similarly,

$$\sigma_b = \sigma \cdot b.$$

Since a and b are unit vectors, the operators σ_a and σ_b have eigenvalues $+1$ and -1 .

2.6 Spin Projectors and Measurement Bases

For $s \in \{\pm 1\}$, let

$$|s_a\rangle$$

denote a normalised eigenvector of σ_a with eigenvalue s :

$$\sigma_a |s_a\rangle = s |s_a\rangle.$$

For $t \in \{\pm 1\}$, let

$$|t_b\rangle$$

denote a normalised eigenvector of σ_b with eigenvalue t :

$$\sigma_b |t_b\rangle = t |t_b\rangle.$$

The corresponding one-qubit projectors are

$$\Pi_s^A(a) = |s_a\rangle\langle s_a| = \frac{1}{2} (I + s\sigma_a),$$

and

$$\Pi_t^B(b) = |t_b\rangle\langle t_b| = \frac{1}{2} (I + t\sigma_b).$$

The joint two-qubit projector for the outcome pair (s, t) is

$$\Pi_{st}(a, b) = \Pi_s^A(a) \otimes \Pi_t^B(b).$$

Equivalently,

$$\Pi_{st}(a, b) = |s_a, t_b\rangle\langle s_a, t_b|,$$

where

$$|s_a, t_b\rangle = |s_a\rangle \otimes |t_b\rangle.$$

These projectors define the system-side measurement basis. The actual readout sectors in LF3 are pointer-sector projectors on the full system-apparatus space. The distinction is important:

$$\Pi_{st}(a, b) \quad \text{acts on} \quad H_{AB},$$

whereas

$$\widehat{M}_{st}^{(a,b)} \quad \text{acts on} \quad H_{SA}^{(N)}.$$

The pointer-sector projectors are defined later by

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \widehat{Q}_s^A \otimes \widehat{Q}_t^B.$$

This ensures that the outcome readout is implemented by apparatus degrees of freedom rather than by merely assigning labels to system eigenvectors.

2.7 Lean Encoding of the Setup

In Lean4, the setup module should collect the following data:

- finite-dimensional pointer spaces K_A and K_B ;
- inner-product-space structure on each pointer space;
- positive and negative readout subspaces for each pointer;
- orthogonality of the two readout subspaces on each wing;
- completeness of the two readout subspaces on each wing;
- detector settings as unit vectors in $\mathbb{R}^3$;
- Pauli-direction operators $\sigma \cdot a$ and $\sigma \cdot b$;
- sign-indexed spin projectors and pointer projectors.

The schematic Lean object is:

```

structure SystemApparatusSetup where
  K_A : Type*
  K_B : Type*
  inst_KA : Fintype K_A
  inst_KB : Fintype K_B
  inst_innerA : InnerProductSpace Complex K_A
  inst_innerB : InnerProductSpace Complex K_B
  Q_plus_A : Submodule Complex K_A
  Q_minus_A : Submodule Complex K_A
  Q_plus_B : Submodule Complex K_B
  Q_minus_B : Submodule Complex K_B
  ortho_A : Q_plus_A inf Q_minus_A = bot
  span_A : Q_plus_A sup Q_minus_A = top
  ortho_B : Q_plus_B inf Q_minus_B = bot
  span_B : Q_plus_B sup Q_minus_B = top
    
```

This is not intended as final Lean syntax. It records the mathematical data that the implementation must provide. In the actual repository, syntax may need to be adjusted to match the available Mathlib interfaces for finite-dimensional Hilbert spaces, tensor products, submodules, and orthogonal projections.

Detector settings can be encoded schematically as:

```

structure DetectorSetting where
  vec : EuclideanSpace Real (Fin 3)
  unit : norm vec = 1
    
```

The setup file should then provide selectors for the sign-indexed pointer sectors:

```

def pointerSectorA (S : SystemApparatusSetup) (s : Bool) :=
  if s then S.Q_plus_A else S.Q_minus_A

def pointerSectorB (S : SystemApparatusSetup) (t : Bool) :=
  if t then S.Q_plus_B else S.Q_minus_B
    
```

Here ‘true’ represents +1, and ‘false’ represents −1. The implementation may later replace this Boolean convention with an explicit two-element sign type if that improves readability.

2.8 Setup Theorem Targets

The setup module should deliver the following elementary theorem targets.

First, detector settings have unit norm by construction:

$$\|a\| = 1, \quad \|b\| = 1.$$

Second, the Pauli-direction operators are self-adjoint:

$$\sigma_a^\dagger = \sigma_a, \quad \sigma_b^\dagger = \sigma_b.$$

Third, the spin projectors are idempotent and self-adjoint:

$$\Pi_s^A(a)^2 = \Pi_s^A(a), \quad \Pi_s^A(a)^\dagger = \Pi_s^A(a),$$

and

$$\Pi_t^B(b)^2 = \Pi_t^B(b), \quad \Pi_t^B(b)^\dagger = \Pi_t^B(b).$$

Fourth, the spin projectors form complete binary resolutions:

$$\Pi_+^A(a) + \Pi_-^A(a) = I, \quad \Pi_+^B(b) + \Pi_-^B(b) = I.$$

Fifth, the pointer readout projectors form complete binary decompositions:

$$\hat{Q}_+^A + \hat{Q}_-^A = I_{K_A}, \quad \hat{Q}_+^B + \hat{Q}_-^B = I_{K_B}.$$

These statements are elementary, but they are load-bearing. They allow the later modules to define the measurement Hamiltonian, prove projector completeness, calculate branch weights, and connect the finite-dimensional pointer readout with the LF2 Born-weight interface.

3 Measurement Hamiltonian and Factorised Evolution

3.1 Purpose of the Hamiltonian Layer

This section defines the measurement Hamiltonian used in LF3 and records the algebraic facts needed by the later branch-separation and pointer-sector arguments. The construction is the finite-dimensional version of the standard von Neumann pointer coupling. It is not introduced as a new physical Hamiltonian. Its role is to give a precise system-apparatus control model for the Bell-singlet measurement in the finite-dimensional setting.

The central object is the setting-dependent readout Hamiltonian

$$\hat{H}_{a,b} = \hat{H}_A(a) + \hat{H}_B(b),$$

acting on

$$H_{SA}^{(N)} = H_{AB} \otimes K_A^{(d_A)} \otimes K_B^{(d_B)}.$$

The two terms represent the two local readout couplings. The A -term couples the A -qubit spin observable to the A -pointer. The B -term couples the B -qubit spin observable to the B -pointer. Since the two couplings act on distinct tensor factors, they commute. This commutation is the algebraic reason the measurement unitary factorises.

3.2 The Impulsive Readout Idealisation

LF3 uses the impulsive-readout idealisation. During the measurement interval, the free Hamiltonian of the system and the free Hamiltonians of the pointers are neglected relative to the readout coupling. Thus the primary derivation sets

$$\hat{H}_{\text{free}} = 0$$

and uses only the interaction Hamiltonian

$$\hat{H}_{a,b} = \hat{H}_A(a) + \hat{H}_B(b).$$

This is a standard approximation in von Neumann measurement models. It isolates the readout step and allows the pointer translation induced by the measurement coupling to be treated directly.

The approximation is not a claim that free evolution is physically absent. It is a controlled modelling choice for the readout interval. Corrections due to finite readout time, imperfect localisation, and finite pointer separation are treated later through the leakage parameters

$$\epsilon_A, \quad \epsilon_B.$$

3.3 Local Readout Terms

Let

$$a, b \in S^2$$

be the two detector settings. Define

$$\sigma_a = \sigma \cdot a, \quad \sigma_b = \sigma \cdot b.$$

The finite-dimensional pointer generators are self-adjoint operators

$$\hat{p}_A : K_A^{(d_A)} \rightarrow K_A^{(d_A)}, \quad \hat{p}_B : K_B^{(d_B)} \rightarrow K_B^{(d_B)}.$$

They play the role of finite-dimensional pointer-momentum operators. The coupling constants are assumed positive:

$$\kappa_A > 0, \quad \kappa_B > 0.$$

The A -wing readout Hamiltonian is

$$\hat{H}_A(a) = \kappa_A(\sigma_a \otimes I_B) \otimes \hat{p}_A \otimes I_{K_B}.$$

The B -wing readout Hamiltonian is

$$\hat{H}_B(b) = \kappa_B(I_A \otimes \sigma_b) \otimes I_{K_A} \otimes \hat{p}_B.$$

Therefore

$$\hat{H}_{a,b} = \kappa_A(\sigma_a \otimes I_B) \otimes \hat{p}_A \otimes I_{K_B} + \kappa_B(I_A \otimes \sigma_b) \otimes I_{K_A} \otimes \hat{p}_B.$$

The physical interpretation is direct. If the system is in an eigenstate of σ_a with eigenvalue s , the A -pointer is translated in the direction associated with s . If the system is in an eigenstate of σ_b with eigenvalue t , the B -pointer is translated in the direction associated with t .

3.4 Self-Adjointness

The readout Hamiltonian must be self-adjoint so that it generates unitary evolution. The Pauli-direction operators are self-adjoint:

$$\sigma_a^\dagger = \sigma_a, \quad \sigma_b^\dagger = \sigma_b.$$

The pointer generators are assumed self-adjoint:

$$\hat{p}_A^\dagger = \hat{p}_A, \quad \hat{p}_B^\dagger = \hat{p}_B.$$

It follows that

$$\hat{H}_A(a)^\dagger = \hat{H}_A(a), \quad \hat{H}_B(b)^\dagger = \hat{H}_B(b),$$

and hence

$$\hat{H}_{a,b}^\dagger = \hat{H}_{a,b}.$$

Thus the finite-dimensional readout Hamiltonian generates a unitary measurement evolution.

3.5 Commutation of the Readout Hamiltonians

The two readout terms commute:

$$[\hat{H}_A(a), \hat{H}_B(b)] = 0.$$

This follows from tensor-factor separation. On the two-qubit system,

$$[\sigma_a \otimes I_B, I_A \otimes \sigma_b] = 0,$$

because the first operator acts non-trivially only on H_A , while the second acts non-trivially only on H_B . On the pointer sectors,

$$[\hat{p}_A \otimes I_{K_B}, I_{K_A} \otimes \hat{p}_B] = 0,$$

because the first operator acts non-trivially only on K_A , while the second acts non-trivially only on K_B .

Therefore the full tensor-product operators commute:

$$\hat{H}_A(a)\hat{H}_B(b) = \hat{H}_B(b)\hat{H}_A(a).$$

Equivalently,

$$[\hat{H}_A(a), \hat{H}_B(b)] = \hat{H}_A(a)\hat{H}_B(b) - \hat{H}_B(b)\hat{H}_A(a) = 0.$$

3.6 Measurement Unitary

For readout time T , the measurement unitary is

$$\hat{U}_{a,b}(T) = \exp\left(-\frac{iT}{\hbar}\hat{H}_{a,b}\right).$$

Using

$$\hat{H}_{a,b} = \hat{H}_A(a) + \hat{H}_B(b),$$

this becomes

$$\hat{U}_{a,b}(T) = \exp\left(-\frac{iT}{\hbar}[\hat{H}_A(a) + \hat{H}_B(b)]\right).$$

Since

$$[\hat{H}_A(a), \hat{H}_B(b)] = 0,$$

the exponential factorises:

$$\hat{U}_{a,b}(T) = \hat{U}_A(a; T)\hat{U}_B(b; T),$$

where

$$\hat{U}_A(a; T) = \exp\left(-\frac{iT}{\hbar}\hat{H}_A(a)\right),$$

and

$$\hat{U}_B(b; T) = \exp\left(-\frac{iT}{\hbar}\hat{H}_B(b)\right).$$

Equivalently,

$$\hat{U}_{a,b}(T) = \hat{U}_B(b; T)\hat{U}_A(a; T),$$

because the two factors commute.

3.7 Action on Joint Measurement Eigenstates

Let

$$|s_a, t_b\rangle = |s_a\rangle \otimes |t_b\rangle$$

be a joint eigenstate of the two measured spin observables, with

$$\sigma_a |s_a\rangle = s |s_a\rangle, \quad \sigma_b |t_b\rangle = t |t_b\rangle, \quad s, t \in \{\pm 1\}.$$

Let the initial pointer states be

$$|\phi_0^A\rangle \in K_A^{(d_A)}, \quad |\phi_0^B\rangle \in K_B^{(d_B)}.$$

Then the action of the measurement unitary on the product state is

$$\hat{U}_{a,b}(T) (|s_a, t_b\rangle \otimes |\phi_0^A\rangle \otimes |\phi_0^B\rangle) = |s_a, t_b\rangle \otimes |\phi_s^A(T)\rangle \otimes |\phi_t^B(T)\rangle,$$

where

$$|\phi_s^A(T)\rangle = \exp\left(-\frac{i}{\hbar} s \kappa_A T \hat{p}_A\right) |\phi_0^A\rangle,$$

and

$$|\phi_t^B(T)\rangle = \exp\left(-\frac{i}{\hbar} t \kappa_B T \hat{p}_B\right) |\phi_0^B\rangle.$$

Thus the pointer displacement is conditional on the relevant spin eigenvalue. The A -pointer displacement depends on s and a , while the B -pointer displacement depends on t and b .

3.8 Parameter Locality at the Hamiltonian-Control Level

At the Hamiltonian-control level, each readout term depends only on its local detector setting:

$$\hat{H}_A(a) \text{ depends on } a \text{ but not on } b,$$

and

$$\hat{H}_B(b) \text{ depends on } b \text{ but not on } a.$$

Formally,

$$\frac{\partial \hat{H}_A(a)}{\partial b} = 0, \quad \frac{\partial \hat{H}_B(b)}{\partial a} = 0.$$

This is a statement about the local control structure of the measurement Hamiltonian. It should not be confused with Bell-local factorisation of outcomes. The later outcome sectors are context-indexed, and the singlet branch weights depend on the joint pair (a, b) .

3.9 Free Evolution and the Impulsive Limit

If free evolution is included, the total Hamiltonian would have the schematic form

$$\hat{H}_{\text{tot}} = \hat{H}_{\text{free}} + \hat{H}_{a,b}.$$

The impulsive idealisation assumes that, during the readout interval,

$$\|\hat{H}_{\text{free}}\|T \ll \|\hat{H}_{a,b}\|T$$

in the operational regime relevant to pointer separation. Therefore the primary LF3 calculation uses

$$\hat{H}_{\text{tot}} \approx \hat{H}_{a,b}.$$

This keeps the formalisation focused on the readout mechanism. Free-evolution corrections can be absorbed into later finite-error estimates, but they are not part of the initial LF3 theorem target.

3.10 Lean Theorem Targets for the Hamiltonian Layer

The Lean module corresponding to this section is intended to be

`CsdLean4/LF3/Hamiltonian.lean`.

It should import the setup module and define the local Hamiltonian terms, the total readout Hamiltonian, and the measurement unitary.

The schematic declarations are:

```
def H_A
  (S : SystemApparatusSetup)
  (kappa_A : Real)
  (a : DetectorSetting)
  (p_A : K_A ->L[Complex] K_A) :
  H_SA S ->L[Complex] H_SA S

def H_B
  (S : SystemApparatusSetup)
  (kappa_B : Real)
  (b : DetectorSetting)
  (p_B : K_B ->L[Complex] K_B) :
  H_SA S ->L[Complex] H_SA S

def H_total
  (S : SystemApparatusSetup)
  (kappa_A kappa_B : Real)
  (a b : DetectorSetting)
  (p_A : K_A ->L[Complex] K_A)
  (p_B : K_B ->L[Complex] K_B) :
  H_SA S ->L[Complex] H_SA S
```

The first core theorem is commutation of the two readout terms:

```
theorem H_AB_commute :
  H_A S kappa_A a p_A * H_B S kappa_B b p_B =
  H_B S kappa_B b p_B * H_A S kappa_A a p_A
```

The second core theorem is factorisation of the measurement unitary:

```
theorem U_measure_factorises :
  U_measure S kappa_A kappa_B a b p_A p_B T =
  U_A S kappa_A a p_A T * U_B S kappa_B b p_B T
```

The third theorem records the action on joint measurement eigenstates:

```
theorem U_measure_on_joint_eigenstate :
  U_measure S kappa_A kappa_B a b p_A p_B T
  (ket_s_a_t_b tensor phi_A0 tensor phi_B0)
  =
  ket_s_a_t_b
  tensor translatedPointerA s kappa_A T p_A phi_A0
  tensor translatedPointerB t kappa_B T p_B phi_B0
```

These theorem targets are intentionally modest. They establish the Hamiltonian-control layer needed for the later branch-separation calculation. The probability kernel, correlation, marginals, and finite-leakage estimates are proved only after the branch-separated state and pointer-sector projectors have been introduced.

4 Branch Separation and Finite-Leakage Bounds

4.1 Purpose of the Branch-Separation Layer

This section records the branch-separation structure produced by the finite-dimensional pointer measurement. The previous section defined the setting-dependent measurement Hamiltonian and its factorised unitary evolution. The present section applies that unitary to the initial singlet-pointer state and obtains the branch-separated form used in the later pointer-sector probability calculation.

The key point is that the measurement interaction correlates each system eigenbranch with a corresponding pair of pointer states. In the ideal strong-readout limit, these pointer states lie entirely inside disjoint readout sectors. At finite pointer separation, however, each translated pointer state may have a small overlap with the wrong readout sector. LF3 therefore includes explicit leakage parameters

$$\epsilon_A, \quad \epsilon_B,$$

and proves that branch distinguishability and branch weights are stable up to errors of order

$$\mathcal{O}(\epsilon_A + \epsilon_B).$$

This layer is essential because it prevents the formalisation from relying only on exact mathematical separation. The ideal projective measurement is recovered as a limiting case, while the finite-dimensional construction remains meaningful when pointer separation is large but not perfect.

4.2 Initial Singlet-Pointer State

The initial system state is the two-qubit singlet

$$|\psi^-\rangle \in H_{AB}.$$

For detector settings

$$a, b \in S^2,$$

let

$$\{|s_a\rangle : s \in \{\pm 1\}\}$$

be the eigenbasis of $\sigma \cdot a$, and let

$$\{|t_b\rangle : t \in \{\pm 1\}\}$$

be the eigenbasis of $\sigma \cdot b$. The corresponding joint measurement basis is

$$|s_a, t_b\rangle = |s_a\rangle \otimes |t_b\rangle.$$

The singlet can be expanded in this basis as

$$|\psi^-\rangle = \sum_{s,t \in \{\pm 1\}} c_{st}(a, b) |s_a, t_b\rangle,$$

where

$$c_{st}(a, b) = \langle s_a, t_b | \psi^-\rangle.$$

The coefficients $c_{st}(a, b)$ encode the singlet amplitudes for the measurement context (a, b) . Their explicit evaluation is deferred to the singlet-specialisation section.

Let the initial pointer states be

$$|\phi_0^A\rangle \in K_A^{(d_A)}, \quad |\phi_0^B\rangle \in K_B^{(d_B)}.$$

The full initial system-apparatus state is

$$|\Psi_0\rangle = |\psi^-\rangle \otimes |\phi_0^A\rangle \otimes |\phi_0^B\rangle.$$

Substituting the singlet expansion gives

$$|\Psi_0\rangle = \sum_{s,t \in \{\pm 1\}} c_{st}(a, b) |s_a, t_b\rangle \otimes |\phi_0^A\rangle \otimes |\phi_0^B\rangle.$$

4.3 Pointer Translation Under Readout

The measurement unitary is

$$\hat{U}_{a,b}(T) = \hat{U}_A(a; T) \hat{U}_B(b; T).$$

Its action on a joint measurement eigenbranch is

$$\hat{U}_{a,b}(T) (|s_a, t_b\rangle \otimes |\phi_0^A\rangle \otimes |\phi_0^B\rangle) = |s_a, t_b\rangle \otimes |\phi_s^A(T)\rangle \otimes |\phi_t^B(T)\rangle.$$

Here

$$|\phi_s^A(T)\rangle = \exp\left(-\frac{i}{\hbar} s \kappa_A T \hat{p}_A\right) |\phi_0^A\rangle,$$

and

$$|\phi_t^B(T)\rangle = \exp\left(-\frac{i}{\hbar} t \kappa_B T \hat{p}_B\right) |\phi_0^B\rangle.$$

Thus the A -pointer state is translated according to the A -wing spin eigenvalue s , while the B -pointer state is translated according to the B -wing spin eigenvalue t .

4.4 Branch-Separated Final State

Applying the measurement unitary to the initial state gives

$$|\Psi_T\rangle = \hat{U}_{a,b}(T) |\Psi_0\rangle.$$

By linearity,

$$|\Psi_T\rangle = \sum_{s,t \in \{\pm 1\}} c_{st}(a, b) |s_a, t_b\rangle \otimes |\phi_s^A(T)\rangle \otimes |\phi_t^B(T)\rangle.$$

This is the finite-dimensional branch-separated form. Each term contains:

1. a two-qubit system eigenbranch $|s_a, t_b\rangle$;
2. an A -pointer state shifted toward the readout sector labelled by s ;
3. a B -pointer state shifted toward the readout sector labelled by t .

The four branch vectors are

$$|s_a, t_b\rangle \otimes |\phi_s^A(T)\rangle \otimes |\phi_t^B(T)\rangle, \quad s, t \in \{\pm 1\}.$$

In the strong-readout regime, these branches are operationally distinguishable because their pointer components lie in disjoint readout sectors.

4.5 Ideal Branch Separation

Let

$$\hat{Q}_s^A$$

be the orthogonal projector onto the A -pointer readout sector labelled by s , and let

$$\hat{Q}_t^B$$

be the orthogonal projector onto the B -pointer readout sector labelled by t .

In the ideal strong-readout limit, the translated pointer states satisfy

$$\hat{Q}_s^A |\phi_s^A(T)\rangle = |\phi_s^A(T)\rangle, \quad \hat{Q}_{-s}^A |\phi_s^A(T)\rangle = 0,$$

and

$$\hat{Q}_t^B |\phi_t^B(T)\rangle = |\phi_t^B(T)\rangle, \quad \hat{Q}_{-t}^B |\phi_t^B(T)\rangle = 0.$$

Equivalently,

$$\langle \phi_s^A(T) | \hat{Q}_s^A | \phi_s^A(T) \rangle = 1, \quad \langle \phi_s^A(T) | \hat{Q}_{-s}^A | \phi_s^A(T) \rangle = 0,$$

and

$$\langle \phi_t^B(T) | \hat{Q}_t^B | \phi_t^B(T) \rangle = 1, \quad \langle \phi_t^B(T) | \hat{Q}_{-t}^B | \phi_t^B(T) \rangle = 0.$$

In this limit, each translated pointer packet lies wholly inside its intended readout sector.

4.6 Leakage Parameters

At finite pointer separation, the translated pointer states need not be perfectly contained in their intended readout sectors. Define the A -pointer leakage parameter by

$$\epsilon_A = \max_{s \in \{\pm 1\}} \langle \phi_s^A(T) | \hat{Q}_{-s}^A | \phi_s^A(T) \rangle.$$

Equivalently,

$$\langle \phi_s^A(T) | \hat{Q}_s^A | \phi_s^A(T) \rangle \geq 1 - \epsilon_A.$$

Define the B -pointer leakage parameter by

$$\epsilon_B = \max_{t \in \{\pm 1\}} \langle \phi_t^B(T) | \hat{Q}_{-t}^B | \phi_t^B(T) \rangle.$$

Equivalently,

$$\langle \phi_t^B(T) | \hat{Q}_t^B | \phi_t^B(T) \rangle \geq 1 - \epsilon_B.$$

The parameters

$$\epsilon_A, \epsilon_B \geq 0$$

are structural inputs to the LF3 model. They encode the compatibility between the chosen pointer generators

$$\hat{p}_A, \hat{p}_B$$

and the chosen readout subspaces

$$K_A^{(+)}, K_A^{(-)}, \quad K_B^{(+)}, K_B^{(-)}.$$

They are not estimates derived in LF3 from a more fundamental pointer construction. A derivation of small ϵ_A and ϵ_B from an explicit finite-dimensional pointer model, such as a cyclic shift on a discrete position lattice, finite Gaussian-like packets on a spin chain, or an analogous construction, is outside the scope of LF3 v1.00 and is assigned to later refinement.

The ideal strong-readout limit is the structural limit

$$\epsilon_A, \epsilon_B \rightarrow 0.$$

4.7 Asymptotic Branch-Separation Lemma

Under the leakage assumptions above, the post-measurement state

$$|\Psi_T\rangle = \sum_{s,t \in \{\pm 1\}} c_{st}(a,b) |s_a, t_b\rangle \otimes |\phi_s^A(T)\rangle \otimes |\phi_t^B(T)\rangle$$

has four operationally distinguishable pointer branches up to total leakage of order

$$\mathcal{O}(\epsilon_A + \epsilon_B).$$

The reason is straightforward. The branch labelled by (s, t) has intended pointer support in

$$K_A^{(s)} \otimes K_B^{(t)}.$$

The probability that the A -pointer lies outside $K_A^{(s)}$ is at most ϵ_A , and the probability that the B -pointer lies outside $K_B^{(t)}$ is at most ϵ_B . Therefore, by the elementary union bound, the probability that at least one pointer is found outside its intended readout sector is bounded by

$$\epsilon_A + \epsilon_B.$$

Higher-order overlap terms are of order

$$\epsilon_A \epsilon_B$$

and are absorbed into

$$\mathcal{O}(\epsilon_A + \epsilon_B)$$

for the finite-leakage estimates used in LF3.

4.8 Pointer-Sector Overlap Functions

For later use, define the pointer-sector overlap functions

$$\alpha_A(r \mid s) = \langle \phi_s^A(T) | \hat{Q}_r^A | \phi_s^A(T) \rangle, \quad r, s \in \{\pm 1\},$$

and

$$\alpha_B(q \mid t) = \langle \phi_t^B(T) | \hat{Q}_q^B | \phi_t^B(T) \rangle, \quad q, t \in \{\pm 1\}.$$

The leakage assumptions imply

$$\alpha_A(s \mid s) \geq 1 - \epsilon_A, \quad \alpha_A(-s \mid s) \leq \epsilon_A,$$

and

$$\alpha_B(t \mid t) \geq 1 - \epsilon_B, \quad \alpha_B(-t \mid t) \leq \epsilon_B.$$

These functions are useful because they make the finite-readout calculation algebraic. The ideal case is recovered when

$$\alpha_A(r \mid s) = \delta_{rs}, \quad \alpha_B(q \mid t) = \delta_{qt}.$$

At finite separation, deviations from the Kronecker-delta form are controlled by ϵ_A and ϵ_B .

4.9 Finite-Separation Branch-Weight Form

Although the pointer-sector projectors are defined formally in the next section, it is useful to record the branch-weight structure here. The pointer-sector probability for readout (s, t) is

$$P_{st}(a, b; T) = \langle \Psi_T | \hat{M}_{st}^{(a,b)} | \Psi_T \rangle,$$

where

$$\hat{M}_{st}^{(a,b)} = I_{AB} \otimes \hat{Q}_s^A \otimes \hat{Q}_t^B.$$

Using the branch-separated state and the orthogonality of the system eigenvectors,

$$\langle u_a, v_b \mid s_a, t_b \rangle = \delta_{us} \delta_{vt},$$

one obtains

$$P_{st}(a, b; T) = \sum_{u, v \in \{\pm 1\}} |c_{uv}(a, b)|^2 \alpha_A(s \mid u) \alpha_B(t \mid v).$$

The ideal branch weight is

$$P_{st}^{\text{ideal}}(a, b) = |c_{st}(a, b)|^2.$$

Define the finite-separation deviation

$$\Delta_{st}(a, b; T) = P_{st}(a, b; T) - P_{st}^{\text{ideal}}(a, b).$$

The leakage bounds imply the explicit estimate

$$|\Delta_{st}(a, b; T)| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

Hence

$$P_{st}(a, b; T) = |c_{st}(a, b)|^2 + \mathcal{O}(\epsilon_A + \epsilon_B).$$

The explicit inequality is the Lean-facing statement. The asymptotic notation records only the leading-order dependence for small leakage.

In the strong-readout limit,

$$\epsilon_A, \epsilon_B \rightarrow 0,$$

and therefore

$$P_{st}(a, b; T) \rightarrow |c_{st}(a, b)|^2.$$

4.10 Interpretation of the Leakage Bounds

The leakage parameters do not modify the singlet probability law. They quantify imperfections in the pointer readout stage. In the exact finite-dimensional construction, they record how well the translated pointer states are resolved by the chosen readout subspaces.

The finite-leakage estimate has three roles in LF3.

First, it separates the algebraic singlet calculation from the physical readout approximation. The coefficients

$$c_{st}(a, b)$$

belong to the system-side singlet expansion, while

$$\epsilon_A, \quad \epsilon_B$$

belong to the apparatus-side readout.

Second, it makes the strong-readout limit explicit:

$$P_{st}(a, b; T) \rightarrow |c_{st}(a, b)|^2.$$

This is the point at which the later singlet calculation becomes the operational pointer-sector probability.

Third, it shows stability. Small imperfections in pointer separation cause only small deviations in the branch weights. This is essential for interpreting the construction as a finite-dimensional approximation to the ideal projective measurement.

4.11 Lean Theorem Targets for Branch Separation

The Lean module corresponding to this section is intended to be

`CsdLean4/LF3/BranchSeparation.lean`.

It should import the setup and Hamiltonian modules, then define the branch-separated state and leakage assumptions.

The schematic declarations are:

```
def branchState
  (a b : DetectorSetting)
  (T : Real)
  (s t : Sign) :
  H_SA S

def finalState
  (a b : DetectorSetting)
  (T : Real) :
  H_SA S
```

The leakage assumptions should be encoded as a structure:

```
structure PointerLeakageBounds where
  epsilon_A : Real
  epsilon_B : Real
  nonneg_A : 0 <= epsilon_A
  nonneg_B : 0 <= epsilon_B
  A_wrong :
    forall s, pointerOverlapA (-s) s <= epsilon_A
  B_wrong :
    forall t, pointerOverlapB (-t) t <= epsilon_B
  A_right :
    forall s, 1 - epsilon_A <= pointerOverlapA s s
  B_right :
    forall t, 1 - epsilon_B <= pointerOverlapB t t
```

The first theorem target is the branch-separated final-state expression:

```

theorem finalState_branch_decomposition :
  finalState a b T =
    sum over s,t of
      c_st a b s t • branchState a b T s t
    
```

The second theorem target is the operational distinguishability estimate:

```

theorem branch_separation_leakage_bound :
  wrongPointerReadoutMass a b T <= epsilon_A + epsilon_B
    
```

The third theorem target is the finite branch-weight estimate:

```

theorem branchWeight_finite_leakage :
  abs (branchWeight a b T s t - normSq (c_st a b s t))
    <= epsilon_A + epsilon_B + epsilon_A * epsilon_B
    
```

These theorem targets do not yet evaluate the singlet coefficients. They provide the finite-readout bridge needed before the later singlet-specialisation theorem proves

$$|c_{st}(a, b)|^2 = \frac{1 - st a \cdot b}{4}.$$

5 Pointer-Sector Projectors and Branch Weights

5.1 Purpose of the Pointer-Sector Layer

This section defines the projectors that represent the actual apparatus readout in LF3. The previous section described how the measurement interaction produces a branch-separated state,

$$|\Psi_T\rangle = \sum_{u,v \in \{\pm 1\}} c_{uv}(a, b) |u_a, v_b\rangle \otimes |\phi_u^A(T)\rangle \otimes |\phi_v^B(T)\rangle.$$

The present section defines the projective measurement on the pointer degrees of freedom and computes the corresponding branch weights.

The central object is the pointer-sector projector

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \widehat{Q}_s^A \otimes \widehat{Q}_t^B, \quad s, t \in \{\pm 1\}.$$

Although this operator has the same algebraic form for each measurement context, the superscript (a, b) is retained because it is applied to the post-measurement state generated by the context-dependent unitary

$$\widehat{U}_{a,b}(T).$$

Thus the pointer-sector readout is evaluated only after a measurement interaction selected by the detector settings a and b .

The purpose of this layer is not to introduce a new probability rule. The probability rule has already been isolated upstream in LF2. LF3 applies that rule to the concrete pointer-sector projectors arising from the TN3 measurement construction. The theorem targets in this section are therefore algebraic: orthogonality, completeness, branch-weight evaluation, and finite-leakage stability.

5.2 Pointer Readout Subspaces

The A -pointer readout is represented by two orthogonal subspaces

$$K_A^{(+)}, \quad K_A^{(-)} \subseteq K_A^{(d_A)}.$$

The corresponding orthogonal projectors are

$$\widehat{Q}_+^A, \quad \widehat{Q}_-^A.$$

They satisfy

$$\widehat{Q}_+^A + \widehat{Q}_-^A = I_{K_A},$$

and

$$\hat{Q}_+^A \hat{Q}_-^A = 0.$$

Similarly, the B -pointer readout is represented by two orthogonal subspaces

$$K_B^{(+)}, \quad K_B^{(-)} \subseteq K_B^{(d_B)}.$$

The corresponding orthogonal projectors are

$$\hat{Q}_+^B, \quad \hat{Q}_-^B.$$

They satisfy

$$\hat{Q}_+^B + \hat{Q}_-^B = I_{K_B},$$

and

$$\hat{Q}_+^B \hat{Q}_-^B = 0.$$

For signs $s, t \in \{\pm 1\}$, define

$$\hat{Q}_s^A = \begin{cases} \hat{Q}_+^A, & s = +1, \\ \hat{Q}_-^A, & s = -1, \end{cases}$$

and

$$\hat{Q}_t^B = \begin{cases} \hat{Q}_+^B, & t = +1, \\ \hat{Q}_-^B, & t = -1. \end{cases}$$

These projectors encode the apparatus-level distinction between the two possible readout values on each wing.

5.3 Definition of Pointer-Sector Projectors

For each outcome pair

$$(s, t) \in \{\pm 1\} \times \{\pm 1\},$$

define the pointer-sector projector on the full system-apparatus Hilbert space by

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \hat{Q}_s^A \otimes \hat{Q}_t^B.$$

Here

$$I_{AB}$$

is the identity operator on the two-qubit system space

$$H_{AB} = \mathbb{C}^2 \otimes \mathbb{C}^2.$$

The operator $\widehat{M}_{st}^{(a,b)}$ does not directly project onto a system eigenstate. It projects onto a pair of pointer readout sectors after the system-apparatus interaction has correlated those pointer sectors with the spin branches.

This distinction is important. The system-side projectors

$$\Pi_{st}(a, b) = \Pi_s^A(a) \otimes \Pi_t^B(b)$$

act on H_{AB} , whereas the pointer-sector projectors

$$\widehat{M}_{st}^{(a,b)}$$

act on $H_{SA}^{(N)}$. The operational outcome is registered by the apparatus, not by assigning a bare label directly to the system state.

5.4 Orthogonality of Pointer-Sector Projectors

The four pointer-sector projectors are mutually orthogonal. For two outcome pairs (s, t) and (s', t') , we have

$$\widehat{M}_{st}^{(a,b)} \widehat{M}_{s't'}^{(a,b)} = \left(I_{AB} \otimes \hat{Q}_s^A \otimes \hat{Q}_t^B \right) \left(I_{AB} \otimes \hat{Q}_{s'}^A \otimes \hat{Q}_{t'}^B \right).$$

Using the tensor-product multiplication rule, this becomes

$$\widehat{M}_{st}^{(a,b)} \widehat{M}_{s't'}^{(a,b)} = I_{AB} \otimes \left(\hat{Q}_s^A \hat{Q}_{s'}^A \right) \otimes \left(\hat{Q}_t^B \hat{Q}_{t'}^B \right).$$

If

$$(s, t) \neq (s', t'),$$

then either $s \neq s'$, or $t \neq t'$, or both. If $s \neq s'$, then

$$\hat{Q}_s^A \hat{Q}_{s'}^A = 0.$$

If $t \neq t'$, then

$$\hat{Q}_t^B \hat{Q}_{t'}^B = 0.$$

Therefore

$$\widehat{M}_{st}^{(a,b)} \widehat{M}_{s't'}^{(a,b)} = 0 \quad \text{whenever} \quad (s, t) \neq (s', t').$$

Thus distinct pointer outcomes correspond to orthogonal apparatus readout sectors.

For the same outcome pair, idempotence follows from idempotence of the pointer projectors:

$$\left(\widehat{M}_{st}^{(a,b)} \right)^2 = \widehat{M}_{st}^{(a,b)}.$$

Self-adjointness follows similarly:

$$\left(\widehat{M}_{st}^{(a,b)} \right)^\dagger = \widehat{M}_{st}^{(a,b)}.$$

Hence each $\widehat{M}_{st}^{(a,b)}$ is an orthogonal projector on $H_{SA}^{(N)}$.

5.5 Completeness of the Pointer-Sector Decomposition

The four pointer-sector projectors form a complete projective decomposition of the system-apparatus readout space. Summing over all signs gives

$$\sum_{s,t \in \{\pm 1\}} \widehat{M}_{st}^{(a,b)} = \sum_{s,t \in \{\pm 1\}} I_{AB} \otimes \hat{Q}_s^A \otimes \hat{Q}_t^B.$$

Since I_{AB} is common to all four terms,

$$\sum_{s,t \in \{\pm 1\}} \widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \left(\sum_{s \in \{\pm 1\}} \hat{Q}_s^A \right) \otimes \left(\sum_{t \in \{\pm 1\}} \hat{Q}_t^B \right).$$

Using the binary completeness relations

$$\hat{Q}_+^A + \hat{Q}_-^A = I_{K_A},$$

and

$$\hat{Q}_+^B + \hat{Q}_-^B = I_{K_B},$$

we obtain

$$\sum_{s,t \in \{\pm 1\}} \widehat{M}_{st}^{(a,b)} = I_{AB} \otimes I_{K_A} \otimes I_{K_B}.$$

Therefore

$$\sum_{s,t \in \{\pm 1\}} \widehat{M}_{st}^{(a,b)} = I_{SA}.$$

The family

$$\left\{ \widehat{M}_{st}^{(a,b)} : s, t \in \{\pm 1\} \right\}$$

is therefore a complete projective measurement on the finite-dimensional system-apparatus Hilbert space.

5.6 Projective Readout Context

For a fixed pair of detector settings (a, b) , the projectors

$$\widehat{M}_{++}^{(a,b)}, \quad \widehat{M}_{+-}^{(a,b)}, \quad \widehat{M}_{-+}^{(a,b)}, \quad \widehat{M}_{--}^{(a,b)}$$

define the four possible pointer readouts.

The superscript (a, b) does not mean that the pointer projectors themselves depend algebraically on the spin directions. The readout subspaces $K_A^{(\pm)}$ and $K_B^{(\pm)}$ are fixed apparatus sectors. The context dependence enters through the state

$$|\Psi_T\rangle = \widehat{U}_{a,b}(T)|\Psi_0\rangle,$$

which is produced by the setting-dependent Hamiltonian.

Thus the operational probability for outcome (s, t) is not assigned to an abstract system projector alone. It is assigned to the pointer-sector projector applied to the final system-apparatus state:

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

This is the projective readout context used in the rest of LF3.

5.7 Operator-Form Branch Weights

For the final system-apparatus state

$$|\Psi_T\rangle = \widehat{U}_{a,b}(T)|\Psi_0\rangle,$$

the pointer-sector branch weight for outcome (s, t) is defined by

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

This is the standard operator-form weight associated with the projective readout

$$\left\{ \widehat{M}_{st}^{(a,b)} : s, t \in \{\pm 1\} \right\}.$$

Within the LF3 dependency structure, this rule is not rederived. It is inherited from the LF2 Born-weight interface and applied to the concrete pointer-sector projectors defined above.

Using the branch-separated state,

$$|\Psi_T\rangle = \sum_{u,v \in \{\pm 1\}} c_{uv}(a, b) |u_a, v_b\rangle \otimes |\phi_u^A(T)\rangle \otimes |\phi_v^B(T)\rangle,$$

we obtain

$$P_{st}(a, b; T) = \left\langle \sum_{u,v} c_{uv}(a, b) |u_a, v_b\rangle \otimes |\phi_u^A(T)\rangle \otimes |\phi_v^B(T)\rangle \left| \widehat{M}_{st}^{(a,b)} \right| \sum_{u',v'} c_{u'v'}(a, b) |u'_a, v'_b\rangle \otimes |\phi_{u'}^A(T)\rangle \otimes |\phi_{v'}^B(T)\rangle \right\rangle.$$

The system eigenvectors form an orthonormal basis:

$$\langle u_a, v_b | u'_a, v'_b \rangle = \delta_{uu'} \delta_{vv'}.$$

Therefore the cross terms vanish, and the expression reduces to

$$P_{st}(a, b; T) = \sum_{u,v \in \{\pm 1\}} |c_{uv}(a, b)|^2 \langle \phi_u^A(T) | \widehat{Q}_s^A | \phi_u^A(T) \rangle \langle \phi_v^B(T) | \widehat{Q}_t^B | \phi_v^B(T) \rangle.$$

This formula separates the calculation into two factors:

1. the system-side coefficients $|c_{uv}(a, b)|^2$, which are fixed by the singlet state and the detector settings;
2. the apparatus-side pointer overlaps, which determine how sharply each translated pointer state is resolved by the readout projectors.

5.8 Pointer-Overlap Notation

Define the A -pointer overlap function by

$$\alpha_A(s \mid u) = \langle \phi_u^A(T) | \hat{Q}_s^A | \phi_u^A(T) \rangle.$$

Define the B -pointer overlap function by

$$\alpha_B(t \mid v) = \langle \phi_v^B(T) | \hat{Q}_t^B | \phi_v^B(T) \rangle.$$

Then the branch-weight formula becomes

$$P_{st}(a, b; T) = \sum_{u, v \in \{\pm 1\}} |c_{uv}(a, b)|^2 \alpha_A(s \mid u) \alpha_B(t \mid v).$$

The leakage assumptions imply

$$\alpha_A(u \mid u) \geq 1 - \epsilon_A, \quad \alpha_A(-u \mid u) \leq \epsilon_A,$$

and

$$\alpha_B(v \mid v) \geq 1 - \epsilon_B, \quad \alpha_B(-v \mid v) \leq \epsilon_B.$$

Thus the intended branch $(u, v) = (s, t)$ contributes dominantly to $P_{st}(a, b; T)$, while the remaining branches contribute only through leakage into the wrong pointer sectors.

5.9 Strong-Readout Branch Weights

In the strong-readout limit,

$$\epsilon_A, \epsilon_B \rightarrow 0,$$

the pointer overlaps become Kronecker deltas:

$$\alpha_A(s \mid u) \rightarrow \delta_{su}, \quad \alpha_B(t \mid v) \rightarrow \delta_{tv}.$$

Therefore

$$P_{st}(a, b; T) \rightarrow \sum_{u, v \in \{\pm 1\}} |c_{uv}(a, b)|^2 \delta_{su} \delta_{tv}.$$

Hence

$$P_{st}(a, b) = |c_{st}(a, b)|^2.$$

This is the strong-readout reduction theorem for the pointer-sector measurement. It states that, once the pointer branches are perfectly separated, the apparatus readout weight equals the squared coefficient of the corresponding system branch.

The next step is not a new probability postulate. It is the explicit singlet calculation:

$$|c_{st}(a, b)|^2 = \frac{1 - st a \cdot b}{4}.$$

That calculation is carried out in the singlet-specialisation section.

5.10 Finite-Leakage Branch-Weight Estimate

At finite pointer separation, the branch weight differs from the ideal strong-readout value by controlled leakage terms. Define

$$P_{st}^{\text{ideal}}(a, b) = |c_{st}(a, b)|^2.$$

Define the finite-separation deviation by

$$\Delta_{st}(a, b; T) = P_{st}(a, b; T) - P_{st}^{\text{ideal}}(a, b).$$

Using the pointer-overlap expression,

$$P_{st}(a, b; T) = \sum_{u, v \in \{\pm 1\}} |c_{uv}(a, b)|^2 \alpha_A(s \mid u) \alpha_B(t \mid v),$$

the ideal contribution comes from the branch $(u, v) = (s, t)$. The remaining terms arise only when one or both translated pointer states leak into the wrong readout sector.

For the intended branch, the deviation is bounded by

$$||c_{st}(a, b)|^2 \alpha_A(s | s) \alpha_B(t | t) - |c_{st}(a, b)|^2| \leq |c_{st}(a, b)|^2 (\epsilon_A + \epsilon_B).$$

For the wrong branches, at least one pointer overlap is a leakage term. Hence their total contribution is bounded by

$$\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

Therefore

$$|\Delta_{st}(a, b; T)| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

Equivalently,

$$P_{st}(a, b; T) = |c_{st}(a, b)|^2 + \mathcal{O}(\epsilon_A + \epsilon_B).$$

The explicit inequality is the Lean-friendly form. The $\mathcal{O}$ -notation is the paper-level summary.

For sufficiently small leakage,

$$\epsilon_A + \epsilon_B \ll 1,$$

the product term $\epsilon_A \epsilon_B$ is second order. The leading deviation is therefore linear in the total leakage:

$$|\Delta_{st}(a, b; T)| = \mathcal{O}(\epsilon_A + \epsilon_B).$$

5.11 Interface with the LF2 Born-Weight Wrapper

LF2 supplies the finite-dimensional Born-weight wrapper. LF3 applies that wrapper to the pointer-sector projectors. For the pure state

$$|\Psi_T\rangle \in H_{SA}^{(N)},$$

and the projective readout

$$\{\widehat{M}_{st}^{(a,b)} : s, t \in \{\pm 1\}\},$$

the LF2 interface gives the operator-form weight

$$w_{st} = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

LF3 identifies this weight with the pointer-sector branch probability:

$$w_{st} = P_{st}(a, b; T).$$

Thus the branch-weight calculation is not an additional probability postulate. It is the LF2 operator-form rule applied to the concrete projective measurement defined by the pointer-sector readout.

This is the precise formal role of the present section. It connects the abstract Born-weight wrapper to the explicit system-apparatus construction without rederiving the Born rule and without importing any new measure-theoretic assumption.

5.12 No Use of Fubini-Study Voronoi Volumes

No Fubini-Study Voronoi-volume construction is used in LF3. The outcome sectors are not defined as nearest-ray cells in projective space. They are defined by physical pointer readout projectors:

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \widehat{Q}_s^A \otimes \widehat{Q}_t^B.$$

Their weights are computed by

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

The Fubini-Study measure remains part of the upstream measure bridge in the CSD programme, but it is not the operational mechanism used to compute the pointer-sector weights in this formalisation layer.

This distinction prevents a common misunderstanding. LF3 is not claiming that the Bell-singlet branch weights are obtained by measuring the Fubini-Study volumes of four projective Voronoi cells. Rather, LF3 formalises a standard finite-dimensional projective measurement on the system-apparatus Hilbert space and then uses the LF2 operator-form weight rule.

5.13 CSD Basin Interpretation

The operator-form calculation is the formal content of LF3. The CSD interpretation adds a structural reading.

Let

$$\pi_{SA}^{(N)} : \Sigma_{SA}^{(N)} \rightarrow \mathbb{CP}^{N-1}$$

be the finite-dimensional system-apparatus projection in the quantum-effective sector. For each measurement context (a, b) , the pointer-sector projector $\widehat{M}_{st}^{(a,b)}$ defines an epistemic outcome sector

$$\Omega_{st}^{\text{ep}}(a, b) \subseteq \mathbb{CP}^{N-1}.$$

The corresponding ontic basin is the inverse image

$$\Omega_{st}^{\Sigma}(a, b) = \left(\pi_{SA}^{(N)} \right)^{-1} (\Omega_{st}^{\text{ep}}(a, b)).$$

This basin notation should not be read as a second probability calculation. The branch weight remains

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

The basin language identifies the ontic sector corresponding to the operational pointer reading. It does not classify all dynamically realisable apparatus partitions, and it does not derive the single-run basin-selection rule. Those are wider programme-level questions.

5.14 Lean Theorem Targets for the Projector Layer

The Lean module corresponding to this section is intended to be

`CsdLean4/LF3/Projectors.lean`.

It should import the setup, Hamiltonian, and branch-separation modules.

The schematic projector definition is:

```
def M_hat
  (S : SystemApparatusSetup)
  (s t : Sign) :
  H_SA S -> L[Complex] H_SA S
```

The first theorem target is completeness:

```
theorem M_complete :
  sum over s,t of M_hat S s t = 1
```

The second theorem target is orthogonality:

```
theorem M_orthogonal :
  (s,t) != (s',t') ->
  M_hat S s t * M_hat S s' t' = 0
```

The branch weight is defined schematically by:

```
def branchWeight
  (Psi_T : H_SA S)
  (s t : Sign) : Real :=
  realPart (inner Psi_T (M_hat S s t Psi_T))
```

The strong-readout theorem target is:

```
theorem branchWeight_strong_readout :
  epsilon_A = 0 ->
  epsilon_B = 0 ->
  branchWeight Psi_T s t = normSq (c_st a b s t)
```

The finite-leakage theorem target is:

```
theorem branchWeight_finite_leakage :
  abs (branchWeight Psi_T s t - normSq (c_st a b s t))
    <= epsilon_A + epsilon_B + epsilon_A * epsilon_B
```

Finally, the LF2 interface theorem is:

```
theorem branchWeight_eq_LF2_Born :
  branchWeight Psi_T s t =
  LF2.bornWeight
    (rankOneState Psi_T)
    (M_hat S s t)
```

These theorem targets complete the projector layer. The next section evaluates the singlet coefficients $c_{st}(a, b)$, which turns the abstract pointer-sector weight

$$P_{st}(a, b) = |c_{st}(a, b)|^2$$

into the explicit Bell-singlet kernel

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

6 Singlet Specialisation

6.1 Purpose of the Singlet Layer

The previous section reduced the pointer-sector branch weights to squared system amplitudes in the strong-readout limit:

$$P_{st}(a, b) = |c_{st}(a, b)|^2.$$

The purpose of the present section is to evaluate these coefficients for the two-qubit singlet state.

The target result is

$$|c_{st}(a, b)|^2 = \frac{1 - st a \cdot b}{4}, \quad s, t \in \{\pm 1\}.$$

Once this identity is established, the Bell-singlet correlation follows by summing over the four pointer-sector outcomes. That calculation is carried out in the next section. The present section is therefore purely the singlet probability-kernel calculation.

The result is standard, but its role in LF3 is specific. LF3 does not derive the Born rule here. It uses the LF2 operator-form probability interface and verifies that the concrete TN3 pointer-sector construction gives the usual singlet branch weights once the amplitudes are evaluated.

6.2 Definition of the Singlet State

Let

$$H_A \cong \mathbb{C}^2, \quad H_B \cong \mathbb{C}^2.$$

In the z -basis, write

$$|+\rangle_z = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |-\rangle_z = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The two-qubit singlet state is

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} (|+\rangle_z \otimes |-\rangle_z - |-\rangle_z \otimes |+\rangle_z).$$

It is normalised:

$$\langle \psi^- | \psi^- \rangle = 1.$$

The associated rank-one density operator is

$$\rho_{\psi^-} = |\psi^-\rangle \langle \psi^-|.$$

A useful operator identity for the singlet projector is

$$\rho_{\psi^-} = \frac{1}{4} (I \otimes I - \sigma_x \otimes \sigma_x - \sigma_y \otimes \sigma_y - \sigma_z \otimes \sigma_z).$$

Equivalently,

$$\rho_{\psi^-} = \frac{1}{4} \left(I \otimes I - \sum_{j=1}^3 \sigma_j \otimes \sigma_j \right),$$

where

$$\sigma_1 = \sigma_x, \quad \sigma_2 = \sigma_y, \quad \sigma_3 = \sigma_z.$$

This identity is the most efficient route to the Lean theorem target because it reduces the singlet probability calculation to Pauli-matrix algebra.

6.3 Detector Directions and Spin Observables

Let

$$a, b \in S^2$$

be unit detector directions. Write

$$a = (a_1, a_2, a_3), \quad b = (b_1, b_2, b_3).$$

Define

$$\sigma_a = \sigma \cdot a = \sum_{j=1}^3 a_j \sigma_j,$$

and

$$\sigma_b = \sigma \cdot b = \sum_{j=1}^3 b_j \sigma_j.$$

Since a and b are unit vectors,

$$\sigma_a^2 = I, \quad \sigma_b^2 = I.$$

Hence the eigenvalues of σ_a and σ_b are

$$+1, \quad -1.$$

For $s, t \in \{\pm 1\}$, the corresponding one-qubit spectral projectors are

$$\Pi_s^A(a) = \frac{1}{2} (I + s \sigma_a),$$

and

$$\Pi_t^B(b) = \frac{1}{2} (I + t \sigma_b).$$

The joint two-qubit projector is

$$\Pi_{st}(a, b) = \Pi_s^A(a) \otimes \Pi_t^B(b).$$

Expanding gives

$$\Pi_{st}(a, b) = \frac{1}{4} (I \otimes I + s \sigma_a \otimes I + t I \otimes \sigma_b + st \sigma_a \otimes \sigma_b).$$

6.4 Rotational Invariance of the Singlet

The singlet state is rotationally invariant under the diagonal action of $SU(2)$. For every $U \in SU(2)$,

$$(U \otimes U)|\psi^-\rangle = |\psi^-\rangle.$$

At the density-operator level, this gives the exact equality

$$(U \otimes U) \rho_{\psi^-} (U^\dagger \otimes U^\dagger) = \rho_{\psi^-}.$$

This invariance implies that the joint probabilities depend on the detector settings a and b only through their scalar product

$$a \cdot b.$$

For the Lean formalisation, however, it is better not to rely solely on this symmetry argument. The direct Pauli identity gives a more explicit proof:

$$\langle \psi^- | \sigma_a \otimes \sigma_b | \psi^- \rangle = -a \cdot b.$$

The remaining one-sided expectations vanish:

$$\langle \psi^- | \sigma_a \otimes I | \psi^- \rangle = 0,$$

and

$$\langle \psi^- | I \otimes \sigma_b | \psi^- \rangle = 0.$$

These three identities are sufficient to evaluate all four joint probabilities.

6.5 Joint Measurement Amplitudes

For each outcome pair

$$(s, t) \in \{\pm 1\} \times \{\pm 1\},$$

define the singlet amplitude

$$c_{st}(a, b) = \langle s_a, t_b | \psi^- \rangle.$$

The corresponding squared amplitude is

$$|c_{st}(a, b)|^2 = \langle \psi^- | \Pi_{st}(a, b) | \psi^- \rangle.$$

Equivalently, using the singlet density operator,

$$|c_{st}(a, b)|^2 = \text{Tr}(\rho_{\psi^-} \Pi_{st}(a, b)).$$

Using the expanded form of the joint projector,

$$\Pi_{st}(a, b) = \frac{1}{4} (I \otimes I + s \sigma_a \otimes I + t I \otimes \sigma_b + st \sigma_a \otimes \sigma_b),$$

we obtain

$$|c_{st}(a, b)|^2 = \frac{1}{4} [\langle I \otimes I \rangle_{\psi^-} + s \langle \sigma_a \otimes I \rangle_{\psi^-} + t \langle I \otimes \sigma_b \rangle_{\psi^-} + st \langle \sigma_a \otimes \sigma_b \rangle_{\psi^-}],$$

where

$$\langle X \rangle_{\psi^-} = \langle \psi^- | X | \psi^- \rangle.$$

The required expectation values are

$$\langle I \otimes I \rangle_{\psi^-} = 1,$$

$$\langle \sigma_a \otimes I \rangle_{\psi^-} = 0,$$

$$\langle I \otimes \sigma_b \rangle_{\psi^-} = 0,$$

and

$$\langle \sigma_a \otimes \sigma_b \rangle_{\psi^-} = -a \cdot b.$$

Substitution gives

$$|c_{st}(a, b)|^2 = \frac{1}{4} (1 - st a \cdot b).$$

Thus

$$\boxed{|c_{st}(a, b)|^2 = \frac{1 - st a \cdot b}{4}}$$

for all

$$s, t \in \{\pm 1\}.$$

6.6 Evaluation of the Singlet Expectations

It remains to record the algebra behind the three expectation identities used above.

First, normalisation gives

$$\langle I \otimes I \rangle_{\psi^-} = \langle \psi^- | \psi^- \rangle = 1.$$

Second, the one-sided spin expectations vanish. Since the singlet state is rotationally invariant and has zero total spin, no single detector direction can define a preferred local spin expectation. Algebraically,

$$\langle \psi^- | \sigma_j \otimes I | \psi^- \rangle = 0, \quad j = 1, 2, 3,$$

and

$$\langle \psi^- | I \otimes \sigma_j | \psi^- \rangle = 0, \quad j = 1, 2, 3.$$

By linearity,

$$\langle \psi^- | \sigma_a \otimes I | \psi^- \rangle = \sum_{j=1}^3 a_j \langle \psi^- | \sigma_j \otimes I | \psi^- \rangle = 0,$$

and

$$\langle \psi^- | I \otimes \sigma_b | \psi^- \rangle = \sum_{k=1}^3 b_k \langle \psi^- | I \otimes \sigma_k | \psi^- \rangle = 0.$$

Third, the two-sided spin expectation is

$$\langle \psi^- | \sigma_j \otimes \sigma_k | \psi^- \rangle = -\delta_{jk}.$$

Therefore

$$\langle \psi^- | \sigma_a \otimes \sigma_b | \psi^- \rangle = \left\langle \psi^- \left| \left(\sum_{j=1}^3 a_j \sigma_j \right) \otimes \left(\sum_{k=1}^3 b_k \sigma_k \right) \right| \psi^- \right\rangle.$$

Expanding by bilinearity gives

$$\langle \psi^- | \sigma_a \otimes \sigma_b | \psi^- \rangle = \sum_{j,k=1}^3 a_j b_k \langle \psi^- | \sigma_j \otimes \sigma_k | \psi^- \rangle.$$

Using

$$\langle \psi^- | \sigma_j \otimes \sigma_k | \psi^- \rangle = -\delta_{jk},$$

we obtain

$$\langle \psi^- | \sigma_a \otimes \sigma_b | \psi^- \rangle = - \sum_{j,k=1}^3 a_j b_k \delta_{jk}.$$

Hence

$$\langle \psi^- | \sigma_a \otimes \sigma_b | \psi^- \rangle = - \sum_{j=1}^3 a_j b_j = -a \cdot b.$$

6.7 Singlet Probability Kernel

Combining the preceding results gives the singlet probability kernel:

$$P_{st}(a, b) = |c_{st}(a, b)|^2 = \frac{1 - st a \cdot b}{4}.$$

The four probabilities are therefore

$$P_{++}(a, b) = \frac{1 - a \cdot b}{4},$$

$$P_{+-}(a, b) = \frac{1 + a \cdot b}{4},$$

$$P_{-+}(a, b) = \frac{1 + a \cdot b}{4},$$

and

$$P_{--}(a, b) = \frac{1 - a \cdot b}{4}.$$

These probabilities are non-negative because

$$-1 \leq a \cdot b \leq 1.$$

They also sum to unity:

$$\sum_{s,t \in \{\pm 1\}} P_{st}(a, b) = 1.$$

Indeed,

$$\sum_{s,t} \frac{1 - st a \cdot b}{4} = \frac{1}{4} \left(\sum_{s,t} 1 - a \cdot b \sum_{s,t} st \right).$$

Since

$$\sum_{s,t} 1 = 4,$$

and

$$\sum_{s,t} st = 0,$$

we have

$$\sum_{s,t} P_{st}(a, b) = 1.$$

Thus the singlet-specialised pointer-sector weights form a normalised probability distribution for every pair of detector settings $a, b \in S^2$.

6.8 Connection to Pointer-Sector Weights

Section 5 established that, in the strong-readout limit, the pointer-sector branch weights reduce to the squared singlet amplitudes:

$$P_{st}(a, b) = |c_{st}(a, b)|^2.$$

The present section has evaluated those amplitudes:

$$|c_{st}(a, b)|^2 = \frac{1 - st a \cdot b}{4}.$$

Therefore the strong-readout pointer-sector probabilities are

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

At finite pointer separation, the branch-weight estimate from Section 5 gives

$$P_{st}(a, b; T) = |c_{st}(a, b)|^2 + \mathcal{O}(\epsilon_A + \epsilon_B).$$

Substituting the singlet kernel gives

$$P_{st}(a, b; T) = \frac{1 - st a \cdot b}{4} + \mathcal{O}(\epsilon_A + \epsilon_B).$$

Thus the finite-dimensional pointer construction reproduces the Bell-singlet probability kernel in the strong-readout limit and remains stable under controlled finite pointer leakage.

This is the central result of the singlet-specialisation layer. The remaining Bell-sector consequences follow by summing these weights. In particular, the next section derives the correlation

$$E(a, b) = -a \cdot b$$

and the marginal identities

$$P_A(s \mid a, b) = \frac{1}{2}, \quad P_B(t \mid a, b) = \frac{1}{2}.$$

6.9 Lean Theorem Targets for the Singlet Layer

The Lean module corresponding to this section is intended to be

`CsdLean4/LF3/Singlet.lean`.

It should import the setup, projector, and branch-weight modules. Its role is to prove the finite-dimensional two-qubit algebra needed to evaluate the singlet branch weights.

The first theorem target is normalisation of the singlet state:

```
theorem singlet_norm :
  normSq singlet = 1
```

The second theorem target is vanishing of the one-sided expectations:

```
theorem singlet_left_pauli_expectation_zero :
  expectation singlet (sigmaDot a tensor I) = 0
```

```
theorem singlet_right_pauli_expectation_zero :
  expectation singlet (I tensor sigmaDot b) = 0
```

The third theorem target is the two-sided Pauli expectation:

```
theorem singlet_pauli_correlation :
  expectation singlet (sigmaDot a tensor sigmaDot b) =
    - dot a.vec b.vec
```

The fourth theorem target is the expanded joint projector identity:

```
theorem joint_spin_projector_expand :
  Pi_st a b s t =
    (1 / 4) •
      (I tensor I
       + sign s • (sigmaDot a tensor I)
       + sign t • (I tensor sigmaDot b)
       + sign s * sign t • (sigmaDot a tensor sigmaDot b))
```

The fifth theorem target is the squared-amplitude identity:

```
theorem cst_squared_eq :
  normSq (c_st a b s t) =
    (1 - sign s * sign t * dot a.vec b.vec) / 4
```

The sixth theorem target is the strong-readout pointer probability:

```
theorem singlet_pointer_probability_strong_readout :
  P_st a b s t =
    (1 - sign s * sign t * dot a.vec b.vec) / 4
```

The seventh theorem target is the finite-leakage pointer probability estimate:

```
theorem singlet_pointer_probability_finite_leakage :
  abs
    (P_st_T a b T s t
     - (1 - sign s * sign t * dot a.vec b.vec) / 4)
  <= epsilon_A + epsilon_B + epsilon_A * epsilon_B
```

These theorem targets complete the singlet layer. They turn the general pointer-sector branch-weight result into the explicit Bell-singlet probability kernel:

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

6.10 Summary

This section established the singlet-specialised probability kernel required by LF3. Starting from the standard singlet state

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} (|+\rangle_z \otimes |-\rangle_z - |-\rangle_z \otimes |+\rangle_z),$$

and the joint spin projectors

$$\Pi_{st}(a, b) = \frac{1}{4} (I \otimes I + s \sigma_a \otimes I + t I \otimes \sigma_b + st \sigma_a \otimes \sigma_b),$$

we used the identities

$$\begin{aligned}\langle \psi^- | \sigma_a \otimes I | \psi^- \rangle &= 0, \\ \langle \psi^- | I \otimes \sigma_b | \psi^- \rangle &= 0,\end{aligned}$$

and

$$\langle \psi^- | \sigma_a \otimes \sigma_b | \psi^- \rangle = -a \cdot b$$

to obtain

$$|c_{st}(a, b)|^2 = \frac{1 - st a \cdot b}{4}.$$

Combined with the strong-readout pointer-sector result from Section 5, this gives

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

Combined with the finite-leakage estimate from Section 5, it gives

$$P_{st}(a, b; T) = \frac{1 - st a \cdot b}{4} + \mathcal{O}(\epsilon_A + \epsilon_B).$$

The next section derives the correlation and marginal consequences of this kernel.

7 Correlation and Marginal Stability

7.1 Purpose of the Correlation Layer

The previous section established the singlet-specialised pointer-sector probability kernel

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}, \quad s, t \in \{\pm 1\}.$$

The purpose of the present section is to derive the two main operational consequences of this kernel.

First, the expectation value of the product of the two readout signs is

$$E(a, b) = -a \cdot b.$$

This is the Bell-singlet correlation.

Second, the local marginals are independent of the remote detector setting:

$$P_A(s | a, b) = \frac{1}{2}, \quad P_B(t | a, b) = \frac{1}{2}.$$

These equalities give operational no-signalling at the level of the pointer-sector readout.

The same calculations are then repeated with finite-leakage errors. If the finite pointer separation is controlled by

$$\epsilon_A, \quad \epsilon_B,$$

then the correlation and marginals remain within controlled errors of order

$$\mathcal{O}(\epsilon_A + \epsilon_B).$$

7.2 Definition of the Correlation Functional

For detector settings

$$a, b \in S^2,$$

define the correlation functional by

$$E(a, b) = \sum_{s, t \in \{\pm 1\}} st P_{st}(a, b).$$

Using the singlet kernel,

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

we have

$$E(a, b) = \sum_{s, t \in \{\pm 1\}} st \frac{1 - st a \cdot b}{4}.$$

Expanding,

$$E(a, b) = \frac{1}{4} \sum_{s, t} st - \frac{a \cdot b}{4} \sum_{s, t} (st)^2.$$

The first sum vanishes:

$$\sum_{s, t \in \{\pm 1\}} st = 0.$$

The second sum is

$$\sum_{s, t \in \{\pm 1\}} (st)^2 = 4,$$

because

$$(st)^2 = 1$$

for all signs s, t . Therefore

$$E(a, b) = -\frac{a \cdot b}{4} \cdot 4.$$

Hence

$$\boxed{E(a, b) = -a \cdot b}.$$

This proves that the strong-readout pointer-sector probabilities reproduce the standard Bell-singlet correlation.

7.3 Finite-Leakage Correlator Bound

At finite pointer separation, the branch probabilities satisfy

$$P_{st}(a, b; T) = P_{st}(a, b) + \delta_{st}(a, b; T),$$

where

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

and

$$|\delta_{st}(a, b; T)| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

Define the finite-time correlation by

$$E_T(a, b) = \sum_{s, t \in \{\pm 1\}} st P_{st}(a, b; T).$$

Then

$$E_T(a, b) - E(a, b) = \sum_{s, t \in \{\pm 1\}} st \delta_{st}(a, b; T).$$

Taking absolute values gives

$$|E_T(a, b) - E(a, b)| \leq \sum_{s, t \in \{\pm 1\}} |\delta_{st}(a, b; T)|.$$

Using the uniform leakage bound,

$$|E_T(a, b) - E(a, b)| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

Since

$$E(a, b) = -a \cdot b,$$

we obtain

$$|E_T(a, b) + a \cdot b| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

Equivalently,

$$E_T(a, b) = -a \cdot b + \mathcal{O}(\epsilon_A + \epsilon_B).$$

This is the finite-leakage correlator estimate. It shows that the singlet correlation is recovered in the strong-readout limit and remains stable under small pointer leakage.

7.4 Marginal Probability at Detector A

The marginal probability for outcome s at detector A is obtained by summing over the remote outcome t :

$$P_A(s \mid a, b) = \sum_{t \in \{\pm 1\}} P_{st}(a, b).$$

Using the singlet kernel,

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

we obtain

$$P_A(s \mid a, b) = \sum_{t \in \{\pm 1\}} \frac{1 - st a \cdot b}{4}.$$

Expanding,

$$P_A(s \mid a, b) = \frac{1}{4} \sum_{t \in \{\pm 1\}} 1 - \frac{s a \cdot b}{4} \sum_{t \in \{\pm 1\}} t.$$

Now

$$\sum_{t \in \{\pm 1\}} 1 = 2,$$

and

$$\sum_{t \in \{\pm 1\}} t = 0.$$

Therefore

$$P_A(s \mid a, b) = \frac{1}{4} \cdot 2 = \frac{1}{2}.$$

Thus

$$\boxed{P_A(s \mid a, b) = \frac{1}{2}}.$$

The marginal probability at detector A is independent of the remote detector setting b . It also does not depend on the local detector setting a , because the singlet state is locally maximally mixed.

7.5 Marginal Probability at Detector B

Similarly, the marginal probability for outcome t at detector B is obtained by summing over the local outcome s :

$$P_B(t \mid a, b) = \sum_{s \in \{\pm 1\}} P_{st}(a, b).$$

Using

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

we have

$$P_B(t \mid a, b) = \sum_{s \in \{\pm 1\}} \frac{1 - st a \cdot b}{4}.$$

Expanding,

$$P_B(t \mid a, b) = \frac{1}{4} \sum_{s \in \{\pm 1\}} 1 - \frac{t a \cdot b}{4} \sum_{s \in \{\pm 1\}} s.$$

Since

$$\sum_{s \in \{\pm 1\}} 1 = 2,$$

and

$$\sum_{s \in \{\pm 1\}} s = 0,$$

we obtain

$$P_B(t \mid a, b) = \frac{1}{2}.$$

Thus

$$P_B(t \mid a, b) = \frac{1}{2}.$$

The marginal probability at detector B is independent of the remote detector setting a . It also does not depend on the local detector setting b , again because the singlet state is locally maximally mixed.

7.6 Operational No-Signalling in the Strong-Readout Limit

The marginal identities

$$P_A(s \mid a, b) = \frac{1}{2}, \quad P_B(t \mid a, b) = \frac{1}{2}$$

imply operational no-signalling in the strong-readout limit.

For detector A , changing the remote setting from b to b' does not change the local marginal:

$$P_A(s \mid a, b) = P_A(s \mid a, b') = \frac{1}{2}.$$

For detector B , changing the remote setting from a to a' does not change the local marginal:

$$P_B(t \mid a, b) = P_B(t \mid a', b) = \frac{1}{2}.$$

Therefore the pointer-sector construction reproduces the standard no-signalling marginals of the Bell-singlet state.

This is an operational statement. It concerns the observable marginal probabilities obtained by summing the pointer-sector branch weights. It does not imply Bell-local factorisation of the joint probabilities. Indeed, the joint distribution remains

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

which depends on the joint pair of detector settings through the scalar product $a \cdot b$.

7.7 Finite-Leakage Marginal Bounds

At finite pointer separation, the branch probabilities are

$$P_{st}(a, b; T) = P_{st}(a, b) + \delta_{st}(a, b; T),$$

with

$$|\delta_{st}(a, b; T)| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

The finite-time marginal at detector A is

$$P_A(s \mid a, b; T) = \sum_{t \in \{\pm 1\}} P_{st}(a, b; T).$$

Substituting the finite-time branch probabilities gives

$$P_A(s \mid a, b; T) = \sum_t P_{st}(a, b) + \sum_t \delta_{st}(a, b; T).$$

The first sum is the ideal marginal:

$$\sum_t P_{st}(a, b) = \frac{1}{2}.$$

Therefore

$$P_A(s \mid a, b; T) - \frac{1}{2} = \sum_t \delta_{st}(a, b; T).$$

Taking absolute values,

$$\left| P_A(s \mid a, b; T) - \frac{1}{2} \right| \leq \sum_t |\delta_{st}(a, b; T)|.$$

Using the uniform leakage bound,

$$\left| P_A(s \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

Hence

$$P_A(s \mid a, b; T) = \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B).$$

Similarly, for detector B ,

$$P_B(t \mid a, b; T) = \sum_{s \in \{\pm 1\}} P_{st}(a, b; T).$$

Thus

$$P_B(t \mid a, b; T) - \frac{1}{2} = \sum_s \delta_{st}(a, b; T),$$

and therefore

$$\left| P_B(t \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

Equivalently,

$$P_B(t \mid a, b; T) = \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B).$$

These estimates show that finite pointer leakage produces only controlled deviations from the ideal no-signalling marginals.

7.8 Finite-Leakage No-Signalling Stability

The finite-leakage bounds imply approximate marginal stability. For detector A , compare two remote settings b and b' . We have

$$P_A(s \mid a, b; T) = \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B),$$

and

$$P_A(s \mid a, b'; T) = \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B).$$

Therefore

$$P_A(s \mid a, b; T) - P_A(s \mid a, b'; T) = \mathcal{O}(\epsilon_A + \epsilon_B).$$

Similarly,

$$P_B(t \mid a, b; T) - P_B(t \mid a', b; T) = \mathcal{O}(\epsilon_A + \epsilon_B).$$

Thus the pointer-sector construction satisfies exact operational no-signalling in the strong-readout limit and approximate no-signalling at finite pointer separation, with deviations controlled by the same leakage parameters that control branch distinguishability.

This statement should be read carefully. The finite-leakage terms describe imperfections in apparatus readout. They do not represent a new physical signalling channel. In the idealised projective limit,

$$\epsilon_A, \epsilon_B \rightarrow 0,$$

the marginal probabilities become exactly independent of the remote settings.

7.9 Relation Between Correlation and Marginals

The singlet kernel has two simultaneous features:

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

$$E(a, b) = -a \cdot b,$$

and

$$P_A(s \mid a, b) = P_B(t \mid a, b) = \frac{1}{2}.$$

The first equation gives the full joint distribution. The second gives the Bell-singlet correlation. The third gives the local marginal distributions.

The dependence on $a \cdot b$ appears only in the joint distribution and in the correlation. It cancels from the local marginals because the sums over the remote signs satisfy

$$\sum_{t \in \{\pm 1\}} t = 0, \quad \sum_{s \in \{\pm 1\}} s = 0.$$

This cancellation is the algebraic reason why the model reproduces Bell correlations without permitting operational signalling.

In LF3, this cancellation is formalised at the pointer-sector level. The probabilities being summed are not abstract hidden-variable assignments. They are the branch weights associated with the four projectors

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \widehat{Q}_s^A \otimes \widehat{Q}_t^B$$

after the context-dependent measurement interaction has produced the final state

$$|\Psi_T\rangle = \widehat{U}_{a,b}(T)|\Psi_0\rangle.$$

7.10 Lean Theorem Targets for the Correlation Layer

The Lean module corresponding to this section may remain part of

`CsdLean4/LF3/Singlet.lean`

or may be separated into a dedicated file

`CsdLean4/LF3/Correlation.lean`.

If the repository follows the seven-module structure, these theorem targets should be placed in `Singlet.lean`, with the Bell-consistency consequences exported later through `ContextMap.lean`.

The first theorem target is the correlation identity:

```
theorem correlation_eq_neg_dot :
  correlation a b =
    - dot a.vec b.vec
```

The second theorem target is the finite-leakage correlator bound:

```
theorem correlation_finite_leakage_bound :
  abs (correlation_T a b T + dot a.vec b.vec)
    <= 4 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)
```

The third theorem target is the A -marginal identity:

```
theorem marginal_A_eq_half :
  marginal_A a b s = 1 / 2
```

The fourth theorem target is the B -marginal identity:

```
theorem marginal_B_eq_half :
  marginal_B a b t = 1 / 2
```

The fifth theorem target is finite-leakage stability of the A -marginal:

```
theorem marginal_A_finite_leakage_bound :
  abs (marginal_A_T a b T s - 1 / 2)
    <= 2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)
```

The sixth theorem target is finite-leakage stability of the B -marginal:

```
theorem marginal_B_finite_leakage_bound :
  abs (marginal_B_T a b T t - 1 / 2)
    <= 2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)
```

The seventh theorem target records operational no-signalling in the strong-readout limit:

```
theorem no_signalling_strong_readout_A :
  marginal_A a b s = marginal_A a b' s
```

```
theorem no_signalling_strong_readout_B :
  marginal_B a b t = marginal_B a' b t
```

These theorem targets complete the correlation and marginal-stability layer.

7.11 Summary

This section derived the operational consequences of the singlet probability kernel. From

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

we obtained

$$E(a, b) = -a \cdot b.$$

We also obtained the local marginals

$$P_A(s \mid a, b) = \frac{1}{2}, \quad P_B(t \mid a, b) = \frac{1}{2}.$$

Thus the finite-dimensional pointer-sector construction reproduces the Bell-singlet correlation while preserving operational no-signalling.

At finite pointer separation, if

$$|\delta_{st}(a, b; T)| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B,$$

then

$$|E_T(a, b) + a \cdot b| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

and

$$\begin{aligned} \left| P_A(s \mid a, b; T) - \frac{1}{2} \right| &\leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B), \\ \left| P_B(t \mid a, b; T) - \frac{1}{2} \right| &\leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B). \end{aligned}$$

Therefore the ideal singlet correlation and no-signalling marginals are recovered in the strong-readout limit and remain stable under controlled finite pointer leakage.

8 Context-Indexed Outcome Maps and Bell Consistency

8.1 Purpose of the Context-Map Layer

The preceding sections established the finite-dimensional pointer-sector construction, the singlet probability kernel, the Bell-singlet correlation, and the no-signalling marginals:

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

$$E(a, b) = -a \cdot b,$$

and

$$P_A(s \mid a, b) = P_B(t \mid a, b) = \frac{1}{2}.$$

The purpose of the present section is to state the corresponding outcome-map structure. This is where LF3 records the distinction between deterministic outcomes for a realised measurement context and an inadmissible global joint assignment across incompatible contexts.

The key point is that LF3 does not define a single context-independent map assigning values to all possible detector settings at once. Instead, for each actually implemented context (a, b) , the measurement Hamiltonian

$$\widehat{H}_{a,b}$$

generates a final state

$$|\Psi_T^{(a,b)}\rangle = \hat{U}_{a,b}(T)|\Psi_0\rangle,$$

and the pointer-sector projectors

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \hat{Q}_s^A \otimes \hat{Q}_t^B$$

define the possible readout sectors for that context.

Thus the relevant outcome map is context-indexed:

$$F^{(a,b)} : \mathcal{D}_{a,b} \rightarrow \{\pm 1\} \times \{\pm 1\},$$

where $\mathcal{D}_{a,b}$ denotes the domain of final system-apparatus states, or the corresponding CSD basin domain, generated by the measurement context (a, b) .

This context-indexing is essential for Bell consistency. LF3 asserts deterministic readout structure per realised context. It does not assert a single global assignment of the form

$$A(a_1), A(a_2), B(b_1), B(b_2)$$

on one common probability space for all incompatible CHSH settings.

8.2 Outcome Maps for a Fixed Measurement Context

Fix detector settings

$$a, b \in S^2.$$

The measurement context is specified by the pair

$$(a, b),$$

the Hamiltonian

$$\hat{H}_{a,b},$$

the unitary

$$\hat{U}_{a,b}(T),$$

and the projective pointer readout

$$\left\{ \widehat{M}_{st}^{(a,b)} : s, t \in \{\pm 1\} \right\}.$$

For this context, the four outcome sectors are

$$\mathcal{R}_{st}^{(a,b)} = \text{Ran} \left(\widehat{M}_{st}^{(a,b)} \right).$$

These sectors are mutually orthogonal:

$$\mathcal{R}_{st}^{(a,b)} \perp \mathcal{R}_{s't'}^{(a,b)} \quad \text{when} \quad (s, t) \neq (s', t'),$$

and complete:

$$H_{SA}^{(N)} = \bigoplus_{s,t \in \{\pm 1\}} \mathcal{R}_{st}^{(a,b)}.$$

An ideal final state belonging to one of these sectors has a definite pointer readout. Thus, for a state

$$|\Phi\rangle \in \mathcal{R}_{st}^{(a,b)},$$

the context-indexed outcome map returns

$$F^{(a,b)}(|\Phi\rangle) = (s, t).$$

For a general superposed or pre-readout state, LF3 does not define an additional collapse rule. The operational weights are instead obtained from the projectors:

$$P_{st}(a, b; T) = \langle \Psi_T^{(a,b)} | \widehat{M}_{st}^{(a,b)} | \Psi_T^{(a,b)} \rangle.$$

In the CSD interpretation, the realised ontic trajectory lies in one outcome basin, but LF3 does not formalise the full Σ -level basin-selection dynamics. Its formal target is the finite-dimensional projective readout structure.

8.3 Epistemic and Ontic Readings

At the Hilbert-space level, the context-indexed readout is represented by the four projectors

$$\widehat{M}_{st}^{(a,b)}.$$

At the projective level, these projectors define epistemic outcome regions

$$\Omega_{st}^{\text{ep}}(a, b) \subseteq \mathbb{CP}^{N-1}.$$

A ray

$$[\Psi] \in \mathbb{CP}^{N-1}$$

belongs to the sector $\Omega_{st}^{\text{ep}}(a, b)$ when it has pointer support in the readout subspace associated with (s, t) , up to the finite-leakage tolerance specified in the previous sections.

At the CSD ontic level, one may write the corresponding basin as

$$\Omega_{st}^{\Sigma}(a, b) = \left(\pi_{SA}^{(N)} \right)^{-1} \left(\Omega_{st}^{\text{ep}}(a, b) \right),$$

where

$$\pi_{SA}^{(N)} : \Sigma_{SA}^{(N)} \rightarrow \mathbb{CP}^{N-1}$$

is the finite-dimensional system-apparatus projection.

The LF3 formalisation does not require the full ontic basin machinery. It records the compatibility between the finite-dimensional pointer projectors and the context-indexed outcome-sector interpretation. The formal calculation remains:

$$P_{st}(a, b; T) = \langle \Psi_T^{(a,b)} | \widehat{M}_{st}^{(a,b)} | \Psi_T^{(a,b)} \rangle.$$

The basin notation is interpretive scaffolding, not an additional source of probability.

8.4 Context Dependence of Pointer-Sector Readout

Although the pointer readout subspaces

$$K_A^{(+)}, K_A^{(-)}, K_B^{(+)}, K_B^{(-)}$$

are fixed apparatus sectors, the measurement context affects which final state is supplied to those sectors. This dependence enters through

$$\widehat{U}_{a,b}(T) = \exp \left(-\frac{iT}{\hbar} \widehat{H}_{a,b} \right).$$

Changing a or b changes the Hamiltonian, the measurement basis, the final branch-separated state, and therefore the joint branch weights.

For context (a, b) , the final state is

$$|\Psi_T^{(a,b)}\rangle = \sum_{s,t} c_{st}(a, b) |s_a, t_b\rangle \otimes |\phi_s^A(T)\rangle \otimes |\phi_t^B(T)\rangle.$$

For a different context (a', b') , the final state is

$$|\Psi_T^{(a',b')}\rangle = \sum_{s,t} c_{st}(a', b') |s_{a'}, t_{b'}\rangle \otimes |\phi_s^A(T)\rangle \otimes |\phi_t^B(T)\rangle.$$

These are different context-generated decompositions. Therefore LF3 does not identify their outcome maps as restrictions of one larger context-independent assignment.

The notation

$$F^{(a,b)}$$

is meant to preserve exactly this distinction. There is a well-defined outcome map for each actually implemented measurement context, but not a single map that assigns all incompatible detector settings simultaneously.

8.5 Absence of Bell-Local Factorisation

The pointer-sector construction gives deterministic readout structure for each realised measurement context. It does not imply Bell-local factorisation.

Bell-local factorisation would require the joint probability to decompose as

$$P_{st}(a, b \mid \lambda) = P_A(s \mid a, \lambda)P_B(t \mid b, \lambda),$$

or, in deterministic form, would require outcome functions

$$A(a, \lambda) \in \{\pm 1\}, \quad B(b, \lambda) \in \{\pm 1\},$$

such that the realised outcome pair is

$$(s, t) = (A(a, \lambda), B(b, \lambda)).$$

The LF3 construction does not supply such functions. Instead, for each joint context (a, b) , there is a context-indexed outcome map

$$F^{(a,b)} : \mathcal{D}_{a,b} \rightarrow \{\pm 1\} \times \{\pm 1\}.$$

The realised pair is determined by the actual system-apparatus evolution associated with the implemented Hamiltonian

$$\hat{H}_{a,b}.$$

Thus the relevant deterministic statement is

$$F^{(a,b)}(x) = (s, t),$$

not

$$A(a, x) = s, \quad B(b, x) = t$$

with A and B defined independently for all settings.

This distinction is essential. The model is deterministic per realised context, but it is not Bell-local in the sense required by the factorisation premise. The singlet probabilities are

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

Their joint dependence on $a \cdot b$ cannot be written as a product of independent local response probabilities for all settings while preserving the observed Bell-violating correlations.

8.6 Bell-Local Outcome Functions Are Not Part of LF3

LF3 does not introduce hidden response functions

$$A(a, \lambda), \quad B(b, \lambda),$$

where λ is assumed to be a context-independent hidden state shared across all possible detector choices. The formal construction uses the context-dependent measurement unitary

$$\hat{U}_{a,b}(T),$$

which generates the context-specific final state

$$|\Psi_T^{(a,b)}\rangle = \hat{U}_{a,b}(T)|\Psi_0\rangle.$$

The outcome sectors are then read from the projectors

$$\hat{M}_{st}^{(a,b)}.$$

Even if the same initial preparation $|\Psi_0\rangle$ is used, different choices of detector settings define different measurement interactions:

$$(a, b) \mapsto \hat{H}_{a,b}.$$

Consequently, the branches generated by (a, b) are not the same objects as the branches generated by (a', b') . The labels s and t are readout labels within a specific context, not pre-existing values assigned simultaneously for all incompatible measurements.

This is why the following implication is rejected:

$$\text{deterministic outcome for each realised context} \quad \Rightarrow \quad \text{global Bell-local outcome functions.}$$

LF3 asserts the first statement. It does not assert the second.

8.7 No Global Joint Assignment Across Incompatible Contexts

In a CHSH setting, each side has two possible detector settings:

$$a_1, a_2 \quad \text{and} \quad b_1, b_2.$$

A global joint assignment would require a single probability space supporting four simultaneously defined random variables:

$$A(a_1), \quad A(a_2), \quad B(b_1), \quad B(b_2).$$

Equivalently, one would require a single measurable map

$$G : \Lambda \rightarrow \{\pm 1\}^4,$$

with

$$G(\lambda) = (A(a_1, \lambda), A(a_2, \lambda), B(b_1, \lambda), B(b_2, \lambda)).$$

LF3 does not construct such a map. Instead, it constructs context-specific maps:

$$F^{(a_i, b_j)} : \mathcal{D}_{a_i, b_j} \rightarrow \{\pm 1\} \times \{\pm 1\}, \quad i, j \in \{1, 2\}.$$

These maps correspond to distinct implemented measurement contexts:

$$(a_1, b_1), \quad (a_1, b_2), \quad (a_2, b_1), \quad (a_2, b_2).$$

The programme does not identify these four context maps as marginals of one global joint assignment.

This is the structural point required by Fine-Bell consistency. Fine's theorem shows that, in the CHSH scenario, the existence of a global joint distribution for all four incompatible observables is equivalent to satisfaction of the Bell inequalities. Since the singlet probabilities violate those inequalities for suitable settings, LF3 must not imply such a global joint distribution.

8.8 Contextual Determinism

The correct LF3 statement is contextual determinism:

$$\forall (a, b), \quad \text{the realised context has a well-defined outcome sector.}$$

It is not global value determinism:

$$\exists G \quad \text{assigning outcomes for all incompatible settings simultaneously.}$$

At the pointer level, contextual determinism is represented by the four-sector projective decomposition associated with the realised context:

$$H_{SA}^{(N)} = \bigoplus_{s, t \in \{\pm 1\}} \mathcal{R}_{st}^{(a, b)}.$$

For a realised branch in $\mathcal{R}_{st}^{(a, b)}$, the readout is definite:

$$F^{(a, b)} = (s, t).$$

However, there is no corresponding statement that the same physical run also has definite values for the unperformed contexts

$$(a', b), \quad (a, b'), \quad (a', b').$$

This is not an ad hoc escape from Bell's theorem. It is the standard distinction between:

1. a deterministic outcome for the measurement actually performed;
2. a joint assignment of outcomes to all measurements that could have been performed.

LF3 formalises the first structure in the two-qubit pointer model. It does not assume the second structure.

8.9 Compatibility with Bell-Singlet Correlations

The context-indexed construction gives the singlet joint probabilities

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

From these probabilities,

$$E(a, b) = -a \cdot b.$$

For the usual CHSH choice of coplanar settings, this gives the standard quantum value

$$S_{\text{CHSH}} = 2\sqrt{2}.$$

This violates the Bell bound

$$|S_{\text{CHSH}}| \leq 2.$$

The violation does not generate a contradiction inside LF3 because LF3 does not assume the Bell-local factorisation premise and does not construct a global joint assignment across incompatible detector settings. The four CHSH contexts are represented by four context-indexed measurement maps:

$$F^{(a_1, b_1)}, \quad F^{(a_1, b_2)}, \quad F^{(a_2, b_1)}, \quad F^{(a_2, b_2)}.$$

These maps are not combined into a single function

$$G : \Lambda \rightarrow \{\pm 1\}^4.$$

Thus LF3 is Bell-consistent in the limited and precise sense required here: it reproduces the singlet correlation and no-signalling marginals while avoiding the global assignment structure that would force Bell inequalities.

8.10 Relation to the Fine-Bell Analysis

The Fine-Bell analysis clarifies why contextual outcome maps are required. In the CHSH scenario, Fine's theorem states that the existence of a single joint probability distribution over

$$A(a_1), \quad A(a_2), \quad B(b_1), \quad B(b_2)$$

is equivalent to satisfaction of the Bell inequalities. Therefore, any model reproducing the Bell-singlet correlations must avoid implying such a global joint distribution.

LF3 respects this constraint by assigning outcome maps only to realised measurement contexts. For each context (a, b) , the formal construction supplies:

$$\widehat{H}_{a,b}, \quad \widehat{U}_{a,b}(T), \quad \left\{ \widehat{M}_{st}^{(a,b)} \right\}_{s,t \in \{\pm 1\}},$$

and hence a context-specific readout structure

$$F^{(a,b)}.$$

No theorem in LF3 asserts that the maps for incompatible contexts are restrictions of one larger map.

The Fine-Bell companion analysis therefore supplies the external structural diagnosis, while LF3 supplies the concrete pointer-sector realisation. The division of labour is:

$$\text{Fine-Bell analysis} \Rightarrow \text{why global assignments are forbidden,}$$

and

$$\text{LF3} \Rightarrow \text{how a context-indexed pointer construction realises the singlet sector.}$$

8.11 Lean Status of the Bell-Consistency Layer

The Bell-consistency layer should be handled carefully in Lean. The direct algebraic results are part of LF3:

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

$$E(a, b) = -a \cdot b,$$

and

$$P_A(s \mid a, b) = P_B(t \mid a, b) = \frac{1}{2}.$$

These are constructive finite-dimensional theorems. They belong directly in the LF3 formalisation.

By contrast, the full Fine theorem should not be reproved inside LF3 unless it is already available in the repository or easy to encode. The intended LF3 target is narrower. It is sufficient to state the global-assignment obstruction as an imported theorem or explicit axiom boundary:

$$\text{Bell-violating singlet correlations} \Rightarrow \text{no global joint assignment.}$$

The Lean file

`CsdLean4/LF3/ContextMap.lean`

should therefore contain two types of result.

First, it should define the context-indexed outcome-map structure:

```
structure MeasurementContext where
  a : DetectorSetting
  b : DetectorSetting

def outcomeMap
  (ctx : MeasurementContext) :
  ContextDomain ctx -> Sign × Sign
```

Second, it should record that LF3 uses context-indexed maps rather than a global CHSH assignment:

```
structure ContextIndexedOutcomeMaps where
  F : MeasurementContext -> ContextDomain ctx -> Sign × Sign
```

A global CHSH assignment, if represented, should be a separate object:

```
structure GlobalCHSHAssignment where
  A1 : HiddenState -> Sign
  A2 : HiddenState -> Sign
  B1 : HiddenState -> Sign
  B2 : HiddenState -> Sign
```

The obstruction theorem can initially be imported or stated as an axiom boundary:

```
axiom fine_bell_no_global_assignment :
  BellViolatingSingletCorrelations settings ->
  Not (ExistsGlobalCHSHAssignment settings)
```

This keeps LF3 focused. The machine-checked contribution of LF3 is the constructive pointer-sector model and its singlet statistics. The Fine-Bell obstruction is part of the surrounding programme architecture, not the central theorem of LF3.

8.12 Context-Map Theorem Targets

The first theorem target is that each realised context has a well-defined four-sector projective readout:

```
theorem context_has_projective_readout :
  IsProjectiveDecomposition
  (fun st => M_hat ctx st)
```

The second theorem target is that each realised context has normalised branch weights:

```
theorem context_branch_weights_sum_one :
  sum over s,t of P_st ctx s t = 1
```

The third theorem target is that the context-indexed branch weights reproduce the singlet kernel:

```

theorem context_singlet_kernel :
  P_st ctx s t =
    (1 - sign s * sign t * dot ctx.a.vec ctx.b.vec) / 4
    
```

The fourth theorem target is the context-indexed correlation:

```

theorem context_correlation_eq_neg_dot :
  correlation ctx =
    - dot ctx.a.vec ctx.b.vec
    
```

The fifth theorem target is operational marginal stability:

```

theorem context_no_signalling_A :
  marginal_A ctx s = 1 / 2

theorem context_no_signalling_B :
  marginal_B ctx t = 1 / 2
    
```

The sixth item is not a theorem target but a structural remark.

The LF3 data structure consists of context-indexed maps

$$F^{(a,b)} : \mathcal{D}_{a,b} \rightarrow \{\pm 1\} \times \{\pm 1\},$$

one per realised measurement context. A global CHSH assignment would instead be a single function

$$G : \Lambda \rightarrow \{\pm 1\}^4.$$

These are distinct data types. The existence of the former is not by itself the construction of the latter. Whether any context-indexed family extends to a single G consistent with the singlet marginals is governed by Fine's theorem, imported in Section 8.11 as

```
fine_bell_no_global_assignment.
```

LF3 does not reprove Fine's theorem. It ensures only that its own outcome-map structure does not assume the global assignment in the first place.

8.13 Summary

This section clarified the Bell-facing interpretation of LF3. The pointer-sector construction gives definite outcome sectors for each realised measurement context:

$$F^{(a,b)} : \mathcal{D}_{a,b} \rightarrow \{\pm 1\} \times \{\pm 1\}.$$

It does not define Bell-local outcome functions

$$A(a, \lambda), \quad B(b, \lambda),$$

nor does it define a global CHSH assignment

$$G : \Lambda \rightarrow \{\pm 1\}^4.$$

The operational results remain:

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

$$E(a, b) = -a \cdot b,$$

and

$$P_A(s \mid a, b) = P_B(t \mid a, b) = \frac{1}{2}.$$

Thus LF3 reproduces the Bell-singlet correlation and no-signalling marginals while preserving the correct Fine-Bell boundary. The model is deterministic per realised context, but it is not Bell-local and does not assume global value assignments across incompatible measurement settings.

9 Lean4 Formalisation Architecture

9.1 Purpose of the Formalisation Architecture

This section records the intended Lean4 structure for LF3. The purpose is not to give final Lean source code, but to specify the repository architecture, module dependencies, theorem targets, and assumption boundaries required for implementation.

LF3 is designed as a narrow formalisation layer. Its task is to verify the finite-dimensional quantum-pointer construction for the Bell-singlet measurement. The implementation should therefore avoid unnecessary generality. It should not attempt to formalise all of geometric quantum mechanics, all of CSD, or all of Bell theory. The correct implementation target is the finite-dimensional chain:

setup $\longrightarrow$ Hamiltonian control $\longrightarrow$ branch separation $\longrightarrow$ pointer projectors $\longrightarrow$ singlet kernel $\longrightarrow$ correlation and marginals $\longrightarrow$ con

The architecture should preserve the logical separation from earlier Lean layers:

$$\text{LF1} \longrightarrow \text{LF2} \longrightarrow \text{LF3}.$$

LF1 supplies the repeated-trial frequency theorem. LF2 supplies the measure bridge and Born-weight wrapper. LF3 applies the LF2 operator-form weight rule to the explicit pointer-sector projectors of the TN3 measurement construction.

9.2 Repository Layout

The intended repository structure is:

```
CsdLean4/
  LF1/
  LF2/
  LF3/
    Setup.lean
    Hamiltonian.lean
    BranchSeparation.lean
    Projectors.lean
    Singlet.lean
    ContextMap.lean
    Interface.lean
  CsdLean4.lean
```

The top-level file should export LF3 through an import such as:

```
import CsdLean4.LF3.Interface
```

The module dependency order should be:

```
Setup.lean  $\rightarrow$  Hamiltonian.lean  $\rightarrow$  BranchSeparation.lean  $\rightarrow$  Projectors.lean  $\rightarrow$  Singlet.lean  $\rightarrow$  ContextMap.lean
```

This order reflects the proof dependency chain. Later files may import earlier files, but earlier files should not import later files.

9.3 Implementation Principles

The implementation should follow five principles.

First, LF3 should remain finite-dimensional. All Hilbert spaces, pointer sectors, operators, and projectors should be encoded in finite-dimensional form. This avoids unbounded-operator theory and continuous spectral analysis.

Second, LF3 should use abstract structures where Mathlib support is incomplete. For example, if finite-dimensional tensor-product operator identities are expensive to prove directly, the relevant algebraic properties may initially be packaged as structure fields or local lemmas, then discharged later.

Third, LF3 should separate exact strong-readout results from finite-leakage estimates. The exact theorem

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}$$

should be proved independently of finite-leakage bounds. The finite-leakage theorem should then wrap the exact result with explicit error estimates.

Fourth, LF3 should make all imported assumptions visible. In particular, the LF2 Born-weight wrapper should be referenced as an upstream interface rather than reproved.

Fifth, LF3 should avoid proving Fine’s theorem unless the required infrastructure is already available. The no-global-assignment statement may be represented as an imported theorem or an explicit axiom boundary.

9.4 Setup.lean

The purpose of `Setup.lean` is to define the finite-dimensional system-apparatus data used throughout LF3.

The file should contain:

- a type or structure for signs $s, t \in \{\pm 1\}$;
- detector settings as unit vectors in $\mathbb{R}^3$;
- the two-qubit system space;
- finite-dimensional pointer spaces K_A and K_B ;
- positive and negative pointer readout subspaces;
- completeness and orthogonality assumptions for the pointer sectors;
- Pauli-direction operators $\sigma \cdot a$;
- spin projectors $\Pi_s(a)$.

A schematic sign type is:

```
inductive Sign where
| plus
| minus
```

The corresponding numerical value is:

```
def Sign.val : Sign -> Real
| Sign.plus  => 1
| Sign.minus => -1
```

A detector setting may be represented as:

```
structure DetectorSetting where
  vec : EuclideanSpace Real (Fin 3)
  unit : norm vec = 1
```

The system-apparatus setup may be represented schematically as:

```
structure SystemApparatusSetup where
  K_A : Type*
  K_B : Type*
  inst_KA : Fintype K_A
  inst_KB : Fintype K_B
  inst_innerA : InnerProductSpace Complex K_A
  inst_innerB : InnerProductSpace Complex K_B
  Q_plus_A : Submodule Complex K_A
  Q_minus_A : Submodule Complex K_A
  Q_plus_B : Submodule Complex K_B
  Q_minus_B : Submodule Complex K_B
  ortho_A : Q_plus_A inf Q_minus_A = bot
  span_A : Q_plus_A sup Q_minus_A = top
  ortho_B : Q_plus_B inf Q_minus_B = bot
  span_B : Q_plus_B sup Q_minus_B = top
```

This schematic declaration records the mathematical data. The final Lean syntax may differ depending on the exact Mathlib interfaces used for finite-dimensional inner-product spaces, submodules, orthogonal complements, and tensor products.

The core theorem targets for `Setup.lean` are:

```

theorem sigmaDot_selfAdjoint :
  IsSelfAdjoint (sigmaDot a)

theorem spin_projector_idempotent :
  Pi_spin s a * Pi_spin s a = Pi_spin s a

theorem spin_projector_selfAdjoint :
  IsSelfAdjoint (Pi_spin s a)

theorem spin_projectors_complete :
  Pi_spin Sign.plus a + Pi_spin Sign.minus a = 1

theorem pointer_A_complete :
  Q_A Sign.plus + Q_A Sign.minus = 1

theorem pointer_B_complete :
  Q_B Sign.plus + Q_B Sign.minus = 1
    
```

These are elementary setup results, but they are required by the projector and branch-weight layers.

9.5 Hamiltonian.lean

The purpose of `Hamiltonian.lean` is to define the local readout Hamiltonians and prove their commutation and unitary factorisation.

The mathematical definitions are:

$$\begin{aligned}\hat{H}_A(a) &= \kappa_A(\sigma_a \otimes I_B) \otimes \hat{p}_A \otimes I_{K_B}, \\ \hat{H}_B(b) &= \kappa_B(I_A \otimes \sigma_b) \otimes I_{K_A} \otimes \hat{p}_B,\end{aligned}$$

and

$$\hat{H}_{a,b} = \hat{H}_A(a) + \hat{H}_B(b).$$

The schematic Lean declarations are:

```

def H_A
  (S : SystemApparatusSetup)
  (kappa_A : Real)
  (a : DetectorSetting)
  (p_A : S.K_A ->L[Complex] S.K_A) :
  H_SA S ->L[Complex] H_SA S

def H_B
  (S : SystemApparatusSetup)
  (kappa_B : Real)
  (b : DetectorSetting)
  (p_B : S.K_B ->L[Complex] S.K_B) :
  H_SA S ->L[Complex] H_SA S

def H_total
  (S : SystemApparatusSetup)
  (kappa_A kappa_B : Real)
  (a b : DetectorSetting)
  (p_A : S.K_A ->L[Complex] S.K_A)
  (p_B : S.K_B ->L[Complex] S.K_B) :
  H_SA S ->L[Complex] H_SA S :=
  H_A S kappa_A a p_A + H_B S kappa_B b p_B
    
```

The principal theorem is commutation:

```
theorem H_AB_commute :
  H_A S kappa_A a p_A * H_B S kappa_B b p_B =
  H_B S kappa_B b p_B * H_A S kappa_A a p_A
```

The corresponding unitary factorisation target is:

```
theorem U_measure_factorises :
  U_measure S kappa_A kappa_B a b p_A p_B T =
  U_A S kappa_A a p_A T *
  U_B S kappa_B b p_B T
```

If the exponential map for finite-dimensional operators is inconvenient in the first implementation, $U_measure$, U_A , and U_B may initially be represented abstractly by unitary operators satisfying the factorisation property:

```
structure MeasurementUnitary where
  U_measure : H_SA S ->L[Complex] H_SA S
  U_A : H_SA S ->L[Complex] H_SA S
  U_B : H_SA S ->L[Complex] H_SA S
  factorises : U_measure = U_A * U_B
```

This is a legitimate first-pass strategy because LF3's main theorem depends on the branch-separated action of the measurement unitary, not on a full formal derivation of the operator exponential.

9.6 BranchSeparation.lean

The purpose of `BranchSeparation.lean` is to encode the branch-separated final state and the finite-leakage assumptions. This module should not evaluate the singlet amplitudes. Its role is to establish that the measurement unitary maps each spin eigenbranch to a corresponding pointer branch.

The mathematical branch state is

$$|B_{st}(a, b; T)\rangle = |s_a, t_b\rangle \otimes |\phi_s^A(T)\rangle \otimes |\phi_t^B(T)\rangle.$$

The final state is

$$|\Psi_T\rangle = \sum_{s,t \in \{\pm 1\}} c_{st}(a, b) |B_{st}(a, b; T)\rangle.$$

A schematic Lean declaration is:

```
def branchState
  (S : SystemApparatusSetup)
  (a b : DetectorSetting)
  (T : Real)
  (s t : Sign) :
  H_SA S
```

```
def finalState
  (S : SystemApparatusSetup)
  (a b : DetectorSetting)
  (T : Real) :
  H_SA S
```

The main branch-decomposition theorem target is:

```
theorem finalState_branch_decomposition :
  finalState S a b T =
  sum over s,t of
    (c_st a b s t) • branchState S a b T s t
```

This theorem records the effect of the measurement unitary on the initial singlet-pointer state. It should be proved from the eigenbranch action of $U_measure$, or treated as a structured assumption if the first implementation abstracts the measurement unitary.

The leakage assumptions should be packaged explicitly. At the paper level, they are

$$\alpha_A(-s \mid s) \leq \epsilon_A, \quad \alpha_B(-t \mid t) \leq \epsilon_B,$$

and

$$\alpha_A(s \mid s) \geq 1 - \epsilon_A, \quad \alpha_B(t \mid t) \geq 1 - \epsilon_B.$$

A schematic Lean structure is:

```
structure PointerLeakageBounds where
  epsilon_A : Real
  epsilon_B : Real
  nonneg_A : 0 <= epsilon_A
  nonneg_B : 0 <= epsilon_B
  A_wrong :
    forall s, pointerOverlapA (-s) s <= epsilon_A
  B_wrong :
    forall t, pointerOverlapB (-t) t <= epsilon_B
  A_right :
    forall s, 1 - epsilon_A <= pointerOverlapA s s
  B_right :
    forall t, 1 - epsilon_B <= pointerOverlapB t t
```

The main leakage theorem target is:

```
theorem branch_separation_leakage_bound :
  wrongPointerReadoutMass S a b T
  <= epsilon_A + epsilon_B
```

The branch-weight estimate may either be proved here or deferred to `Projectors.lean`. The preferred architecture is to define leakage here and prove the branch-weight estimate in `Projectors.lean`, because the estimate depends on the pointer-sector projectors.

9.7 Projectors.lean

The purpose of `Projectors.lean` is to define the four pointer-sector projectors and prove their orthogonality, completeness, and branch-weight consequences.

The mathematical definition is

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \widehat{Q}_s^A \otimes \widehat{Q}_t^B.$$

A schematic Lean declaration is:

```
def M_hat
  (S : SystemApparatusSetup)
  (s t : Sign) :
  H_SA S ->L[Complex] H_SA S
```

The first theorem target is idempotence:

```
theorem M_idempotent :
  M_hat S s t * M_hat S s t = M_hat S s t
```

The second theorem target is self-adjointness:

```
theorem M_selfAdjoint :
  IsSelfAdjoint (M_hat S s t)
```

The third theorem target is orthogonality:

```
theorem M_orthogonal :
  (s,t) != (s',t') ->
  M_hat S s t * M_hat S s' t' = 0
```

The fourth theorem target is completeness:

```
theorem M_complete :
  sum over s,t of M_hat S s t = 1
```

The branch weight is then defined by:

```
def branchWeight
  (S : SystemApparatusSetup)
  (Psi_T : H_SA S)
  (s t : Sign) : Real :=
  realPart (inner Psi_T (M_hat S s t Psi_T))
```

The strong-readout branch-weight theorem is:

```
theorem branchWeight_strong_readout :
  epsilon_A = 0 ->
  epsilon_B = 0 ->
  branchWeight S Psi_T s t =
    normSq (c_st a b s t)
```

The finite-leakage theorem is:

```
theorem branchWeight_finite_leakage :
  abs
    (branchWeight S Psi_T s t
     - normSq (c_st a b s t))
  <= epsilon_A + epsilon_B + epsilon_A * epsilon_B
```

This module is the formal point at which pointer-sector readout becomes an explicit projective measurement. It should not evaluate `c_st`. That belongs to `Singlet.lean`.

9.8 Singlet.lean

The purpose of `Singlet.lean` is to prove the two-qubit singlet identities. This module converts the abstract branch-weight formula

$$P_{st}(a, b) = |c_{st}(a, b)|^2$$

into the explicit singlet kernel

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

The first theorem target is singlet normalisation:

```
theorem singlet_norm :
  normSq singlet = 1
```

The second theorem target is the vanishing of local expectations:

```
theorem singlet_left_pauli_expectation_zero :
  expectation singlet (sigmaDot a tensor I) = 0
```

```
theorem singlet_right_pauli_expectation_zero :
  expectation singlet (I tensor sigmaDot b) = 0
```

The third theorem target is the singlet correlation identity:

```
theorem singlet_pauli_correlation :
  expectation singlet (sigmaDot a tensor sigmaDot b) =
    - dot a.vec b.vec
```

The fourth theorem target is expansion of the joint spin projector:


```

theorem joint_spin_projector_expand :
  Pi_st a b s t =
    (1 / 4) •
      (I tensor I
       + sign s • (sigmaDot a tensor I)
       + sign t • (I tensor sigmaDot b)
       + sign s * sign t •
         (sigmaDot a tensor sigmaDot b))

```

The central theorem target is:

```

theorem cst_squared_eq :
  normSq (c_st a b s t) =
    (1 - sign s * sign t * dot a.vec b.vec) / 4

```

This theorem is the algebraic core of LF3. Once it is available, the correlation and marginal theorems become finite sums over the two-element sign type.

The correlation theorem target is:

```

theorem correlation_eq_neg_dot :
  correlation a b =
    - dot a.vec b.vec

```

The marginal theorem targets are:

```

theorem marginal_A_eq_half :
  marginal_A a b s = 1 / 2

```

```

theorem marginal_B_eq_half :
  marginal_B a b t = 1 / 2

```

The finite-leakage versions are:

```

theorem singlet_pointer_probability_finite_leakage :
  abs
    (P_st_T a b T s t
     - (1 - sign s * sign t * dot a.vec b.vec) / 4)
  <= epsilon_A + epsilon_B + epsilon_A * epsilon_B

theorem correlation_finite_leakage_bound :
  abs (correlation_T a b T + dot a.vec b.vec)
  <= 4 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)

theorem marginal_A_finite_leakage_bound :
  abs (marginal_A_T a b T s - 1 / 2)
  <= 2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)

theorem marginal_B_finite_leakage_bound :
  abs (marginal_B_T a b T t - 1 / 2)
  <= 2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)

```

These results complete the finite-dimensional singlet calculation required by LF3.

9.9 ContextMap.lean

The purpose of ContextMap.lean is to encode the distinction between context-indexed outcome maps and global CHSH assignments. This file should not attempt to reprove Fine's theorem. Its role is to represent the LF3 measurement structure correctly.

A measurement context is a pair of detector settings:

$$(a, b) \in S^2 \times S^2.$$

A schematic Lean declaration is:

```
structure MeasurementContext where
  a : DetectorSetting
  b : DetectorSetting
```

For each context, LF3 supplies a projective readout:

$$\left\{ \widehat{M}_{st}^{(a,b)} : s, t \in \{\pm 1\} \right\}.$$

The corresponding outcome map is context-indexed:

$$F^{(a,b)} : \mathcal{D}_{a,b} \rightarrow \{\pm 1\} \times \{\pm 1\}.$$

A schematic Lean declaration is:

```
def outcomeMap
  (ctx : MeasurementContext) :
  ContextDomain ctx -> Sign × Sign
```

The file should also define the structure of context-indexed outcome maps:

```
structure ContextIndexedOutcomeMaps where
  F :
    (ctx : MeasurementContext) ->
    ContextDomain ctx ->
    Sign × Sign
```

This should be kept separate from a global CHSH assignment. A global assignment would have the form:

$$G : \Lambda \rightarrow \{\pm 1\}^4,$$

assigning simultaneous values to

$$A(a_1), \quad A(a_2), \quad B(b_1), \quad B(b_2).$$

A schematic Lean declaration is:

```
structure GlobalCHSHAssignment where
  A1 : HiddenState -> Sign
  A2 : HiddenState -> Sign
  B1 : HiddenState -> Sign
  B2 : HiddenState -> Sign
```

The important architectural point is that a family of context-indexed maps is not the same data structure as a global CHSH assignment. LF3 records this distinction definitionally, not as a Bell theorem.

At the Lean level, the appropriate statement is a type-level separation:

ContextIndexedOutcomeMaps

and

GlobalCHSHAssignment

are distinct structures with different fields and different domains. This observation has no independent Bell-theorem content. The Bell-facing obstruction is instead imported, when needed, through the Fine-style theorem boundary:

```
axiom fine_bell_no_global_assignment :
  BellViolatingSingletCorrelations settings ->
  Not (ExistsGlobalCHSHAssignment settings)
```

If the Fine-Bell obstruction is included, it should initially be imported or stated as an explicit axiom boundary:

```
axiom fine_bell_no_global_assignment :
  BellViolatingSingletCorrelations settings ->
  Not (ExistsGlobalCHSHAssignment settings)
```

The constructive LF3 theorem targets in this file are:

```
theorem context_has_projective_readout :
  IsProjectiveDecomposition
  (fun st => M_hat ctx st)

theorem context_branch_weights_sum_one :
  sum over s,t of P_st ctx s t = 1

theorem context_singlet_kernel :
  P_st ctx s t =
    (1 - sign s * sign t * dot ctx.a.vec ctx.b.vec) / 4

theorem context_correlation_eq_neg_dot :
  correlation ctx =
    - dot ctx.a.vec ctx.b.vec

theorem context_no_signalling_A :
  marginal_A ctx s = 1 / 2

theorem context_no_signalling_B :
  marginal_B ctx t = 1 / 2
```

These are the results that should be proved directly in LF3. The broader no-global-assignment theorem should remain external unless a separate formalisation of Fine’s theorem is added later.

9.10 Interface.lean

The purpose of `Interface.lean` is to gather the preceding LF3 modules and export the main theorem package. It should import:

```
import CsdLean4.LF3.Setup
import CsdLean4.LF3.Hamiltonian
import CsdLean4.LF3.BranchSeparation
import CsdLean4.LF3.Projectors
import CsdLean4.LF3.Singlet
import CsdLean4.LF3.ContextMap
```

The main theorem package should state that the finite-dimensional pointer construction gives:

1. a complete projective pointer readout;
2. the singlet probability kernel;
3. the Bell-singlet correlation;
4. no-signalling marginals;
5. finite-leakage stability;
6. context-indexed outcome maps rather than global CHSH assignments.

A schematic main theorem is:

```
theorem LF3_main_theorem :
  ProjectivePointerReadout ctx
  and SingletKernel ctx
  and CorrelationNegDot ctx
  and NoSignallingMarginals ctx
  and FiniteLeakageStable ctx
  and ContextIndexedReadout ctx
```

A more explicit version is:

```
theorem LF3_main_theorem :
  (forall s t,
    P_st ctx s t =
      (1 - sign s * sign t * dot ctx.a.vec ctx.b.vec) / 4)
  and
  correlation ctx =
    - dot ctx.a.vec ctx.b.vec
  and
  (forall s, marginal_A ctx s = 1 / 2)
  and
  (forall t, marginal_B ctx t = 1 / 2)
```

The finite-leakage version may be exported separately:

```
theorem LF3_finite_leakage_theorem :
  (forall s t,
    abs
      (P_st_T ctx T s t
        - (1 - sign s * sign t * dot ctx.a.vec ctx.b.vec) / 4)
    <= epsilon_A + epsilon_B + epsilon_A * epsilon_B)
  and
  abs (correlation_T ctx T + dot ctx.a.vec ctx.b.vec)
    <= 4 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)
  and
  (forall s,
    abs (marginal_A_T ctx T s - 1 / 2)
      <= 2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B))
  and
  (forall t,
    abs (marginal_B_T ctx T t - 1 / 2)
      <= 2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B))
```

The interface file should be the only file imported by later LF modules unless those modules need internal details.

9.11 Mathlib Dependencies

The LF3 implementation is expected to require Mathlib support for:

- finite types;
- finite sums;
- real and complex numbers;
- finite-dimensional vector spaces;
- inner-product spaces;
- matrices over $\mathbb{C}$;
- linear maps and continuous linear maps;
- submodules;
- tensor products, if the implementation uses Mathlib tensor products directly;
- elementary norm and inner-product identities.

The most delicate dependency is the tensor-product operator algebra. If this becomes a bottleneck, the first implementation may abstract the required tensor-factor identities as local structure fields. This is acceptable for LF3 v1.00, provided the paper states the boundary clearly.

For example, instead of proving all tensor-factor commutation results from first principles, one may initially encode:

```

structure TensorFactorReadoutAlgebra where
  H_A : H_SA S ->L[Complex] H_SA S
  H_B : H_SA S ->L[Complex] H_SA S
  commute : H_A * H_B = H_B * H_A
    
```

Similarly, if projectors on submodules are awkward to construct explicitly, one may package their algebraic properties:

```

structure BinaryPointerProjectors where
  Q_plus : K ->L[Complex] K
  Q_minus : K ->L[Complex] K
  plus_idem : Q_plus * Q_plus = Q_plus
  minus_idem : Q_minus * Q_minus = Q_minus
  orthogonal : Q_plus * Q_minus = 0
  complete : Q_plus + Q_minus = 1
    
```

This keeps the first version focused on the LF3 theorem chain rather than on low-level library engineering.

9.12 Imported Results and Axiom Boundary

LF3 should distinguish proved internal results from imported or assumed results.

The following should be proved internally:

- orthogonality and completeness of pointer-sector projectors;
- branch-weight reduction in the strong-readout limit;
- finite-leakage branch-weight bounds;
- singlet squared-amplitude identity;
- correlation identity $E(a, b) = -a \cdot b$;
- marginal identities $P_A = P_B = \frac{1}{2}$;
- finite-leakage correlation and marginal bounds.

The following may be imported or treated as explicit boundaries:

- the LF2 Born-weight wrapper;
- the full Fine theorem;
- the Ashtekar-Schilling correspondence;
- the Σ -level ontic basin interpretation;
- general POVM, mixed-state, and subsystem-reduction machinery.

The paper-level claim should match the Lean boundary. LF3 proves the finite-dimensional pointer-sector singlet layer. It does not claim to formalise the entire CSD programme.

9.13 Top-Level Export Structure

The top-level LF3 export should expose two principal results:

LF3_main_theorem

and

LF3_finite_leakage_theorem.

The first records the ideal strong-readout result:

$$P_{st}(a, b) = \frac{1 - st\, a \cdot b}{4},$$

$$E(a, b) = -a \cdot b,$$

and

$$P_A(s \mid a, b) = P_B(t \mid a, b) = \frac{1}{2}.$$

The second records the finite-leakage stability result:

$$P_{st}(a, b; T) = \frac{1 - st a \cdot b}{4} + \mathcal{O}(\epsilon_A + \epsilon_B),$$

$$E_T(a, b) = -a \cdot b + \mathcal{O}(\epsilon_A + \epsilon_B),$$

and

$$P_A(s \mid a, b; T) = \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B),$$

$$P_B(t \mid a, b; T) = \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B).$$

At the Lean level, the $\mathcal{O}$ -notation should be replaced by explicit inequalities:

$$\left| P_{st}(a, b; T) - \frac{1 - st a \cdot b}{4} \right| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B,$$

$$|E_T(a, b) + a \cdot b| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

and

$$\left| P_A(s \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

$$\left| P_B(t \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

9.14 Summary

This section specified the Lean4 architecture for LF3. The recommended structure is:

Setup $\rightarrow$ Hamiltonian $\rightarrow$ BranchSeparation $\rightarrow$ Projectors $\rightarrow$ Singlet $\rightarrow$ ContextMap $\rightarrow$ Interface.

This structure mirrors the mathematical dependency chain of the paper.

The main formal target is not a general reconstruction of quantum measurement. It is the narrower finite-dimensional theorem that the TN3 pointer-sector construction, when combined with the LF2 Born-weight interface, yields the Bell-singlet probability kernel, the correlation

$$E(a, b) = -a \cdot b,$$

the no-signalling marginals

$$P_A = P_B = \frac{1}{2},$$

and controlled finite-leakage stability.

10 Relation to the CSD Dependency Stack

10.1 Purpose of this Section

This section situates LF3 within the wider Constraint-Surface Dynamics dependency stack. LF3 is not an isolated formalisation. It depends on earlier layers and prepares later layers. Its role is to machine-check the finite-dimensional pointer-sector construction for the two-qubit singlet measurement, while preserving the logical boundaries of the surrounding programme.

The relevant dependency chain is

$$\text{LF1} \longrightarrow \text{LF2} \longrightarrow \text{LF3} \longrightarrow \text{LF4} \longrightarrow \text{LF5}.$$

In this sequence, LF3 occupies the first explicitly system-apparatus measurement layer. It takes the Born-weight interface from LF2 and applies it to the TN3 pointer-sector construction. It does not rederive the frequency theorem of LF1, the measure bridge of LF2, or the later mixed-state and update layers assigned to LF4 and LF5.

At the paper level, the closest dependencies are

$$\text{Paper B}, \quad \text{TN2}, \quad \text{TN3}, \quad \text{LF1}, \quad \text{LF2}.$$

The closest downstream targets are

$$\text{TN4}, \quad \text{TN6}, \quad \text{LF4}, \quad \text{LF5}.$$

10.2 Relation to Paper B

Paper B supplies the finite-dimensional Born-weight layer. Its purpose is to fix the admissible probability assignment on finite-dimensional projective state space using symmetry and operational consistency. LF3 does not repeat this argument.

In LF3, the relevant probability rule is used in operator form:

$$w(P \mid \Psi) = \langle \Psi | P | \Psi \rangle.$$

For the pointer-sector projectors,

$$P = \widehat{M}_{st}^{(a,b)},$$

and for the final system-apparatus state,

$$|\Psi\rangle = |\Psi_T\rangle,$$

this gives

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

Thus LF3 depends on Paper B and LF2 for the admissibility of the operator-form branch weights. LF3 adds the concrete measurement-control structure:

$$\widehat{H}_{a,b}, \quad \widehat{U}_{a,b}(T), \quad \widehat{M}_{st}^{(a,b)},$$

and verifies that these objects reproduce the Bell-singlet kernel.

The division of labour is therefore:

$$\text{Paper B} \Rightarrow \text{why the operator-form weight is admissible,}$$

whereas

$$\text{LF3} \Rightarrow \text{how that weight is realised in the explicit pointer model.}$$

10.3 Relation to TN2

TN2 verifies the two-qubit singlet correlation non-constructively from the finite-dimensional Born-weight layer. It starts from the singlet state, applies the admissible Born-form probability rule, and obtains

$$P_{st}(a, b) = \frac{1 - st \, a \cdot b}{4},$$

and hence

$$E(a, b) = -a \cdot b.$$

LF3 overlaps with TN2 in the final algebraic singlet calculation. However, LF3 has a different purpose. TN2 is a non-constructive singlet specialisation. LF3 is the machine-verified formalisation of the constructive pointer-sector model.

The distinction is:

$$\text{TN2} \Rightarrow \text{singlet correlation from the Born-weight layer,}$$

whereas

$$\text{LF3} \Rightarrow \text{singlet correlation from a finite-dimensional pointer readout construction.}$$

In particular, LF3 adds the following structures that TN2 does not need to formalise:

$$H_{SA}^{(N)}, \quad \widehat{H}_{a,b}, \quad \widehat{U}_{a,b}(T), \quad |\Psi_T\rangle, \quad \widehat{M}_{st}^{(a,b)},$$

and the finite-leakage bounds

$$\mathcal{O}(\epsilon_A + \epsilon_B).$$

Thus TN2 supplies the non-constructive probability check, while LF3 supplies the constructive formal measurement layer.

10.4 Relation to TN3

TN3 is the direct paper-level source for LF3. It provides the quantum-pointer Hamiltonian realisation of the Bell-singlet correlation. LF3 is the Lean formalisation counterpart of that construction.

The TN3 construction begins with the finite-dimensional system-apparatus Hilbert space

$$H_{SA}^{(N)} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes K_A^{(d_A)} \otimes K_B^{(d_B)}, \quad N = 4d_A d_B.$$

It defines the readout Hamiltonian

$$\hat{H}_{a,b} = \kappa_A(\sigma \cdot a \otimes I) \otimes \hat{p}_A + \kappa_B(I \otimes \sigma \cdot b) \otimes \hat{p}_B,$$

and the pointer-sector projectors

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \hat{Q}_s^A \otimes \hat{Q}_t^B.$$

It then shows that the pointer-sector weights reproduce

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

with no-signalling marginals

$$P_A(s \mid a, b) = \frac{1}{2}, \quad P_B(t \mid a, b) = \frac{1}{2}.$$

LF3 formalises this sequence. Its module structure mirrors the TN3 proof structure:

$$\text{Setup} \rightarrow \text{Hamiltonian} \rightarrow \text{BranchSeparation} \rightarrow \text{Projectors} \rightarrow \text{Singlet} \rightarrow \text{ContextMap}.$$

The purpose is to make the TN3 constructive layer auditable at the theorem level.

10.5 Relation to LF1

LF1 formalises the repeated-trial frequency theorem. Its role is to show that, under a repeated-preparation model, empirical frequencies converge almost surely to the corresponding normalised weights.

LF3 does not use this theorem directly in its internal algebraic calculations. LF3 proves statements about finite-dimensional branch weights:

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

LF1 is required only when these weights are interpreted as long-run empirical frequencies under repeated trials.

The relationship is:

$$\text{LF3} \quad \Rightarrow \quad \text{what the branch weights are,}$$

while

$$\text{LF1} \quad \Rightarrow \quad \text{why repeated-trial frequencies converge to those weights.}$$

Thus the full empirical chain is:

$$\text{LF3 pointer weights} \longrightarrow \text{LF1 repeated-trial convergence} \longrightarrow \text{observed long-run frequencies.}$$

LF3 itself remains a finite-dimensional branch-weight formalisation.

10.6 Relation to LF2

LF2 is the immediate upstream formalisation layer for LF3. Its role is to isolate the measure bridge and Born-weight wrapper in finite dimension. LF2 does not formalise an explicit measurement apparatus. Instead, it supplies the operator-form probability interface used by later reconstruction layers.

LF3 uses this interface in the following form. Given a normalised finite-dimensional state

$$|\Psi\rangle \in H_{SA}^{(N)},$$

and a projective outcome operator

$$P,$$

the corresponding weight is

$$w(P \mid \Psi) = \langle \Psi | P | \Psi \rangle.$$

For the LF3 pointer-sector construction, the relevant projectors are

$$P = \widehat{M}_{st}^{(a,b)}.$$

Thus the LF2 interface gives

$$w_{st} = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

LF3 identifies this with the pointer-sector branch weight:

$$w_{st} = P_{st}(a, b; T).$$

The relationship between LF2 and LF3 is therefore:

$$\text{LF2} \Rightarrow \text{admissible operator-form weights,}$$

and

$$\text{LF3} \Rightarrow \text{explicit pointer-sector projectors and singlet weights.}$$

This separation is important. LF3 does not strengthen, modify, or rederive LF2. It applies LF2 to a concrete finite-dimensional system-apparatus construction.

10.7 Forward Relation to LF4

LF4 is assigned to the mixed-state, POVM, and subsystem-reduction layer. Its paper-level source is TN4. LF3 prepares LF4 by providing a formally checked projective measurement model on a finite-dimensional system-apparatus space.

The transition from LF3 to LF4 is natural. LF3 treats projective pointer readout:

$$\left\{ \widehat{M}_{st}^{(a,b)} \right\}_{s,t \in \{\pm 1\}}.$$

LF4 should generalise this operational machinery to mixed preparations, generalised measurements, and subsystem reductions. At the paper level, this means moving from pure-state projective weights of the form

$$\langle \Psi | P | \Psi \rangle$$

to density-operator and POVM weights of the form

$$\text{Tr}(\rho E_k).$$

LF3 therefore supplies a useful base case. It verifies the projective readout layer before LF4 extends the formalisation to the more general finite-dimensional operational dictionary.

The relationship is:

$$\text{LF3} \Rightarrow \text{pure-state projective pointer readout,}$$

and

$$\text{LF4} \Rightarrow \text{mixed states, POVMs, and subsystem reduction.}$$

10.8 Forward Relation to LF5

LF5 is assigned to the post-measurement update and sequential-experiment layer. Its paper-level source is TN6. LF3 prepares LF5 by formalising the outcome sectors that later update rules must condition on.

In LF3, the outcome sectors are

$$\widehat{M}_{st}^{(a,b)}, \quad s, t \in \{\pm 1\}.$$

These projectors define the readout outcomes for the realised measurement context. A later sequential-update formalisation must describe what happens after one of these outcomes is registered.

At the projective level, the ideal post-measurement update associated with outcome (s, t) has the schematic form

$$|\Psi_T\rangle \mapsto \frac{\widehat{M}_{st}^{(a,b)} |\Psi_T\rangle}{\left\| \widehat{M}_{st}^{(a,b)} |\Psi_T\rangle \right\|}.$$

At the density-operator level, the corresponding Lüders-type update is

$$\rho \mapsto \frac{\widehat{M}_{st}^{(a,b)} \rho \widehat{M}_{st}^{(a,b)}}{\text{Tr}(\rho \widehat{M}_{st}^{(a,b)})}.$$

LF3 does not formalise these updates. It provides the projectors and branch weights that LF5 will later use.

The relationship is:

$$\text{LF3} \Rightarrow \text{outcome sectors,}$$

and

$$\text{LF5} \Rightarrow \text{outcome-conditioned update and sequential statistics.}$$

10.9 What LF3 Adds to the Formalisation Branch

LF3 adds a new kind of result to the formalisation branch. LF1 and LF2 are largely measure-theoretic and probability-interface layers. LF3 is the first layer that formalises an explicit finite-dimensional measurement-control model.

The new objects introduced by LF3 are:

$$H_{SA}^{(N)}, \quad \widehat{H}_{a,b}, \quad \widehat{U}_{a,b}(T), \quad |\Psi_T\rangle, \quad \widehat{M}_{st}^{(a,b)}.$$

The new theorem package is:

$$\begin{aligned} P_{st}(a, b) &= \frac{1 - st \, a \cdot b}{4}, \\ E(a, b) &= -a \cdot b, \\ P_A(s \mid a, b) &= \frac{1}{2}, \quad P_B(t \mid a, b) = \frac{1}{2}, \end{aligned}$$

and the finite-leakage estimates

$$\begin{aligned} P_{st}(a, b; T) &= \frac{1 - st \, a \cdot b}{4} + \mathcal{O}(\epsilon_A + \epsilon_B), \\ E_T(a, b) &= -a \cdot b + \mathcal{O}(\epsilon_A + \epsilon_B), \\ P_A(s \mid a, b; T) &= \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B), \\ P_B(t \mid a, b; T) &= \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B). \end{aligned}$$

Thus LF3 converts the Bell-singlet part of the programme from a paper-level constructive model into a machine-checkable theorem layer.

10.10 Dependency Boundary

The following dependencies are upstream of LF3:

$$\text{LF1}, \quad \text{LF2}, \quad \text{Paper B}, \quad \text{TN2}, \quad \text{TN3}.$$

The following are downstream of LF3:

$$\text{LF4}, \quad \text{LF5}, \quad \text{TN4}, \quad \text{TN6}.$$

LF3 should not be judged as a complete formalisation of CSD. Its claim is narrower:

LF3 formalises the finite-dimensional pointer-sector singlet measurement layer.

It does not formalise:

typicality, the measure bridge, Born-weight uniqueness, mixed states, POVMs, sequential update, full ontic

This boundary is a strength rather than a weakness. It makes the formalisation tractable and keeps each theorem layer auditable.

10.11 Summary

LF3 sits between the Born-weight interface and the later operational-reconstruction layers. Its role is:

$$\begin{aligned} \text{LF2} &\Rightarrow \text{operator-form weights,} \\ \text{LF3} &\Rightarrow \text{finite-dimensional pointer-sector singlet measurement,} \\ \text{LF4} &\Rightarrow \text{mixed states, POVMs, and reduction,} \\ \text{LF5} &\Rightarrow \text{post-measurement update and sequential experiments.} \end{aligned}$$

Within this sequence, LF3 establishes the constructive Bell-singlet measurement layer. It shows that the TN3 pointer-sector construction can be represented as a finite-dimensional projective readout model with the correct singlet probability kernel, correlation, no-signalling marginals, and finite-leakage stability.

11 Scope, Assumptions, and Non-Claims

11.1 Purpose of this Section

This section records the scope and assumption boundary of LF3. The purpose is to prevent the formalisation from being read as a broader claim than it supports. LF3 is a finite-dimensional Lean formalisation of the quantum-pointer construction for the Bell-singlet measurement. It is not a complete formalisation of Constraint-Surface Dynamics, nor is it a derivation of quantum mechanics from first principles.

The main formal target is the theorem package

$$\begin{aligned} P_{st}(a, b) &= \frac{1 - st a \cdot b}{4}, \\ E(a, b) &= -a \cdot b, \\ P_A(s \mid a, b) &= \frac{1}{2}, \quad P_B(t \mid a, b) = \frac{1}{2}, \end{aligned}$$

together with finite-leakage estimates of the form

$$P_{st}(a, b; T) = \frac{1 - st a \cdot b}{4} + \mathcal{O}(\epsilon_A + \epsilon_B).$$

These results are established inside a finite-dimensional system-apparatus setting, using the LF2 operator-form Born-weight interface and the TN3 pointer-sector construction.

The scope is therefore intentionally narrow. LF3 verifies that a concrete finite-dimensional measurement model realises the Bell-singlet sector. It does not attempt to solve all remaining foundational, geometric, or continuum-level questions in the CSD programme.

11.2 Finite-Dimensional Scope

LF3 is restricted to finite-dimensional Hilbert spaces. The system-apparatus Hilbert space is

$$H_{SA}^{(N)} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes K_A^{(d_A)} \otimes K_B^{(d_B)}, \quad N = 4d_A d_B.$$

The pointer spaces

$$K_A^{(d_A)}, \quad K_B^{(d_B)}$$

are finite-dimensional truncations. The corresponding projective state space is

$$\mathbb{CP}^{N-1}.$$

This restriction is important for Lean formalisation. It avoids:

- unbounded operators;
- domain subtleties for momentum operators;
- continuous pointer spectra;
- infinite-dimensional spectral theory;
- measure-theoretic complications associated with continuous readout spaces.

The finite-dimensional truncation should not be read as a claim that physical pointers are literally finite-dimensional in ordinary laboratory descriptions. Rather, it is the controlled finite model used by LF3. Continuum-like limits and truncation comparison belong to other parts of the CSD programme.

11.3 Pure Singlet-State Restriction

LF3 treats the two-qubit singlet state

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} (|+\rangle_z \otimes |-\rangle_z - |-\rangle_z \otimes |+\rangle_z).$$

It does not formalise arbitrary two-qubit pure states, mixed states, multipartite states, GHZ states, W states, or general entangled states.

The reason is methodological. The singlet state is the canonical Bell-sector test case. It is sufficient to verify that the TN3 pointer construction yields

$$E(a, b) = -a \cdot b$$

and the no-signalling marginals

$$P_A = P_B = \frac{1}{2}.$$

General state classes would require additional infrastructure and are outside the LF3 target.

11.4 Projective Pointer-Readout Restriction

LF3 treats projective pointer readout. The measurement outcomes are represented by four orthogonal projectors

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \widehat{Q}_s^A \otimes \widehat{Q}_t^B, \quad s, t \in \{\pm 1\}.$$

These projectors satisfy

$$\widehat{M}_{st}^{(a,b)} \widehat{M}_{s't'}^{(a,b)} = 0 \quad \text{when} \quad (s, t) \neq (s', t'),$$

and

$$\sum_{s,t \in \{\pm 1\}} \widehat{M}_{st}^{(a,b)} = I_{SA}.$$

LF3 does not formalise general POVMs. It also does not formalise non-projective measurement instruments. Those belong to LF4, which is assigned to the mixed-state, POVM, and subsystem-reduction layer.

11.5 No Formalisation of the Ashtekar-Schilling Correspondence

The TN3 paper-level construction can be interpreted geometrically through the standard correspondence between finite-dimensional Schrödinger evolution and Hamiltonian flow on complex projective space. LF3 does not formalise that correspondence.

At the Lean level, LF3 works directly with finite-dimensional Hilbert-space objects:

$$|\Psi\rangle, \quad \widehat{H}_{a,b}, \quad \widehat{U}_{a,b}(T), \quad \widehat{M}_{st}^{(a,b)}.$$

It does not prove that the Schrödinger flow on $H_{SA}^{(N)}$ corresponds to a Hamiltonian flow on

$$\mathbb{CP}^{N-1}.$$

This is a deliberate scope boundary. Formalising geometric quantum mechanics would be valuable, but it is not required for the LF3 theorem package. LF3 verifies the finite-dimensional pointer-sector calculation at the operator level.

11.6 No Formalisation of Full Ontic Basin Dynamics on Σ

LF3 does not formalise the full ontic basin dynamics on the CSD configuration surface

$$\Sigma.$$

The paper-level interpretation allows one to associate pointer-sector outcome regions with epistemic sectors in

$$\mathbb{CP}^{N-1}$$

and, by inverse image, with ontic basins in

$$\Sigma_{SA}^{(N)}.$$

Schematic notation for this relation is

$$\Omega_{st}^\Sigma(a, b) = \left(\pi_{SA}^{(N)} \right)^{-1} (\Omega_{st}^{\text{ep}}(a, b)).$$

However, LF3 does not prove a general theorem classifying which ontic basins are dynamically realised by physical apparatus Hamiltonians. It also does not derive the single-run basin-selection rule from deeper Σ -level principles.

The formal content of LF3 remains the finite-dimensional pointer-sector calculation:

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

The Σ -level basin language is interpretive scaffolding. It connects LF3 to the wider CSD ontology, but it is not part of the machine-checked LF3 theorem package.

11.7 No General POVM or Mixed-State Extension

LF3 treats pure-state projective pointer readout. It does not formalise mixed states, density operators, POVMs, dilation theorems, or subsystem reduction.

The LF3 probability rule is used in the form

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

The more general finite-dimensional operational form

$$P(k | \rho) = \text{Tr}(\rho E_k)$$

belongs to the later mixed-state and POVM layer.

This boundary is important. LF3 should not be read as the full operational reconstruction of finite-dimensional quantum mechanics. It is the projective singlet-pointer case. The general operational dictionary is assigned to LF4.

11.8 No Sequential-Update Formalisation

LF3 does not formalise post-measurement update or sequential experiments. It defines the pointer-sector projectors

$$\widehat{M}_{st}^{(a,b)}$$

and their branch weights, but it does not formalise the conditional state update

$$|\Psi_T\rangle \mapsto \frac{\widehat{M}_{st}^{(a,b)} |\Psi_T\rangle}{\left\| \widehat{M}_{st}^{(a,b)} |\Psi_T\rangle \right\|}.$$

Nor does it formalise the density-operator update

$$\rho \mapsto \frac{\widehat{M}_{st}^{(a,b)} \rho \widehat{M}_{st}^{(a,b)}}{\text{Tr}(\rho \widehat{M}_{st}^{(a,b)})}.$$

Those update rules belong to the later sequential-measurement layer. LF3 provides the outcome projectors and branch weights that such an update theory would use, but does not itself formalise update dynamics.

11.9 No General Multipartite Bell Formalisation

LF3 treats the two-qubit singlet only. It does not formalise arbitrary bipartite Bell scenarios, multipartite Bell inequalities, GHZ-type contradictions, Hardy-type arguments, or device-independent protocols.

The relevant detector settings are

$$a, b \in S^2,$$

and the relevant outcome signs are

$$s, t \in \{\pm 1\}.$$

The central probability kernel is

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

This is sufficient for the Bell-singlet correlation

$$E(a, b) = -a \cdot b.$$

A more general Bell formalisation would require a separate framework for arbitrary measurement-setting sets, outcome alphabets, joint distributions, compatibility structures, and no-go theorem statements. LF3 does not attempt that broader formalisation.

11.10 No Derivation of the Born Rule

LF3 does not derive the Born rule. It uses the operator-form weight interface supplied upstream by LF2.

The formal branch weight is

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

This expression is not introduced in LF3 as a new primitive. It is the LF2 Born-weight wrapper applied to the concrete projective pointer readout defined by

$$\widehat{M}_{st}^{(a,b)}.$$

The LF3 contribution begins after the probability interface has been supplied. It shows that the TN3 system-apparatus construction produces the projectors, branch-separated state, singlet coefficients, correlation, marginals, and finite-leakage bounds required for the Bell-singlet sector.

11.11 No Claim of Bell Locality

LF3 does not claim Bell locality. The construction is deterministic per realised measurement context, but it is not Bell-local.

Bell-local determinism would require outcome functions

$$A(a, \lambda), \quad B(b, \lambda),$$

such that

$$(s, t) = (A(a, \lambda), B(b, \lambda))$$

for all relevant settings. LF3 does not construct such functions.

Instead, LF3 uses context-indexed outcome maps

$$F^{(a,b)} : \mathcal{D}_{a,b} \rightarrow \{\pm 1\} \times \{\pm 1\}.$$

The outcome map is indexed by the realised measurement context (a, b) . This preserves the distinction between deterministic readout for the measurement actually performed and a forbidden global assignment of outcomes to all incompatible measurements simultaneously.

Thus LF3 is compatible with Bell violation because it does not assume the factorisation premise required by Bell-local hidden-variable models.

11.12 No Global CHSH Assignment

LF3 does not define a global CHSH assignment. In a CHSH experiment, such an assignment would require simultaneous values for

$$A(a_1), \quad A(a_2), \quad B(b_1), \quad B(b_2).$$

Equivalently, one would require a map

$$G : \Lambda \rightarrow \{\pm 1\}^4.$$

LF3 instead defines context-indexed readout structures:

$$F^{(a_1, b_1)}, \quad F^{(a_1, b_2)}, \quad F^{(a_2, b_1)}, \quad F^{(a_2, b_2)}.$$

These are not assumed to be restrictions of a single global map. This is the correct Fine-Bell boundary for a model reproducing Bell-singlet correlations.

11.13 Assumptions Used Internally

The internal LF3 theorem package depends on the following assumptions:

1. finite-dimensional system and pointer Hilbert spaces;
2. self-adjoint Pauli-direction spin observables;
3. finite-dimensional pointer generators;
4. orthogonal and complete pointer readout projectors;
5. the impulsive-readout approximation;
6. branch-separated final-state form;
7. structural pointer-readout compatibility bounds ϵ_A, ϵ_B , axiomatised in LF3 rather than derived from a concrete microscopic pointer construction;
8. the LF2 operator-form Born-weight interface;
9. standard two-qubit singlet identities.

Under these assumptions, LF3 proves the pointer-sector singlet theorem package. The assumptions are explicit so that the formalisation can be audited and later strengthened if deeper derivations are added.

11.14 What LF3 Establishes

Within its stated scope, LF3 establishes the following:

$$\widehat{M}_{st}^{(a,b)}$$

forms a complete orthogonal pointer-sector projective readout;

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle$$

is the corresponding operator-form branch weight;

in the strong-readout limit,

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4};$$

the correlation is

$$E(a, b) = -a \cdot b;$$

the local marginals are

$$P_A(s | a, b) = P_B(t | a, b) = \frac{1}{2};$$

and the finite-leakage corrections obey explicit bounds of order

$$\mathcal{O}(\epsilon_A + \epsilon_B).$$

This is the intended LF3 result. It is narrow, finite-dimensional, and constructive.

The finite-leakage result should therefore be read conditionally: given pointer-readout compatibility bounds ϵ_A, ϵ_B , LF3 verifies stability of the singlet kernel, correlation, and marginals. LF3 v1.00 does not derive those bounds from an independently specified microscopic pointer model.

11.15 Summary

LF3 should be read as a machine-verified finite-dimensional measurement layer, not as a complete foundational theory. It formalises the TN3 quantum-pointer construction for the Bell-singlet sector and connects it to the LF2 Born-weight interface.

Its non-claims are equally important. LF3 does not derive the Born rule, formalise full Σ -level basin dynamics, prove the Ashtekar-Schilling correspondence, handle general POVMs, treat mixed states, formalise sequential update, or construct global Bell-local outcome functions.

The result is therefore precise:

LF3 verifies the finite-dimensional pointer-sector realisation of the Bell-singlet measurement.

12 Conclusion

12.1 Summary of Formalised Results

LF3 formalises the finite-dimensional quantum-pointer construction for the two-qubit Bell-singlet measurement. Its role is to connect the LF2 Born-weight interface to an explicit system-apparatus readout model. The formalisation is deliberately narrow: it treats a pure singlet state, finite-dimensional pointer sectors, projective pointer readout, and controlled finite-leakage corrections.

The core system-apparatus space is

$$H_{SA}^{(N)} = \mathbb{C}^2 \otimes \mathbb{C}^2 \otimes K_A^{(d_A)} \otimes K_B^{(d_B)}, \quad N = 4d_A d_B.$$

For detector settings

$$a, b \in S^2,$$

the measurement Hamiltonian is

$$\hat{H}_{a,b} = \kappa_A(\sigma \cdot a \otimes I) \otimes \hat{p}_A + \kappa_B(I \otimes \sigma \cdot b) \otimes \hat{p}_B.$$

The two readout terms commute because they act on distinct tensor factors. Hence the measurement unitary factorises into local pointer-coupling evolutions.

The interaction maps the initial singlet-pointer state

$$|\Psi_0\rangle = |\psi^-\rangle \otimes |\phi_0^A\rangle \otimes |\phi_0^B\rangle$$

to a branch-separated final state

$$|\Psi_T\rangle = \sum_{s,t \in \{\pm 1\}} c_{st}(a,b) |s_a, t_b\rangle \otimes |\phi_s^A(T)\rangle \otimes |\phi_t^B(T)\rangle.$$

The pointer-sector projectors are

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \widehat{Q}_s^A \otimes \widehat{Q}_t^B.$$

They form a complete orthogonal projective readout:

$$\widehat{M}_{st}^{(a,b)} \widehat{M}_{s't'}^{(a,b)} = 0 \quad \text{for } (s,t) \neq (s',t'),$$

and

$$\sum_{s,t \in \{\pm 1\}} \widehat{M}_{st}^{(a,b)} = I_{SA}.$$

Applying the LF2 operator-form weight interface gives

$$P_{st}(a,b;T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

In the strong-readout limit, the pointer-sector weights reduce to the squared singlet amplitudes:

$$P_{st}(a,b) = |c_{st}(a,b)|^2.$$

The singlet calculation gives

$$|c_{st}(a,b)|^2 = \frac{1 - st a \cdot b}{4}.$$

Therefore

$$P_{st}(a,b) = \frac{1 - st a \cdot b}{4}.$$

From this kernel, LF3 derives the Bell-singlet correlation

$$E(a,b) = -a \cdot b,$$

and the no-signalling marginals

$$P_A(s | a,b) = \frac{1}{2}, \quad P_B(t | a,b) = \frac{1}{2}.$$

LF3 also records finite-leakage stability. If ϵ_A and ϵ_B quantify imperfect pointer separation, then

$$P_{st}(a,b;T) = \frac{1 - st a \cdot b}{4} + \mathcal{O}(\epsilon_A + \epsilon_B),$$

$$E_T(a, b) = -a \cdot b + \mathcal{O}(\epsilon_A + \epsilon_B),$$

and

$$P_A(s \mid a, b; T) = \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B),$$

$$P_B(t \mid a, b; T) = \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B).$$

At the Lean level, these statements are represented by explicit inequalities rather than asymptotic notation.

12.2 Contribution to the CSD Formalisation Programme

LF3 is the third formalisation layer in the current CSD sequence. LF1 formalises repeated-trial frequency convergence. LF2 formalises the measure bridge and Born-weight wrapper. LF3 formalises the first explicit measurement-control layer.

Its contribution is therefore not another derivation of probability. Its contribution is the machine-checkable construction of a finite-dimensional pointer measurement whose branch weights reproduce the Bell-singlet statistics. This moves the Bell-sector analysis from a paper-level constructive model into the Lean formalisation branch.

In programme terms, LF3 supplies the formal bridge:

LF2 Born-weight interface $\longrightarrow$ TN3 pointer-sector measurement model $\longrightarrow$ Bell-singlet correlation and no-signalling margin

This bridge matters because it verifies that the upstream probability machinery can be applied to a concrete system-apparatus construction, not merely to abstract projectors.

12.3 Bell-Consistency Boundary

LF3 reproduces Bell-singlet correlations, but it does not claim Bell locality. The outcome maps are context-indexed:

$$F^{(a,b)} : \mathcal{D}_{a,b} \rightarrow \{\pm 1\} \times \{\pm 1\}.$$

They are not global assignments of the form

$$A(a, \lambda), \quad B(b, \lambda)$$

defined simultaneously for all incompatible settings. Nor does LF3 construct a single CHSH assignment

$$G : \Lambda \rightarrow \{\pm 1\}^4.$$

This distinction is essential. The construction is deterministic per realised measurement context, but it does not satisfy the Bell-local factorisation premise. Thus the formalised pointer-sector model is compatible with the Fine-Bell boundary: it gives definite outcomes for performed measurements without implying a global joint distribution for all unperformed incompatible measurements.

12.4 Next Formalisation Targets

The immediate next formalisation target is LF4. Its role should be to formalise the mixed-state, POVM, and subsystem-reduction layer associated with TN4. LF4 should generalise the projective pure-state readout used in LF3 to the density-operator and effect-operator form

$$P(k \mid \rho) = \text{Tr}(\rho E_k).$$

This would extend the formalisation branch from projective singlet readout to the broader finite-dimensional operational dictionary.

A later target is LF5, corresponding to the TN6 sequential-update layer. LF5 should use the outcome-sector projectors formalised in LF3 as inputs for outcome-conditioned update and sequential experiments. In particular, it should formalise ideal projective updates of the schematic form

$$\rho \mapsto \frac{\widehat{M}_{st} \rho \widehat{M}_{st}}{\text{Tr}(\rho \widehat{M}_{st})}.$$

Thus LF3 prepares the formal ground for two later extensions:

LF4 : mixed states, POVMs, and subsystem reduction,

and

LF5 : post-measurement update and sequential experiments.

12.5 Final Statement

LF3 establishes a precise finite-dimensional result. Given the LF2 operator-form weight interface and the TN3 quantum-pointer construction, the Bell-singlet measurement admits a Lean-formalisable projective readout model with:

- complete orthogonal pointer-sector projectors;
- branch-separated final states;
- controlled finite-leakage bounds;
- singlet branch weights

$$P_{st}(a, b) = \frac{1 - st\, a \cdot b}{4};$$

- Bell-singlet correlation

$$E(a, b) = -a \cdot b;$$

- no-signalling marginals

$$P_A = P_B = \frac{1}{2};$$

- context-indexed outcome maps rather than global CHSH assignments.

Accordingly, LF3 closes the formal measurement-control layer for the finite-dimensional Bell-singlet sector. It does not complete the full CSD programme, but it gives a sharply bounded and machine-checkable result: the constructive quantum-pointer realisation of the Bell-singlet correlation can be represented as a finite-dimensional theorem package with explicit assumptions, explicit projectors, explicit weights, and explicit leakage control.

A Lean Theorem Map

A.1 Purpose of the Theorem Map

This appendix records the principal Lean theorem targets for LF3. The purpose is to give an implementation-facing map of the formalisation without requiring the main text to carry all repository-level details.

The theorem map follows the LF3 module structure:

Setup → Hamiltonian → BranchSeparation → Projectors → Singlet → ContextMap → Interface.

Each group below identifies the intended mathematical content of the corresponding Lean file. The declarations are schematic. They are not final Lean syntax. Their role is to define the target statements that the implementation should realise.

A.2 Setup Theorems

The setup layer defines the finite-dimensional system-apparatus data. It includes detector settings, Pauli-direction operators, spin projectors, pointer spaces, and pointer readout sectors.

The first group of theorem targets concerns detector settings:

```
theorem detectorSetting_unit :
  norm a.vec = 1
```

The second group concerns Pauli-direction operators:

```
theorem sigmaDot_selfAdjoint :
  IsSelfAdjoint (sigmaDot a)
```

```
theorem sigmaDot_square :
  sigmaDot a * sigmaDot a = 1
```

The identity

$$(\sigma \cdot a)^2 = I$$

uses the unit-norm condition

$$\|a\| = 1.$$

It implies that the eigenvalues of $\sigma \cdot a$ are ± 1 .

The third group concerns spin projectors:

```
theorem spin_projector_idempotent :
  Pi_spin s a * Pi_spin s a = Pi_spin s a

theorem spin_projector_selfAdjoint :
  IsSelfAdjoint (Pi_spin s a)

theorem spin_projectors_orthogonal :
  s != s' ->
  Pi_spin s a * Pi_spin s' a = 0

theorem spin_projectors_complete :
  Pi_spin Sign.plus a + Pi_spin Sign.minus a = 1
```

These correspond to the mathematical statements

$$\begin{aligned}\Pi_s(a)^2 &= \Pi_s(a), \\ \Pi_s(a)^\dagger &= \Pi_s(a), \\ \Pi_s(a)\Pi_{s'}(a) &= 0 \quad \text{for } s \neq s',\end{aligned}$$

and

$$\Pi_+(a) + \Pi_-(a) = I.$$

The fourth group concerns pointer readout sectors:

```
theorem pointer_A_projector_idempotent :
  Q_A s * Q_A s = Q_A s

theorem pointer_B_projector_idempotent :
  Q_B t * Q_B t = Q_B t

theorem pointer_A_projectors_orthogonal :
  s != s' ->
  Q_A s * Q_A s' = 0

theorem pointer_B_projectors_orthogonal :
  t != t' ->
  Q_B t * Q_B t' = 0

theorem pointer_A_complete :
  Q_A Sign.plus + Q_A Sign.minus = 1

theorem pointer_B_complete :
  Q_B Sign.plus + Q_B Sign.minus = 1
```

These results are the algebraic inputs for the pointer-sector projectors in `Projectors.lean`.

A.3 Hamiltonian Theorems

The Hamiltonian layer defines the local readout Hamiltonians

$$\hat{H}_A(a), \quad \hat{H}_B(b),$$

and the total readout Hamiltonian

$$\hat{H}_{a,b} = \hat{H}_A(a) + \hat{H}_B(b).$$

The first theorem target is self-adjointness:

```

theorem H_A_selfAdjoint :
  IsSelfAdjoint (H_A S kappa_A a p_A)

theorem H_B_selfAdjoint :
  IsSelfAdjoint (H_B S kappa_B b p_B)

theorem H_total_selfAdjoint :
  IsSelfAdjoint
    (H_total S kappa_A kappa_B a b p_A p_B)

```

The second theorem target is commutation:

```

theorem H_AB_commute :
  H_A S kappa_A a p_A * H_B S kappa_B b p_B =
  H_B S kappa_B b p_B * H_A S kappa_A a p_A

```

This corresponds to

$$[\hat{H}_A(a), \hat{H}_B(b)] = 0.$$

The third theorem target is factorisation of the measurement unitary:

```

theorem U_measure_factorises :
  U_measure S kappa_A kappa_B a b p_A p_B T =
  U_A S kappa_A a p_A T *
  U_B S kappa_B b p_B T

```

If operator exponentials are not implemented in the first pass, this theorem may be represented as a field in an abstract measurement-unitary structure:

```

structure MeasurementUnitary where
  U_measure : H_SA S ->L[Complex] H_SA U
  U_A : H_SA S ->L[Complex] H_SA S
  U_B : H_SA S ->L[Complex] H_SA S
  factorises : U_measure = U_A * U_B

```

The fourth theorem target is the action on joint eigenbranches:

```

theorem U_measure_on_joint_eigenstate :
  U_measure S kappa_A kappa_B a b p_A p_B T
  (ket_s_a_t_b tensor phi_A0 tensor phi_B0)
=
  ket_s_a_t_b
  tensor translatedPointerA s kappa_A T p_A phi_A0
  tensor translatedPointerB t kappa_B T p_B phi_B0

```

This theorem supplies the transition from Hamiltonian control to branch separation.

A.4 Branch-Separation Theorems

The branch-separation layer defines the branch states

$$|B_{st}(a, b; T)\rangle = |s_a, t_b\rangle \otimes |\phi_s^A(T)\rangle \otimes |\phi_t^B(T)\rangle,$$

and the final state

$$|\Psi_T\rangle = \sum_{s,t \in \{\pm 1\}} c_{st}(a, b) |B_{st}(a, b; T)\rangle.$$

The first theorem target is the branch decomposition:

```

theorem finalState_branch_decomposition :
  finalState S a b T =
  sum over s,t of
    (c_st a b s t) • branchState S a b T s t

```

The second group records the pointer-overlap definitions:

```
def pointerOverlapA
  (s u : Sign) : Real :=
  realPart
    (inner (phi_A u T)
      (Q_A s (phi_A u T)))

def pointerOverlapB
  (t v : Sign) : Real :=
  realPart
    (inner (phi_B v T)
      (Q_B t (phi_B v T)))
```

At the mathematical level, these are

$$\alpha_A(s \mid u) = \langle \phi_u^A(T) | \hat{Q}_s^A | \phi_u^A(T) \rangle,$$

and

$$\alpha_B(t \mid v) = \langle \phi_v^B(T) | \hat{Q}_t^B | \phi_v^B(T) \rangle.$$

The leakage assumptions are packaged as:

```
structure PointerLeakageBounds where
  epsilon_A : Real
  epsilon_B : Real
  nonneg_A : 0 <= epsilon_A
  nonneg_B : 0 <= epsilon_B
  A_wrong :
    forall s, pointerOverlapA (-s) s <= epsilon_A
  B_wrong :
    forall t, pointerOverlapB (-t) t <= epsilon_B
  A_right :
    forall s, 1 - epsilon_A <= pointerOverlapA s s
  B_right :
    forall t, 1 - epsilon_B <= pointerOverlapB t t
```

This corresponds to

$$\alpha_A(-s \mid s) \leq \epsilon_A, \quad \alpha_B(-t \mid t) \leq \epsilon_B,$$

and

$$\alpha_A(s \mid s) \geq 1 - \epsilon_A, \quad \alpha_B(t \mid t) \geq 1 - \epsilon_B.$$

The main branch-separation estimate is:

```
theorem branch_separation_leakage_bound :
  wrongPointerReadoutMass S a b T
  <= epsilon_A + epsilon_B
```

This theorem records the union-bound estimate that the probability of at least one pointer being found in the wrong readout sector is bounded by

$$\epsilon_A + \epsilon_B.$$

A.5 Projector Theorems

The projector layer defines the four pointer-sector projectors

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \hat{Q}_s^A \otimes \hat{Q}_t^B.$$

The schematic Lean definition is:

```
def M_hat
  (S : SystemApparatusSetup)
  (s t : Sign) :
  H_SA S ->L[Complex] H_SA S
```

The first theorem target is idempotence:

```
theorem M_idempotent :
  M_hat S s t * M_hat S s t = M_hat S s t
```

The second theorem target is self-adjointness:

```
theorem M_selfAdjoint :
  IsSelfAdjoint (M_hat S s t)
```

The third theorem target is orthogonality:

```
theorem M_orthogonal :
  (s,t) != (s',t') ->
  M_hat S s t * M_hat S s' t' = 0
```

The fourth theorem target is completeness:

```
theorem M_complete :
  sum over s,t of M_hat S s t = 1
```

Together, these establish that

$$\left\{ \widehat{M}_{st}^{(a,b)} : s, t \in \{\pm 1\} \right\}$$

is a complete orthogonal projective readout on $H_{SA}^{(N)}$.

The branch weight is defined by:

```
def branchWeight
  (S : SystemApparatusSetup)
  (Psi_T : H_SA S)
  (s t : Sign) : Real :=
  realPart (inner Psi_T (M_hat S s t Psi_T))
```

The strong-readout theorem target is:

```
theorem branchWeight_strong_readout :
  epsilon_A = 0 ->
  epsilon_B = 0 ->
  branchWeight S Psi_T s t =
    normSq (c_st a b s t)
```

The finite-leakage theorem target is:

```
theorem branchWeight_finite_leakage :
  abs
    (branchWeight S Psi_T s t
     - normSq (c_st a b s t))
  <= epsilon_A + epsilon_B + epsilon_A * epsilon_B
```

At the mathematical level, this states

$$\left| P_{st}(a, b; T) - |c_{st}(a, b)|^2 \right| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

The LF2 interface theorem is:

```
theorem branchWeight_eq_LF2_Born :
  branchWeight S Psi_T s t =
  LF2.bornWeight
    (rankOneState Psi_T)
    (M_hat S s t)
```

This theorem records that the pointer-sector branch weight is the LF2 operator-form Born weight applied to the concrete pointer-sector projector.

A.6 Singlet Theorems

The singlet layer evaluates the coefficients

$$c_{st}(a, b) = \langle s_a, t_b \mid \psi^- \rangle.$$

The first theorem target is normalisation:

```
theorem singlet_norm :
  normSq singlet = 1
```

The second group gives the one-sided expectation identities:

```
theorem singlet_left_pauli_expectation_zero :
  expectation singlet (sigmaDot a tensor I) = 0
```

```
theorem singlet_right_pauli_expectation_zero :
  expectation singlet (I tensor sigmaDot b) = 0
```

These correspond to

$$\langle \psi^- \mid \sigma_a \otimes I \mid \psi^- \rangle = 0,$$

and

$$\langle \psi^- \mid I \otimes \sigma_b \mid \psi^- \rangle = 0.$$

The third theorem target is the two-sided Pauli correlation:

```
theorem singlet_pauli_correlation :
  expectation singlet (sigmaDot a tensor sigmaDot b) =
    - dot a.vec b.vec
```

This corresponds to

$$\langle \psi^- \mid \sigma_a \otimes \sigma_b \mid \psi^- \rangle = -a \cdot b.$$

The fourth theorem target expands the joint spin projector:

```
theorem joint_spin_projector_expand :
  Pi_st a b s t =
    (1 / 4) •
      (I tensor I
       + sign s • (sigmaDot a tensor I)
       + sign t • (I tensor sigmaDot b)
       + sign s * sign t •
         (sigmaDot a tensor sigmaDot b))
```

At the mathematical level,

$$\Pi_{st}(a, b) = \frac{1}{4} (I \otimes I + s \sigma_a \otimes I + t I \otimes \sigma_b + st \sigma_a \otimes \sigma_b).$$

The central singlet theorem is:

```
theorem cst_squared_eq :
  normSq (c_st a b s t) =
    (1 - sign s * sign t * dot a.vec b.vec) / 4
```

This corresponds to

$$|c_{st}(a, b)|^2 = \frac{1 - st a \cdot b}{4}.$$

The strong-readout pointer probability theorem is:

```
theorem singlet_pointer_probability_strong_readout :
  P_st a b s t =
    (1 - sign s * sign t * dot a.vec b.vec) / 4
```

The finite-leakage probability theorem is:

```
theorem singlet_pointer_probability_finite_leakage :
  abs
    (P_st_T a b T s t
     - (1 - sign s * sign t * dot a.vec b.vec) / 4)
  <= epsilon_A + epsilon_B + epsilon_A * epsilon_B
```

These theorems complete the singlet probability-kernel layer.

A.7 Correlation and Marginal Theorems

The correlation layer derives the Bell-singlet correlation and the no-signalling marginals from the singlet probability kernel

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

The first theorem target is the correlation identity:

```
theorem correlation_eq_neg_dot :
  correlation a b =
    - dot a.vec b.vec
```

At the mathematical level, this is

$$E(a, b) = \sum_{s, t \in \{\pm 1\}} st P_{st}(a, b) = -a \cdot b.$$

The second theorem target is the finite-leakage correlator bound:

```
theorem correlation_finite_leakage_bound :
  abs (correlation_T a b T + dot a.vec b.vec)
  <= 4 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)
```

At the mathematical level,

$$|E_T(a, b) + a \cdot b| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

The third theorem target is the A -marginal identity:

```
theorem marginal_A_eq_half :
  marginal_A a b s = 1 / 2
```

The fourth theorem target is the B -marginal identity:

```
theorem marginal_B_eq_half :
  marginal_B a b t = 1 / 2
```

These correspond to

$$P_A(s \mid a, b) = \sum_{t \in \{\pm 1\}} P_{st}(a, b) = \frac{1}{2},$$

and

$$P_B(t \mid a, b) = \sum_{s \in \{\pm 1\}} P_{st}(a, b) = \frac{1}{2}.$$

The finite-leakage marginal theorem targets are:

```
theorem marginal_A_finite_leakage_bound :
  abs (marginal_A_T a b T s - 1 / 2)
  <= 2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)
```

```
theorem marginal_B_finite_leakage_bound :
  abs (marginal_B_T a b T t - 1 / 2)
  <= 2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)
```


At the mathematical level,

$$\left| P_A(s \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

and

$$\left| P_B(t \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

The strong-readout no-signalling theorem targets are:

```
theorem no_signalling_strong_readout_A :
  marginal_A a b s = marginal_A a b' s
```

```
theorem no_signalling_strong_readout_B :
  marginal_B a b t = marginal_B a' b t
```

These record that the local marginals do not depend on the remote detector setting.

A.8 Context-Map Theorems

The context-map layer records the distinction between context-indexed readout maps and global CHSH assignments.

A measurement context is represented by:

```
structure MeasurementContext where
  a : DetectorSetting
  b : DetectorSetting
```

For each context, the pointer-sector readout is:

$$\left\{ \widehat{M}_{st}^{(a,b)} : s, t \in \{\pm 1\} \right\}.$$

The first theorem target is that every context has a projective readout:

```
theorem context_has_projective_readout :
  IsProjectiveDecomposition
  (fun st => M_hat ctx st)
```

At the mathematical level, this packages orthogonality and completeness:

$$\widehat{M}_{st}^{(a,b)} \widehat{M}_{s't'}^{(a,b)} = 0 \quad \text{for } (s, t) \neq (s', t'),$$

and

$$\sum_{s, t \in \{\pm 1\}} \widehat{M}_{st}^{(a,b)} = I_{SA}.$$

The second theorem target is normalisation of the branch weights:

```
theorem context_branch_weights_sum_one :
  sum over s, t of P_st ctx s t = 1
```

At the mathematical level,

$$\sum_{s, t \in \{\pm 1\}} P_{st}(a, b) = 1.$$

The third theorem target is the context-indexed singlet kernel:

```
theorem context_singlet_kernel :
  P_st ctx s t =
    (1 - sign s * sign t * dot ctx.a.vec ctx.b.vec) / 4
```

The fourth theorem target is the context-indexed correlation:

```
theorem context_correlation_eq_neg_dot :
  correlation ctx =
    - dot ctx.a.vec ctx.b.vec
```

The fifth theorem target is operational marginal stability:

```
theorem context_no_signalling_A :
  marginal_A ctx s = 1 / 2

theorem context_no_signalling_B :
  marginal_B ctx t = 1 / 2
```

The file may also define a context-indexed outcome-map structure:

```
structure ContextIndexedOutcomeMaps where
  F :
    (ctx : MeasurementContext) ->
      ContextDomain ctx ->
        Sign × Sign
```

A global CHSH assignment should be represented separately:

```
structure GlobalCHSHAssignment where
  A1 : HiddenState -> Sign
  A2 : HiddenState -> Sign
  B1 : HiddenState -> Sign
  B2 : HiddenState -> Sign
```

There is no separate theorem target asserting that context-indexed maps do not automatically define a global CHSH assignment. The point is instead recorded as a structural distinction between two definitions.

The context-indexed structure is:

```
structure ContextIndexedOutcomeMaps where
  F :
    (ctx : MeasurementContext) ->
      ContextDomain ctx ->
        Sign × Sign
```

A global CHSH assignment, if represented, is a different structure:

```
structure GlobalCHSHAssignment where
  A1 : HiddenState -> Sign
  A2 : HiddenState -> Sign
  B1 : HiddenState -> Sign
  B2 : HiddenState -> Sign
```

This type-level distinction has no independent Bell-theorem content. If a Fine-Bell obstruction is needed, it should be imported or stated as an explicit axiom boundary:

```
axiom fine_bell_no_global_assignment :
  BellViolatingSingletCorrelations settings ->
    Not (ExistsGlobalCHSHAssignment settings)
```

A.9 Interface Theorems

The interface layer gathers the theorem package exported by LF3. The main theorem should expose the strong-readout result:

```
theorem LF3_main_theorem :
  (forall s t,
    P_st ctx s t =
```

```

        (1 - sign s * sign t * dot ctx.a.vec ctx.b.vec) / 4)
    and
    correlation ctx =
        - dot ctx.a.vec ctx.b.vec
    and
    (forall s, marginal_A ctx s = 1 / 2)
    and
    (forall t, marginal_B ctx t = 1 / 2)

```

At the mathematical level, this exports:

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

$$E(a, b) = -a \cdot b,$$

and

$$P_A(s \mid a, b) = P_B(t \mid a, b) = \frac{1}{2}.$$

The finite-leakage theorem should be exported separately:

```

theorem LF3_finite_leakage_theorem :
  (forall s t,
    abs
      (P_st_T ctx T s t
        - (1 - sign s * sign t * dot ctx.a.vec ctx.b.vec) / 4)
    <= epsilon_A + epsilon_B + epsilon_A * epsilon_B)
  and
  abs (correlation_T ctx T + dot ctx.a.vec ctx.b.vec)
    <= 4 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)
  and
  (forall s,
    abs (marginal_A_T ctx T s - 1 / 2)
    <= 2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B))
  and
  (forall t,
    abs (marginal_B_T ctx T t - 1 / 2)
    <= 2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B))

```

At the mathematical level, this exports:

$$\left| P_{st}(a, b; T) - \frac{1 - st a \cdot b}{4} \right| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B,$$

$$|E_T(a, b) + a \cdot b| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

$$\left| P_A(s \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

and

$$\left| P_B(t \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

A.10 Summary

The theorem map identifies the intended formal deliverables of LF3. The core theorem chain is:

$$\begin{aligned}
 & \widehat{M}_{st}^{(a,b)} \quad \text{complete and orthogonal} \\
 \implies & P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle \\
 \implies & P_{st}(a, b) = \frac{1 - st a \cdot b}{4} \\
 \implies & E(a, b) = -a \cdot b
 \end{aligned}$$

$$\implies P_A(s \mid a, b) = P_B(t \mid a, b) = \frac{1}{2}.$$

The finite-leakage theorem chain adds explicit stability bounds controlled by

$$\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

This appendix should be treated as the implementation checklist for LF3.

B Finite-Leakage Estimate Details

B.1 Purpose of this Appendix

This appendix records the finite-leakage estimates used in LF3. The purpose is to separate the ideal strong-readout calculation from the finite-separation error analysis.

In the ideal strong-readout limit, each translated pointer state lies entirely inside its intended readout sector. In that case,

$$P_{st}(a, b) = |c_{st}(a, b)|^2.$$

At finite pointer separation, each translated pointer state may have a small component in the wrong readout sector. LF3 records this by introducing leakage parameters

$$\epsilon_A, \quad \epsilon_B.$$

The goal is to show that finite pointer leakage perturbs the branch weights, correlation, and marginals by controlled amounts.

The estimates in this appendix use only:

- orthogonality and completeness of the pointer readout projectors;
- the branch-separated final state;
- non-negativity of pointer-sector overlaps;
- the leakage bounds defining ϵ_A and ϵ_B ;
- finite summation over $s, t \in \{\pm 1\}$.

No continuum pointer analysis is required.

B.2 Pointer Leakage Definitions

Let

$$\hat{Q}_+^A, \quad \hat{Q}_-^A$$

be the two orthogonal readout projectors for the A -pointer, and let

$$\hat{Q}_+^B, \quad \hat{Q}_-^B$$

be the corresponding readout projectors for the B -pointer.

For a translated A -pointer state

$$|\phi_u^A(T)\rangle, \quad u \in \{\pm 1\},$$

define

$$\alpha_A(s \mid u) = \langle \phi_u^A(T) | \hat{Q}_s^A | \phi_u^A(T) \rangle.$$

For a translated B -pointer state

$$|\phi_v^B(T)\rangle, \quad v \in \{\pm 1\},$$

define

$$\alpha_B(t \mid v) = \langle \phi_v^B(T) | \hat{Q}_t^B | \phi_v^B(T) \rangle.$$

Since the readout projectors are positive and complete,

$$0 \leq \alpha_A(s \mid u) \leq 1, \quad 0 \leq \alpha_B(t \mid v) \leq 1,$$

and

$$\alpha_A(+ \mid u) + \alpha_A(- \mid u) = 1,$$

$$\alpha_B(+ | v) + \alpha_B(- | v) = 1.$$

The leakage parameters are defined by the wrong-sector overlaps:

$$\epsilon_A = \max_{u \in \{\pm 1\}} \alpha_A(-u | u),$$

and

$$\epsilon_B = \max_{v \in \{\pm 1\}} \alpha_B(-v | v).$$

Thus

$$\alpha_A(-u | u) \leq \epsilon_A, \quad \alpha_B(-v | v) \leq \epsilon_B.$$

By completeness,

$$\alpha_A(u | u) = 1 - \alpha_A(-u | u) \geq 1 - \epsilon_A,$$

and

$$\alpha_B(v | v) = 1 - \alpha_B(-v | v) \geq 1 - \epsilon_B.$$

The ideal strong-readout limit is

$$\epsilon_A, \epsilon_B \rightarrow 0.$$

In that limit,

$$\alpha_A(s | u) \rightarrow \delta_{su}, \quad \alpha_B(t | v) \rightarrow \delta_{tv}.$$

B.3 Finite-Readout Branch Weights

The branch-separated final state is

$$|\Psi_T\rangle = \sum_{u,v \in \{\pm 1\}} c_{uv}(a, b) |u_a, v_b\rangle \otimes |\phi_u^A(T)\rangle \otimes |\phi_v^B(T)\rangle.$$

The pointer-sector projector for readout (s, t) is

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \widehat{Q}_s^A \otimes \widehat{Q}_t^B.$$

The corresponding finite-readout branch weight is

$$P_{st}(a, b; T) = \langle \Psi_T | \widehat{M}_{st}^{(a,b)} | \Psi_T \rangle.$$

Using orthogonality of the system measurement basis,

$$\langle u_a, v_b | u'_a, v'_b \rangle = \delta_{uu'} \delta_{vv'},$$

the cross terms vanish. Therefore

$$P_{st}(a, b; T) = \sum_{u,v \in \{\pm 1\}} |c_{uv}(a, b)|^2 \alpha_A(s | u) \alpha_B(t | v).$$

Define

$$p_{uv} = |c_{uv}(a, b)|^2.$$

Then

$$p_{uv} \geq 0, \quad \sum_{u,v \in \{\pm 1\}} p_{uv} = 1.$$

The finite-readout branch weight becomes

$$P_{st}(a, b; T) = \sum_{u,v} p_{uv} \alpha_A(s | u) \alpha_B(t | v).$$

The ideal strong-readout branch weight is

$$P_{st}^{\text{ideal}}(a, b) = p_{st} = |c_{st}(a, b)|^2.$$

The finite-leakage deviation is

$$\Delta_{st}(a, b; T) = P_{st}(a, b; T) - p_{st}.$$

B.4 Branch-Weight Error Bound

We now bound

$$|P_{st}(a, b; T) - p_{st}|.$$

Write

$$\begin{aligned} P_{st}(a, b; T) &= p_{st}\alpha_A(s | s)\alpha_B(t | t) + p_{s,-t}\alpha_A(s | s)\alpha_B(t | -t) \\ &\quad + p_{-s,t}\alpha_A(s | -s)\alpha_B(t | t) + p_{-s,-t}\alpha_A(s | -s)\alpha_B(t | -t). \end{aligned}$$

The intended branch (s, t) contributes

$$p_{st}\alpha_A(s | s)\alpha_B(t | t).$$

Its deviation from the ideal contribution p_{st} is

$$p_{st} [1 - \alpha_A(s | s)\alpha_B(t | t)].$$

Since

$$\alpha_A(s | s) \geq 1 - \epsilon_A, \quad \alpha_B(t | t) \geq 1 - \epsilon_B,$$

we have

$$\alpha_A(s | s)\alpha_B(t | t) \geq (1 - \epsilon_A)(1 - \epsilon_B).$$

Therefore

$$1 - \alpha_A(s | s)\alpha_B(t | t) \leq \epsilon_A + \epsilon_B - \epsilon_A\epsilon_B.$$

The intended-branch deficit is bounded by

$$p_{st} (\epsilon_A + \epsilon_B - \epsilon_A\epsilon_B).$$

There are three wrong-branch contributions. For the branch $(s, -t)$, the A -pointer is correctly aligned but the B -pointer is in the wrong readout sector:

$$p_{s,-t}\alpha_A(s | s)\alpha_B(t | -t) \leq p_{s,-t}\epsilon_B.$$

For the branch $(-s, t)$, the B -pointer is correctly aligned but the A -pointer is in the wrong readout sector:

$$p_{-s,t}\alpha_A(s | -s)\alpha_B(t | t) \leq p_{-s,t}\epsilon_A.$$

For the branch $(-s, -t)$, both pointer factors are in wrong readout sectors:

$$p_{-s,-t}\alpha_A(s | -s)\alpha_B(t | -t) \leq p_{-s,-t}\epsilon_A\epsilon_B.$$

Combining the intended and wrong-branch estimates gives

$$|P_{st}(a, b; T) - p_{st}| \leq p_{st} (\epsilon_A + \epsilon_B - \epsilon_A\epsilon_B) + p_{s,-t}\epsilon_B + p_{-s,t}\epsilon_A + p_{-s,-t}\epsilon_A\epsilon_B.$$

Using

$$p_{st} + p_{s,-t} + p_{-s,t} + p_{-s,-t} = 1,$$

and non-negativity of all p -terms, this implies the uniform bound

$$|P_{st}(a, b; T) - p_{st}| \leq \epsilon_A + \epsilon_B + \epsilon_A\epsilon_B.$$

Since

$$p_{st} = |c_{st}(a, b)|^2,$$

we obtain

$$|P_{st}(a, b; T) - |c_{st}(a, b)|^2| \leq \epsilon_A + \epsilon_B + \epsilon_A\epsilon_B.$$

For the singlet kernel,

$$|c_{st}(a, b)|^2 = \frac{1 - st a \cdot b}{4}.$$

Therefore

$$\left| P_{st}(a, b; T) - \frac{1 - st a \cdot b}{4} \right| \leq \epsilon_A + \epsilon_B + \epsilon_A\epsilon_B.$$

Equivalently,

$$P_{st}(a, b; T) = \frac{1 - st a \cdot b}{4} + \mathcal{O}(\epsilon_A + \epsilon_B).$$

B.5 Correlator Error Bound

The ideal correlation is

$$E(a, b) = \sum_{s, t \in \{\pm 1\}} st P_{st}(a, b).$$

The finite-readout correlation is

$$E_T(a, b) = \sum_{s, t \in \{\pm 1\}} st P_{st}(a, b; T).$$

Let

$$\Delta_{st}(a, b; T) = P_{st}(a, b; T) - P_{st}(a, b).$$

Then

$$E_T(a, b) - E(a, b) = \sum_{s, t \in \{\pm 1\}} st \Delta_{st}(a, b; T).$$

Taking absolute values,

$$|E_T(a, b) - E(a, b)| \leq \sum_{s, t \in \{\pm 1\}} |\Delta_{st}(a, b; T)|.$$

Using the branch-weight bound,

$$|\Delta_{st}(a, b; T)| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B,$$

we obtain

$$|E_T(a, b) - E(a, b)| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

Since

$$E(a, b) = -a \cdot b,$$

this gives

$$\boxed{|E_T(a, b) + a \cdot b| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).}$$

Equivalently,

$$E_T(a, b) = -a \cdot b + \mathcal{O}(\epsilon_A + \epsilon_B).$$

B.6 Marginal Error Bounds

The ideal marginal at detector A is

$$P_A(s | a, b) = \sum_{t \in \{\pm 1\}} P_{st}(a, b) = \frac{1}{2}.$$

The finite-readout marginal is

$$P_A(s | a, b; T) = \sum_{t \in \{\pm 1\}} P_{st}(a, b; T).$$

Hence

$$P_A(s | a, b; T) - \frac{1}{2} = \sum_{t \in \{\pm 1\}} \Delta_{st}(a, b; T).$$

Taking absolute values,

$$\left| P_A(s | a, b; T) - \frac{1}{2} \right| \leq \sum_{t \in \{\pm 1\}} |\Delta_{st}(a, b; T)|.$$

Using the uniform branch-weight bound gives

$$\boxed{\left| P_A(s | a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).}$$

Similarly, the ideal marginal at detector B is

$$P_B(t | a, b) = \sum_{s \in \{\pm 1\}} P_{st}(a, b) = \frac{1}{2}.$$

The finite-readout marginal is

$$P_B(t \mid a, b; T) = \sum_{s \in \{\pm 1\}} P_{st}(a, b; T).$$

Therefore

$$P_B(t \mid a, b; T) - \frac{1}{2} = \sum_{s \in \{\pm 1\}} \Delta_{st}(a, b; T).$$

Taking absolute values gives

$$\left| P_B(t \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

Thus

$$P_A(s \mid a, b; T) = \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B),$$

and

$$P_B(t \mid a, b; T) = \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B).$$

B.7 Finite-Leakage No-Signalling Stability

The finite-leakage marginal estimates imply approximate operational no-signalling. For detector A , compare two remote settings b and b' . We have

$$P_A(s \mid a, b; T) = \frac{1}{2} + \delta_A(s \mid a, b; T),$$

and

$$P_A(s \mid a, b'; T) = \frac{1}{2} + \delta_A(s \mid a, b'; T),$$

where

$$|\delta_A(s \mid a, b; T)| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

and

$$|\delta_A(s \mid a, b'; T)| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

Therefore

$$|P_A(s \mid a, b; T) - P_A(s \mid a, b'; T)| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

Similarly, for detector B ,

$$|P_B(t \mid a, b; T) - P_B(t \mid a', b; T)| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

Thus finite leakage can produce small apparent marginal deviations, but these deviations are controlled by the same pointer-separation error parameters. In the strong-readout limit,

$$\epsilon_A, \epsilon_B \rightarrow 0,$$

the marginal differences vanish:

$$P_A(s \mid a, b; T) - P_A(s \mid a, b'; T) \rightarrow 0,$$

and

$$P_B(t \mid a, b; T) - P_B(t \mid a', b; T) \rightarrow 0.$$

The ideal pointer-sector model therefore satisfies exact operational no-signalling, while the finite-readout model satisfies no-signalling stability up to controlled apparatus-readout error.

B.8 Strong-Readout Limit

The strong-readout limit is defined by

$$\epsilon_A, \epsilon_B \rightarrow 0.$$

In this limit, the pointer-overlap functions converge to Kronecker deltas:

$$\alpha_A(s \mid u) \rightarrow \delta_{su}, \quad \alpha_B(t \mid v) \rightarrow \delta_{tv}.$$

Therefore

$$P_{st}(a, b; T) = \sum_{u, v} p_{uv} \alpha_A(s \mid u) \alpha_B(t \mid v) \rightarrow p_{st}.$$

Since

$$p_{st} = |c_{st}(a, b)|^2,$$

and

$$|c_{st}(a, b)|^2 = \frac{1 - st a \cdot b}{4},$$

we recover

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

The finite-readout correlation converges to

$$E(a, b) = -a \cdot b.$$

The finite-readout marginals converge to

$$P_A(s \mid a, b) = \frac{1}{2},$$

and

$$P_B(t \mid a, b) = \frac{1}{2}.$$

Thus the strong-readout limit recovers the exact Bell-singlet probability kernel, correlation, and no-signalling marginals.

B.9 Lean Theorem Targets for Finite Leakage

The finite-leakage estimates should be represented in Lean using explicit inequalities rather than asymptotic notation.

The branch-weight estimate is:

```
theorem branchWeight_leakage_bound :
  abs
    (P_st_T a b T s t
     - (1 - sign s * sign t * dot a.vec b.vec) / 4)
  <=
    epsilon_A + epsilon_B + epsilon_A * epsilon_B
```

The correlation estimate is:

```
theorem correlation_leakage_bound :
  abs
    (E_T a b T + dot a.vec b.vec)
  <=
    4 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)
```

The A -marginal estimate is:

```
theorem marginal_A_leakage_bound :
  abs
    (P_A_T a b T s - 1 / 2)
  <=
    2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)
```

The B -marginal estimate is:

```

theorem marginal_B_leakage_bound :
  abs
    (P_B_T a b T t - 1 / 2)
  <=
    2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)

```

The strong-readout branch-weight theorem may be stated as:

```

theorem strong_readout_limit :
  epsilon_A = 0 ->
  epsilon_B = 0 ->
  P_st_T a b T s t =
    (1 - sign s * sign t * dot a.vec b.vec) / 4

```

Similarly, the correlation and marginal strong-readout limits are:

```

theorem correlation_strong_readout :
  epsilon_A = 0 ->
  epsilon_B = 0 ->
  E_T a b T =
    - dot a.vec b.vec

```

```

theorem marginal_A_strong_readout :
  epsilon_A = 0 ->
  epsilon_B = 0 ->
  P_A_T a b T s = 1 / 2

```

```

theorem marginal_B_strong_readout :
  epsilon_A = 0 ->
  epsilon_B = 0 ->
  P_B_T a b T t = 1 / 2

```

These theorem targets make the finite-error layer explicit and auditable.

B.10 Consistency Note for the Main Text

The main text may use the compact notation

$$\mathcal{O}(\epsilon_A + \epsilon_B)$$

when discussing the physical interpretation of finite pointer leakage. However, formal theorem statements should use explicit inequalities.

The recommended explicit branch-weight statement is

$$\left| P_{st}(a, b; T) - \frac{1 - st a \cdot b}{4} \right| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

The corresponding correlator and marginal bounds are

$$|E_T(a, b) + a \cdot b| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

$$\left| P_A(s \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

and

$$\left| P_B(t \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

This keeps LF3 aligned with the explicit TN3 finite-separation estimate while preserving the physical conclusion: finite pointer leakage perturbs the ideal singlet statistics only by controlled small errors.

B.11 Summary

This appendix supplied the finite-leakage estimates underlying LF3. Starting from the finite-readout branch-weight formula

$$P_{st}(a, b; T) = \sum_{u, v} p_{uv} \alpha_A(s \mid u) \alpha_B(t \mid v),$$

and the leakage assumptions

$$\alpha_A(-u \mid u) \leq \epsilon_A, \quad \alpha_B(-v \mid v) \leq \epsilon_B,$$

we obtained

$$|P_{st}(a, b; T) - P_{st}(a, b)| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

For the singlet kernel,

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

this gives

$$P_{st}(a, b; T) = \frac{1 - st a \cdot b}{4} + \mathcal{O}(\epsilon_A + \epsilon_B).$$

The corresponding correlation and marginal estimates are

$$E_T(a, b) = -a \cdot b + \mathcal{O}(\epsilon_A + \epsilon_B),$$

$$P_A(s \mid a, b; T) = \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B),$$

and

$$P_B(t \mid a, b; T) = \frac{1}{2} + \mathcal{O}(\epsilon_A + \epsilon_B).$$

Thus the Bell-singlet correlation and no-signalling marginals are stable under controlled finite pointer leakage and are recovered exactly in the strong-readout limit.

C Repository Implementation Order

C.1 Purpose of this Appendix

This appendix gives the recommended implementation order for LF3 in the Lean repository. The purpose is practical. The paper-level theorem structure is already defined in the main text and Appendix A. The present appendix translates that structure into a staged build plan.

The recommended order is:

`Setup.lean` $\rightarrow$ `Projectors.lean` $\rightarrow$ `Singlet.lean` $\rightarrow$ `ContextMap.lean` $\rightarrow$ `Interface.lean`

with

`Hamiltonian.lean` and `BranchSeparation.lean`

introduced either before or after the first algebraic singlet theorem, depending on implementation difficulty.

This ordering is slightly different from the conceptual order of the paper. The reason is that the fastest route to a compiling LF3 theorem package is to prove the algebraic projective-readout and singlet results first, then enrich the formalisation with the Hamiltonian and leakage layers.

C.2 Initial Compilation Target

The initial compilation target should be a minimal LF3 skeleton:

`CsdLean4/LF3/Setup.lean`,
`CsdLean4/LF3/Projectors.lean`,
`CsdLean4/LF3/Singlet.lean`,
`CsdLean4/LF3/Interface.lean`.

The first milestone is not to prove every theorem. The first milestone is to obtain a compiling repository structure with named definitions, theorem statements, and explicit sorry boundaries. This allows the dependency graph to stabilise before the hardest algebraic proofs are attempted.

The minimal export should be:

```
import CsdLean4.LF3.Interface
```

The first compiling interface theorem may be stated schematically as:

```
theorem LF3_main_theorem :
  (forall s t,
    P_st ctx s t =
      (1 - sign s * sign t * dot ctx.a.vec ctx.b.vec) / 4)
  and
  correlation ctx =
    - dot ctx.a.vec ctx.b.vec
  and
  (forall s, marginal_A ctx s = 1 / 2)
  and
  (forall t, marginal_B ctx t = 1 / 2)
```

At the initial stage, this theorem may depend on earlier theorem targets as assumptions or local lemmas. The objective is to establish the formal endpoint first, then progressively discharge the assumptions.

C.3 Module Dependency Order

The recommended dependency order is as follows.

First:

`Setup.lean`

This file defines the sign type, detector settings, Pauli-direction operators, spin projectors, pointer projectors, and system-apparatus setup.

Second:

`Projectors.lean`

This file defines the pointer-sector projectors

$$\widehat{M}_{st}^{(a,b)}$$

and proves, or initially assumes, their orthogonality and completeness.

Third:

`Singlet.lean`

This file proves the singlet kernel

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4},$$

then derives

$$E(a, b) = -a \cdot b,$$

and

$$P_A(s \mid a, b) = P_B(t \mid a, b) = \frac{1}{2}.$$

Fourth:

`Hamiltonian.lean`

This file defines the local readout Hamiltonians and proves commutation:

$$[\widehat{H}_A(a), \widehat{H}_B(b)] = 0.$$

If operator exponentials are difficult, the unitary factorisation may initially be encoded as a structure field.

Fifth:

`BranchSeparation.lean`

This file defines the branch-separated state and the pointer leakage assumptions. It should provide the bridge from measurement dynamics to the branch-weight expressions used in `Projectors.lean`.

Sixth:

`ContextMap.lean`

This file defines context-indexed outcome maps and keeps them separate from global CHSH assignments.

Seventh:

`Interface.lean`

This file exports the final LF3 theorem package.

C.4 Suggested First-Pass sorry Boundary

The first implementation should allow `sorry` only at clearly identified mathematical boundaries. The recommended first-pass boundaries are:

1. tensor-product operator algebra;
2. construction of finite-dimensional pointer projectors from submodules;
3. operator exponential and unitary factorisation;
4. branch-separated action of the measurement unitary;
5. finite-leakage branch-weight estimate;
6. Fine-Bell no-global-assignment obstruction.

The following should not remain as long-term `sorry` targets:

1. finite sums over $\{\pm 1\}$;
2. normalisation of the four singlet probabilities;
3. derivation of $E(a, b) = -a \cdot b$ from the singlet kernel;
4. derivation of $P_A = P_B = \frac{1}{2}$ from the singlet kernel;
5. orthogonality and completeness of an explicitly packaged four-projector family.

These latter results are finite algebraic checks and should be discharged early.

C.5 Recommended First Theorems to Prove

The first theorem to prove should be the sign-sum identity:

$$\sum_{s,t \in \{\pm 1\}} 1 = 4.$$

The second should be

$$\sum_{s,t \in \{\pm 1\}} st = 0.$$

The third should be

$$\sum_{s,t \in \{\pm 1\}} (st)^2 = 4.$$

These elementary identities immediately support normalisation and correlation.

Next prove that the singlet kernel is normalised:

$$\sum_{s,t \in \{\pm 1\}} \frac{1 - st a \cdot b}{4} = 1.$$

Then prove the correlation:

$$\sum_{s,t \in \{\pm 1\}} st \frac{1 - st a \cdot b}{4} = -a \cdot b.$$

Then prove the marginals:

$$\sum_{t \in \{\pm 1\}} \frac{1 - st a \cdot b}{4} = \frac{1}{2},$$

and

$$\sum_{s \in \{\pm 1\}} \frac{1 - st a \cdot b}{4} = \frac{1}{2}.$$

These are the easiest high-value theorem targets. They give LF3 an early successful endpoint before the more difficult Hamiltonian and tensor-product formalisation is attempted.

C.6 Implementation Stage 1: Algebraic Kernel

The first implementation stage should focus on the algebraic singlet kernel rather than on the full Hamiltonian model. This is the fastest route to a useful LF3 milestone.

The target theorem is

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

Once this theorem is available, the correlation and marginal results are finite sums.

The recommended Lean path is:

1. define the two-element sign type;
2. define $\text{sgn}(s) \in \{-1, +1\}$;
3. define the kernel

$$K(a, b, s, t) = \frac{1 - \text{sgn}(s) \text{sgn}(t) a \cdot b}{4};$$

4. prove kernel normalisation;
5. prove the correlation identity;
6. prove the marginal identities.

This gives a compiling mathematical core even before the full pointer-sector construction is implemented.

A schematic kernel definition is:

```
def singletKernel
  (a b : DetectorSetting)
  (s t : Sign) : Real :=
  (1 - sign s * sign t * dot a.vec b.vec) / 4
```

The first theorem is normalisation:

```
theorem singletKernel_sum_one :
  sum over s,t of singletKernel a b s t = 1
```

The second theorem is the correlation identity:

```
theorem singletKernel_correlation :
  sum over s,t of
    sign s * sign t * singletKernel a b s t
  =
    - dot a.vec b.vec
```

The third and fourth theorems are the marginals:

```
theorem singletKernel_marginal_A :
  sum over t of singletKernel a b s t = 1 / 2
```

```
theorem singletKernel_marginal_B :
  sum over s of singletKernel a b s t = 1 / 2
```

This stage proves the observable Bell-sector consequences of the kernel. It does not yet prove that the kernel arises from the pointer Hamiltonian. That connection is added in later stages.

C.7 Implementation Stage 2: Pointer Projectors

The second implementation stage should introduce the pointer-sector projectors

$$\widehat{M}_{st}^{(a,b)} = I_{AB} \otimes \widehat{Q}_s^A \otimes \widehat{Q}_t^B.$$

The immediate target is to prove that these projectors form a complete orthogonal readout.

If constructing the projectors from subspaces is difficult in the first pass, define an abstract structure:

```
structure FourOutcomeProjectiveReadout where
  M : Sign -> Sign -> Operator H
  idem :
    forall s t, M s t * M s t = M s t
  selfAdjoint :
    forall s t, IsSelfAdjoint (M s t)
  orthogonal :
    forall s t s' t',
      (s,t) != (s',t') ->
        M s t * M s' t' = 0
  complete :
    sum over s,t of M s t = 1
```

This structure lets the formalisation proceed while preserving the exact assumptions needed from the pointer readout.

The later refinement can replace the abstract projective readout with an explicit construction from

$$\widehat{Q}_+^A, \widehat{Q}_-^A, \widehat{Q}_+^B, \widehat{Q}_-^B.$$

The first theorem in this stage should be:

```
theorem pointer_readout_complete :
  sum over s,t of M_hat S s t = 1
```

The second should be:

```
theorem pointer_readout_orthogonal :
  (s,t) != (s',t') ->
    M_hat S s t * M_hat S s' t' = 0
```

The third should connect projectors to weights:

```
theorem pointer_branch_weight_def :
  branchWeight Psi s t =
    realPart (inner Psi (M_hat S s t Psi))
```

This stage establishes the projective measurement layer independently of the Hamiltonian details.

C.8 Implementation Stage 3: Branch Weights

The third stage connects the pointer-sector projectors to the singlet kernel.

The target is:

$$P_{st}(a,b) = |c_{st}(a,b)|^2.$$

Then, using the singlet theorem,

$$|c_{st}(a,b)|^2 = \frac{1 - st a \cdot b}{4},$$

one obtains

$$P_{st}(a,b) = \frac{1 - st a \cdot b}{4}.$$

The recommended theorem is:

```

theorem branchWeight_eq_cst_normSq :
  branchWeight Psi_T s t =
    normSq (c_st a b s t)

```

Then:

```

theorem branchWeight_eq_singletKernel :
  branchWeight Psi_T s t =
    singletKernel a b s t

```

This stage may initially assume the branch-separated final-state form:

$$|\Psi_T\rangle = \sum_{s,t} c_{st}(a,b) |B_{st}(a,b;T)\rangle.$$

That assumption is later discharged by the Hamiltonian and branch-separation modules.

C.9 Implementation Stage 4: Hamiltonian Control

The fourth stage introduces the readout Hamiltonian:

$$\hat{H}_{a,b} = \hat{H}_A(a) + \hat{H}_B(b).$$

The key theorem is:

$$[\hat{H}_A(a), \hat{H}_B(b)] = 0.$$

This gives unitary factorisation:

$$\hat{U}_{a,b}(T) = \hat{U}_A(a;T) \hat{U}_B(b;T).$$

The first-pass theorem target is:

```

theorem H_AB_commute :
  H_A S kappa_A a p_A * H_B S kappa_B b p_B =
  H_B S kappa_B b p_B * H_A S kappa_A a p_A

```

If operator exponentials are not convenient, use an abstract unitary factorisation structure:

```

structure FactorisedMeasurementUnitary where
  U : Operator H
  U_A : Operator H
  U_B : Operator H
  unitary_U : IsUnitary U
  unitary_A : IsUnitary U_A
  unitary_B : IsUnitary U_B
  factorises : U = U_A * U_B

```

This is acceptable for LF3 v1.00 if the abstraction is clearly stated.

C.10 Implementation Stage 5: Branch Separation and Leakage

The fifth stage formalises branch separation and finite leakage.

The ideal branch-separation theorem is:

```

theorem finalState_branch_decomposition :
  finalState S a b T =
    sum over s,t of
      (c_st a b s t) • branchState S a b T s t

```

The finite-leakage assumptions are packaged as:


```

structure PointerLeakageBounds where
  epsilon_A : Real
  epsilon_B : Real
  nonneg_A : 0 <= epsilon_A
  nonneg_B : 0 <= epsilon_B
  branch_bound :
    forall s t,
      abs
        (P_st_T a b T s t - P_st a b s t)
      <=
        epsilon_A + epsilon_B + epsilon_A * epsilon_B
    
```

The finite-leakage theorem then becomes:

```

theorem LF3_finite_leakage_theorem :
  abs
    (P_st_T a b T s t
     - singletKernel a b s t)
  <=
    epsilon_A + epsilon_B + epsilon_A * epsilon_B
    
```

The correlation and marginal bounds follow by finite summation:

```

theorem correlation_leakage_bound :
  abs (correlation_T a b T + dot a.vec b.vec)
  <= 4 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)

theorem marginal_A_leakage_bound :
  abs (marginal_A_T a b T s - 1 / 2)
  <= 2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)

theorem marginal_B_leakage_bound :
  abs (marginal_B_T a b T t - 1 / 2)
  <= 2 * (epsilon_A + epsilon_B + epsilon_A * epsilon_B)
    
```

This stage completes the finite-error layer.

C.11 Criteria for LF3 v1.00 Completion

LF3 v1.00 should be considered complete when the repository exports the following results.

First, the projective readout theorem:

$$\left\{ \widehat{M}_{st}^{(a,b)} : s, t \in \{\pm 1\} \right\}$$

is complete and orthogonal.

Second, the singlet kernel theorem:

$$P_{st}(a, b) = \frac{1 - st a \cdot b}{4}.$$

Third, the correlation theorem:

$$E(a, b) = -a \cdot b.$$

Fourth, the marginal theorems:

$$P_A(s \mid a, b) = \frac{1}{2}, \quad P_B(t \mid a, b) = \frac{1}{2}.$$

Fifth, the finite-leakage theorem:

$$\left| P_{st}(a, b; T) - \frac{1 - st a \cdot b}{4} \right| \leq \epsilon_A + \epsilon_B + \epsilon_A \epsilon_B.$$

Sixth, the finite-leakage correlation and marginal bounds:

$$|E_T(a, b) + a \cdot b| \leq 4(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

$$\left| P_A(s \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B),$$

and

$$\left| P_B(t \mid a, b; T) - \frac{1}{2} \right| \leq 2(\epsilon_A + \epsilon_B + \epsilon_A \epsilon_B).$$

Seventh, the context-indexed readout structure:

$$F^{(a,b)} : \mathcal{D}_{a,b} \rightarrow \{\pm 1\} \times \{\pm 1\}.$$

These results are sufficient for LF3 v1.00. Full derivation of tensor-product operator identities, explicit operator exponentials, and the Fine-Bell theorem may remain deferred if their boundaries are stated clearly.

C.12 Summary

The recommended implementation strategy is to build LF3 from the algebraic kernel outward. The fastest successful path is:

sign sums $\rightarrow$ singlet kernel $\rightarrow$ correlation and marginals $\rightarrow$ pointer projectors $\rightarrow$ branch weights $\rightarrow$ finite leakage $\rightarrow$ Hamiltonian con

This order gives useful verified results early while preserving the full TN3-inspired architecture. It also reduces implementation risk because the hardest library-dependent components, such as tensor products and operator exponentials, do not block the first formal theorem package.

LF3 v1.00 therefore provides a complete, compiling theorem interface with explicit assumption boundaries. Later repository refinements may reduce abstraction in the tensor-product and operator-exponential layers, but they do not alter the exported theorem package.

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